

When Neutron Stars Collide: Unravelling one of the Most Extreme Dynamos in the Universe

James Beattie

1st year NASA Hubble Fellow | Member

Institute for Advanced Study
NHFP Symposium, 2026

Collaborators for this work:

Elias Most (Caltech), Amitava Bhattacharjee (PU), Bart Ripperda (CITA), Alexander Philippov (Stanford), Neco Kriel (UBonn), Ralf Klessen (UHeidelberg), Salvatore Cielo (LRZ)

A little about my science interests

- Fundamentals of astrophysical / space turbulence
 - Host of the world's largest supersonic turbulence simulations
- Magnetic dynamos (more on this today)
- Laboratory astrophysics (HEDLA)
- Magnetogenesis
- Visco-resistive (causal) relativistic MHD for plasma around compact objects

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- Fundamentals of astrophysical / s
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- plasma around compact objects

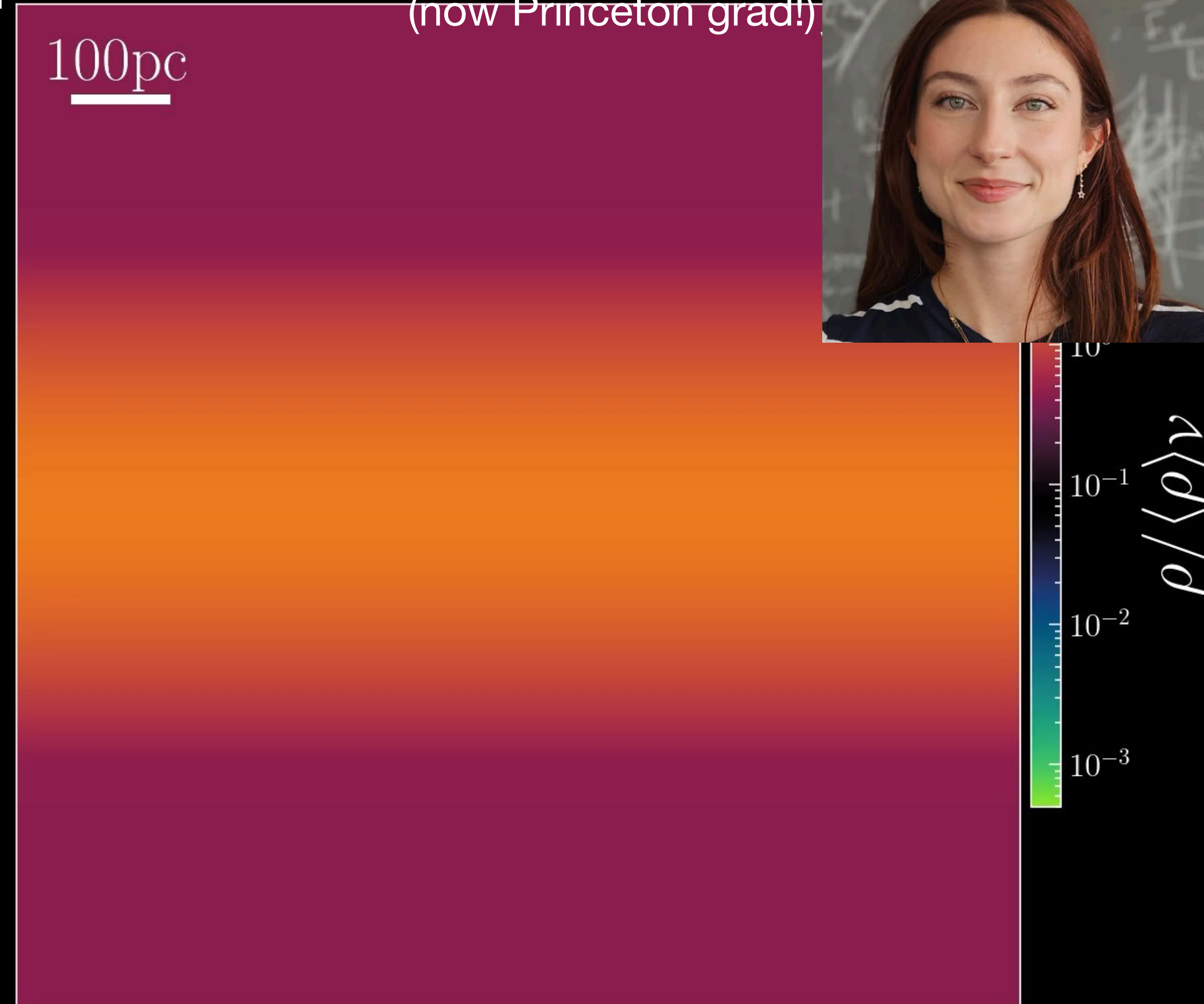
6PB (160Mcore-h) turbulence dataset,
now utilizing for doing detailed
comparisons with MMS (Earth's magnetosheath)



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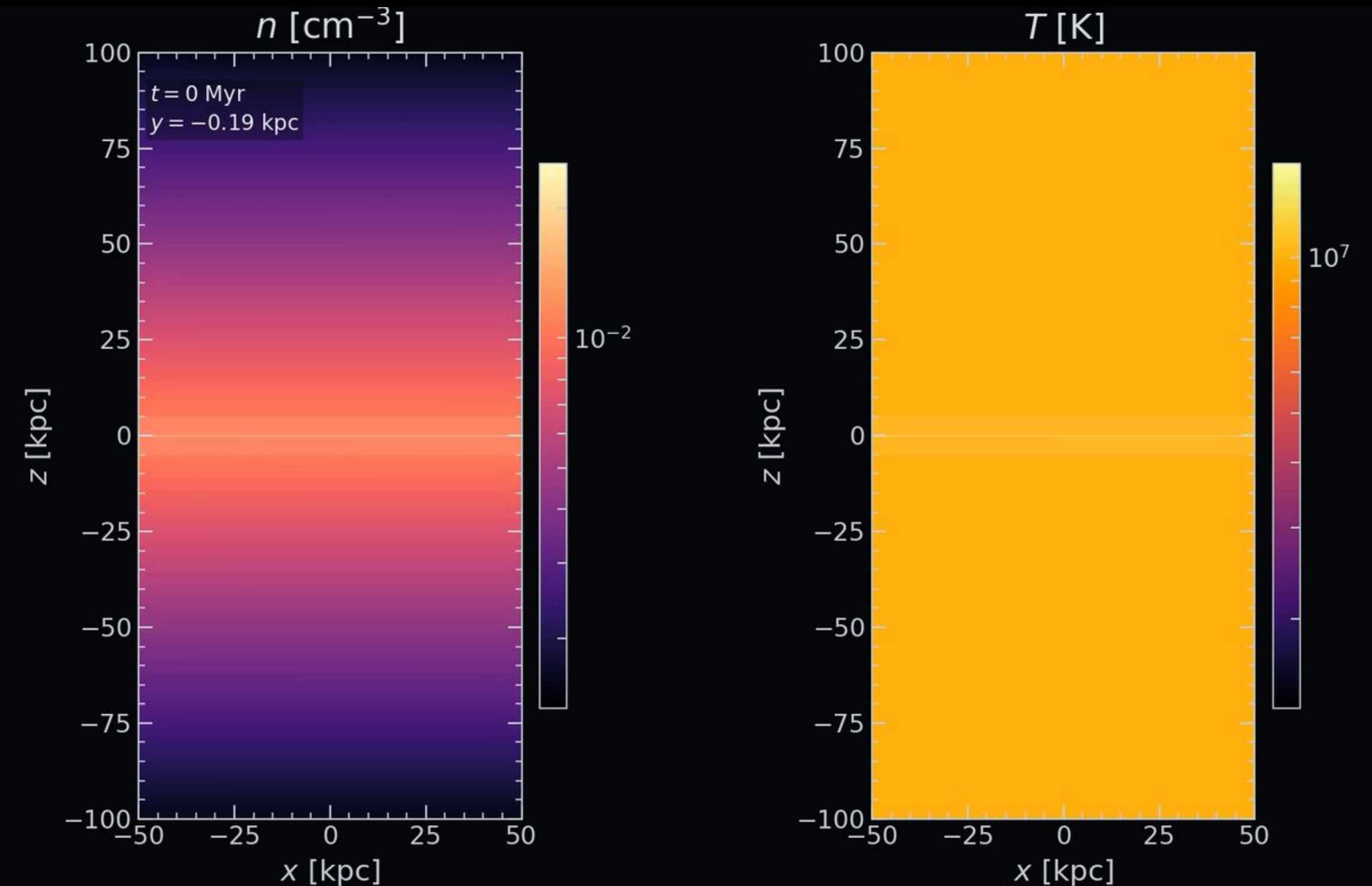
Izzy Connor
(now Princeton grad!)



ISM / supernova-driven turbulence — completely different from classical models! w. Isabelle Connor (prev. UG, UCSC; now PU grad.)

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CGM / ICM turbulence — with stable entropy profiles!!
w. William Grant (UG; Amherst College)

Astrophysics simulations on quantum computers (wishful thinking!)
... but future-proofing numerical astrophysics :P



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OPEN ACCESS

Compressible Navier-Stokes flow in Schrödinger-type variables

[James R. Beattie](#) ^{1,2,*}, [Max Sokolova](#) ³, [Khush Negandhi](#) ³, and [Bart Ripperda](#) ^{1,3,4,5}

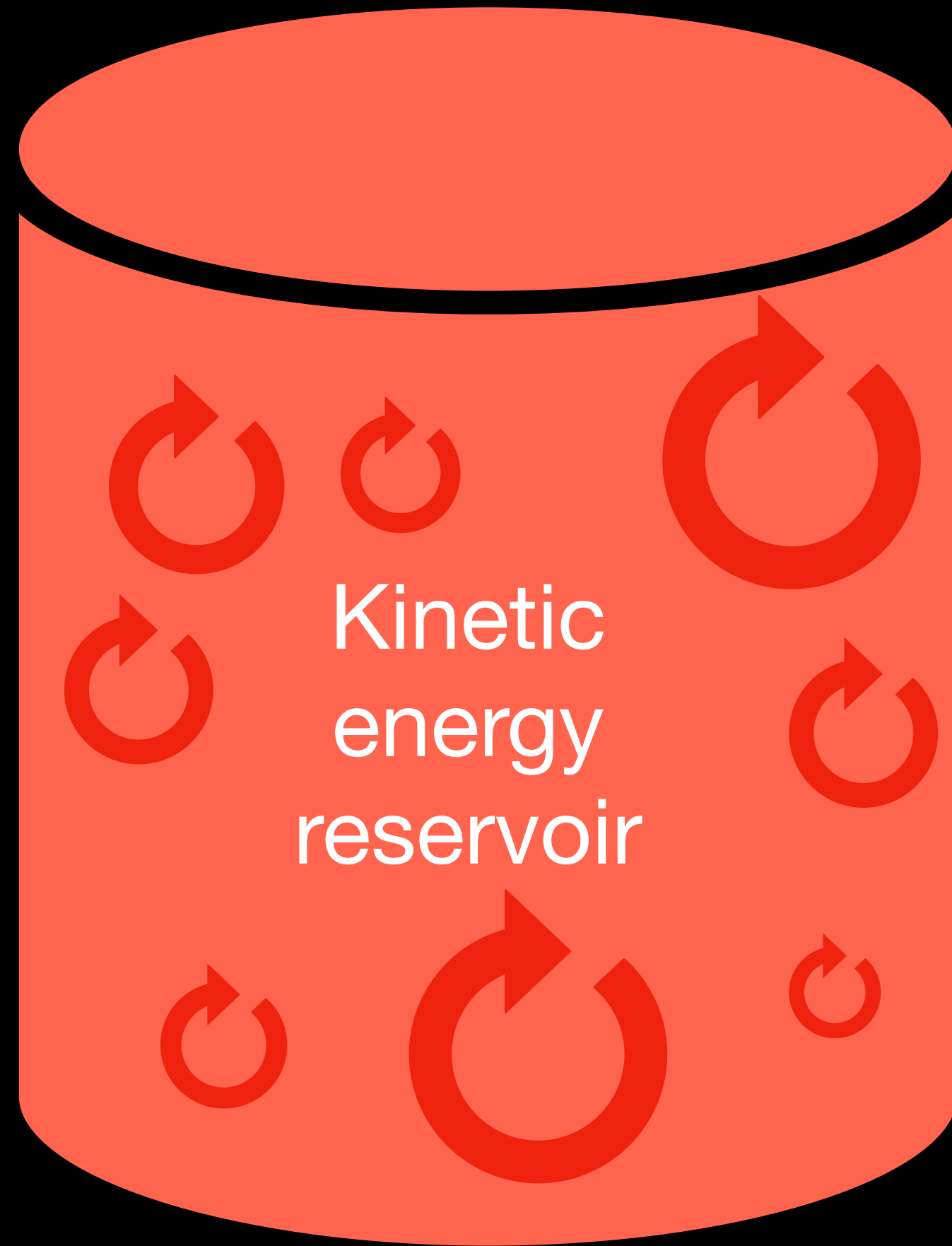
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Phys. Rev. Research **8**, 033383 – **Published 30 September, 2026**

DOI: <https://doi.org/10.1103/6gnr-vkx3>

First things first: what is a magnetic dynamo?

Starting with a seed magnetic field



Kinetic
energy
reservoir

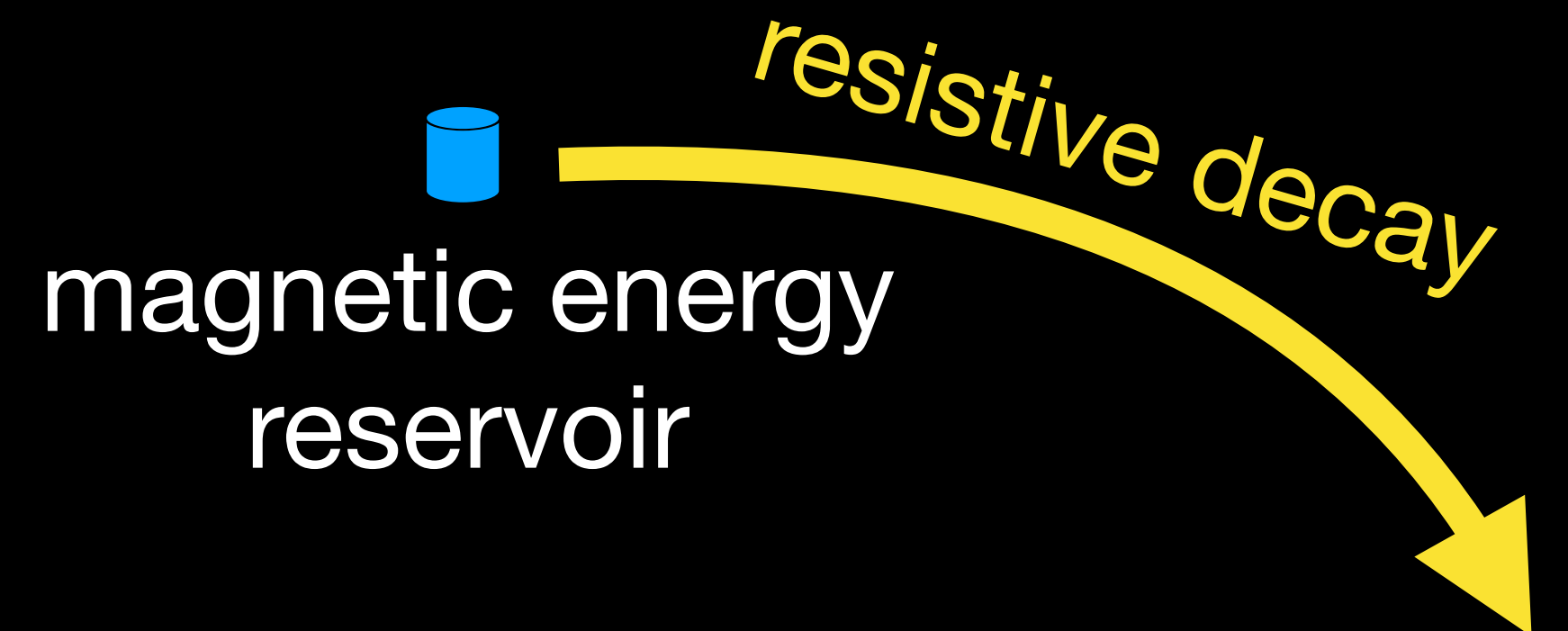
 U

characteristic velocity

 L

characteristic size-scale

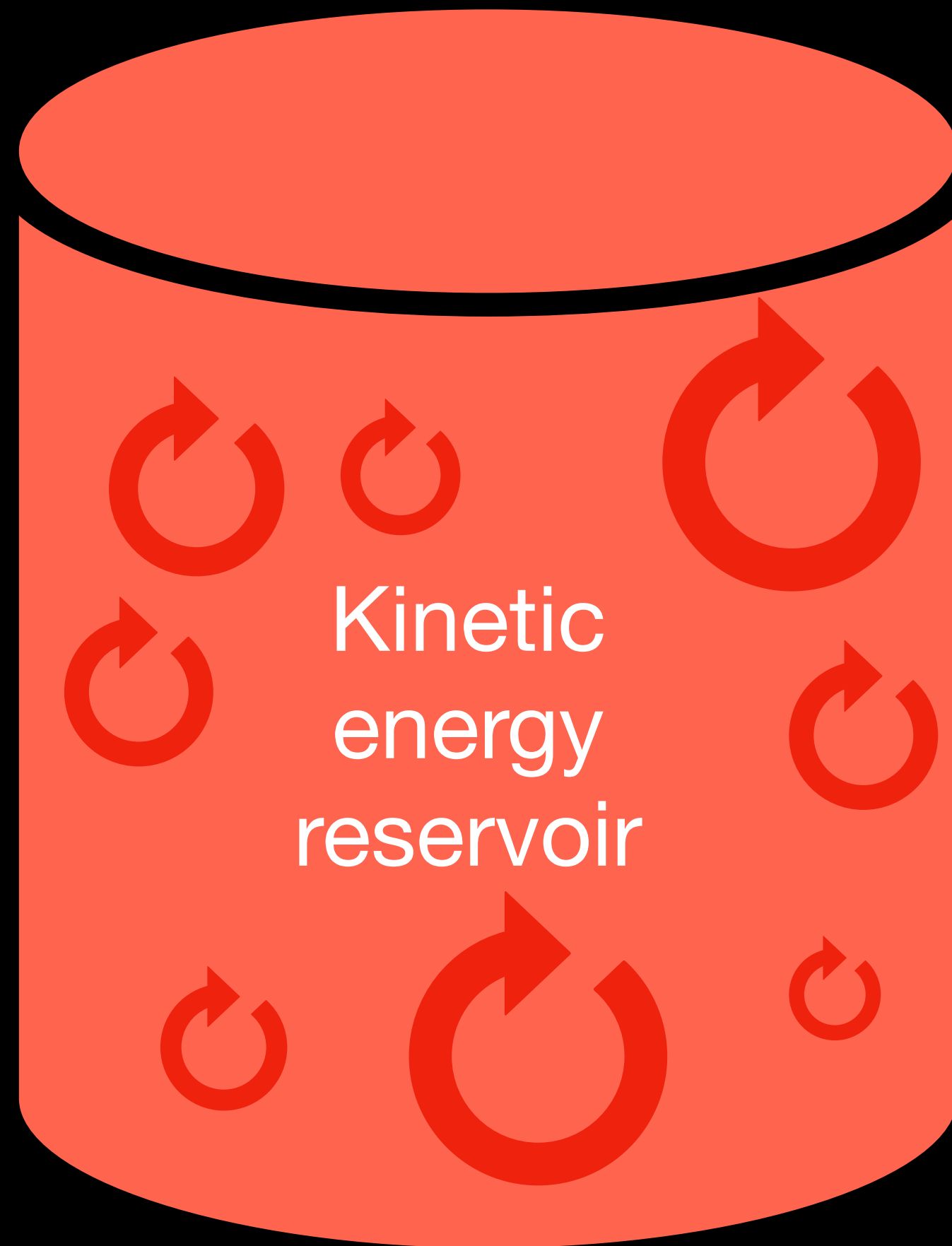
$$U^2 \gg B^2$$



magnetic energy
reservoir

What is a magnetic dynamo?

Starting with a seed magnetic field



Weibel instability, (Zhou + 2023, 2024)

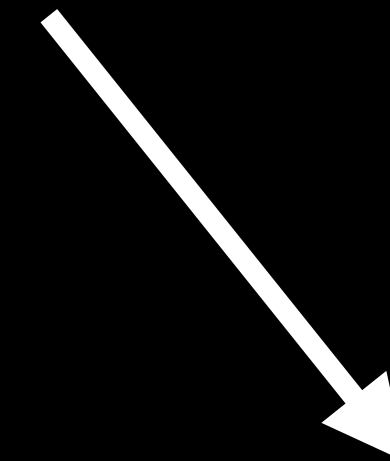
Biermann battery,

Unstable cooling shockwaves, (Beattie+ in prep.)

Dust battery (Soliman+, 2024)

...

come to IAS and discuss :)



magnetic energy
reservoir

resistive decay



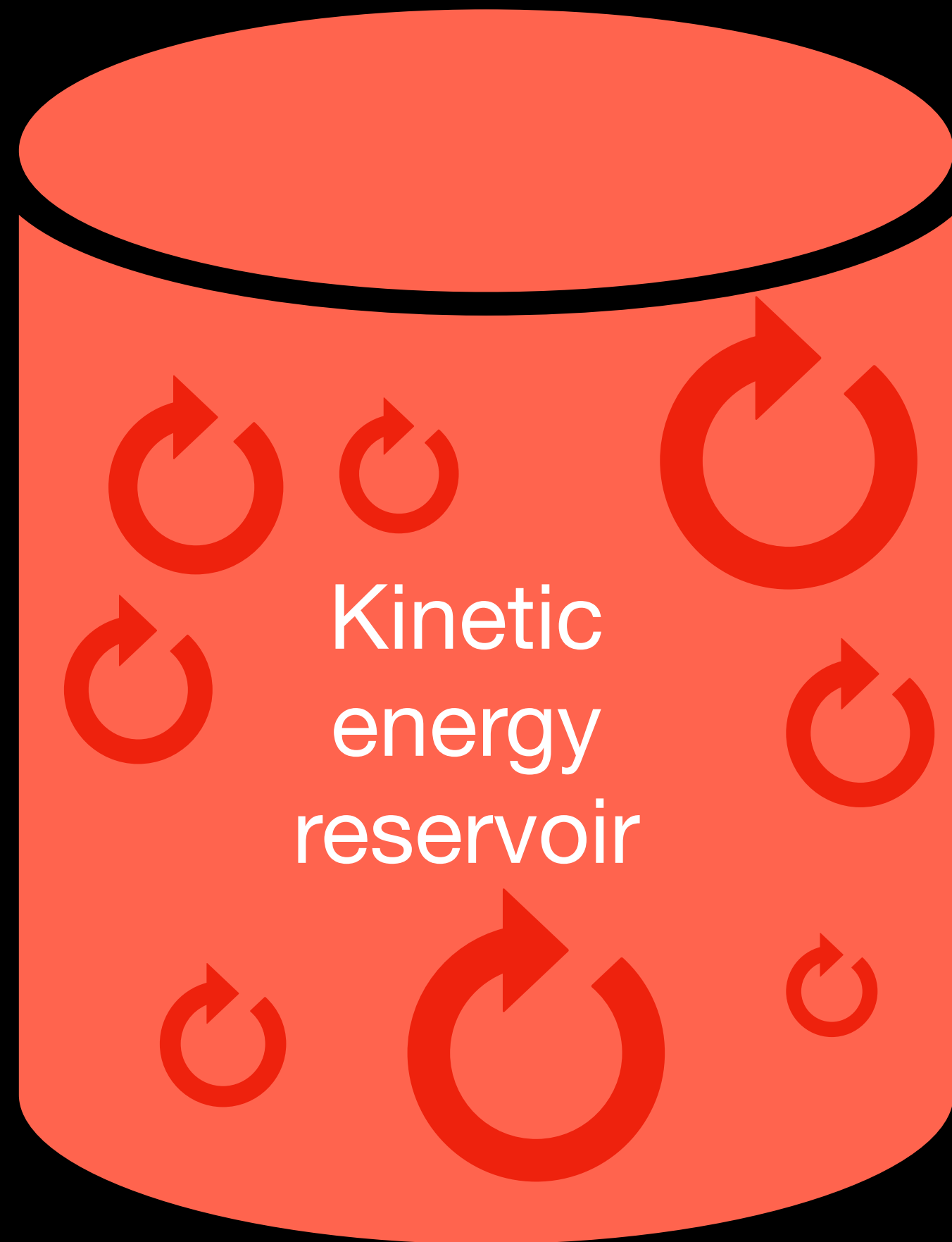
$$U^2 \gg B^2$$

What is a magnetic dynamo?

Kinematic growth

$$Rm \sim \frac{UL}{\eta} = \frac{\text{induction}}{\text{diffusion}} > Rm_{\text{crit}}$$

can't be too resistive



energy flux in

$$B_i B_j \partial_j u_i$$

$$\eta B_i \partial_j \partial_j B_i$$

resistive decay

magnetic energy
reservoir

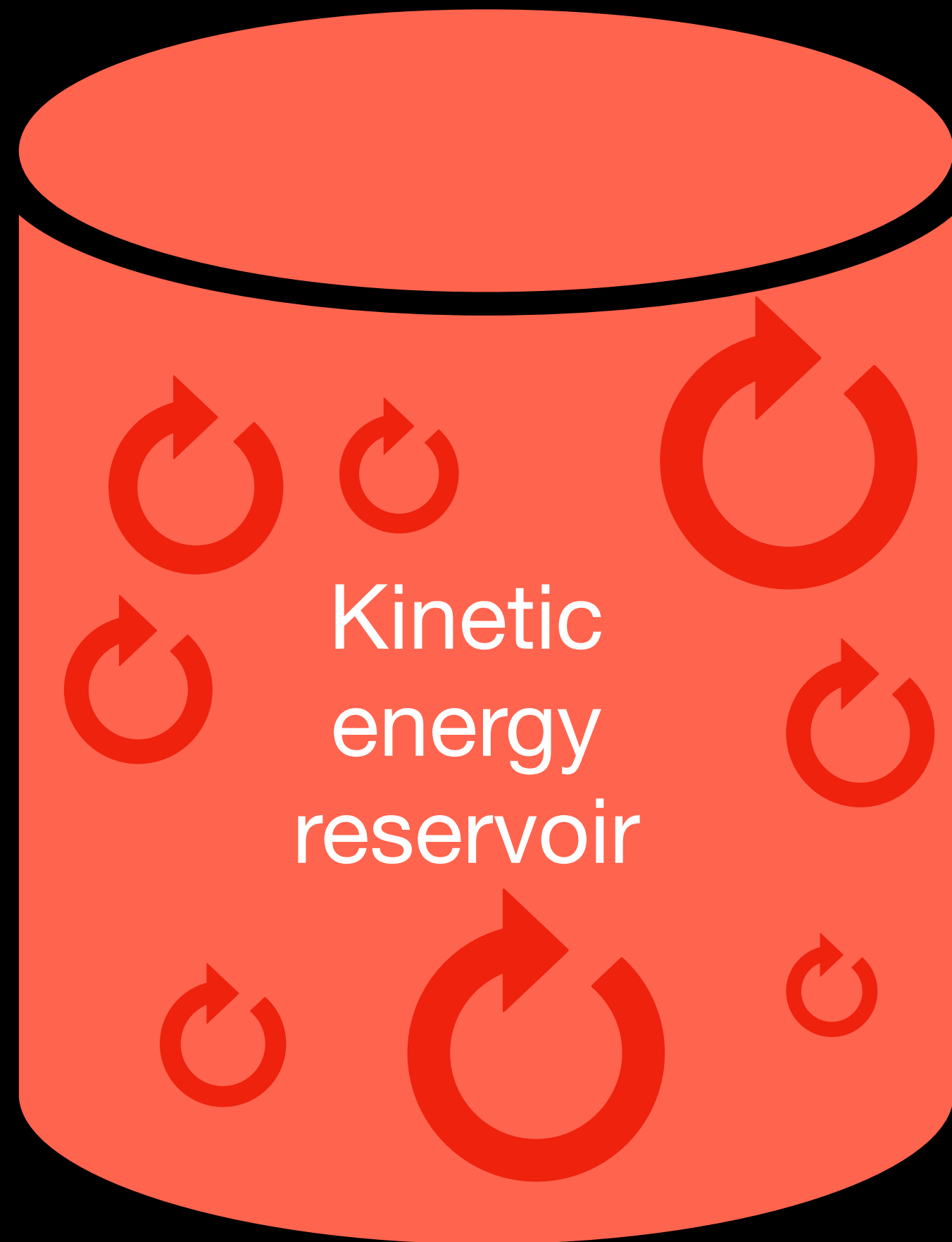
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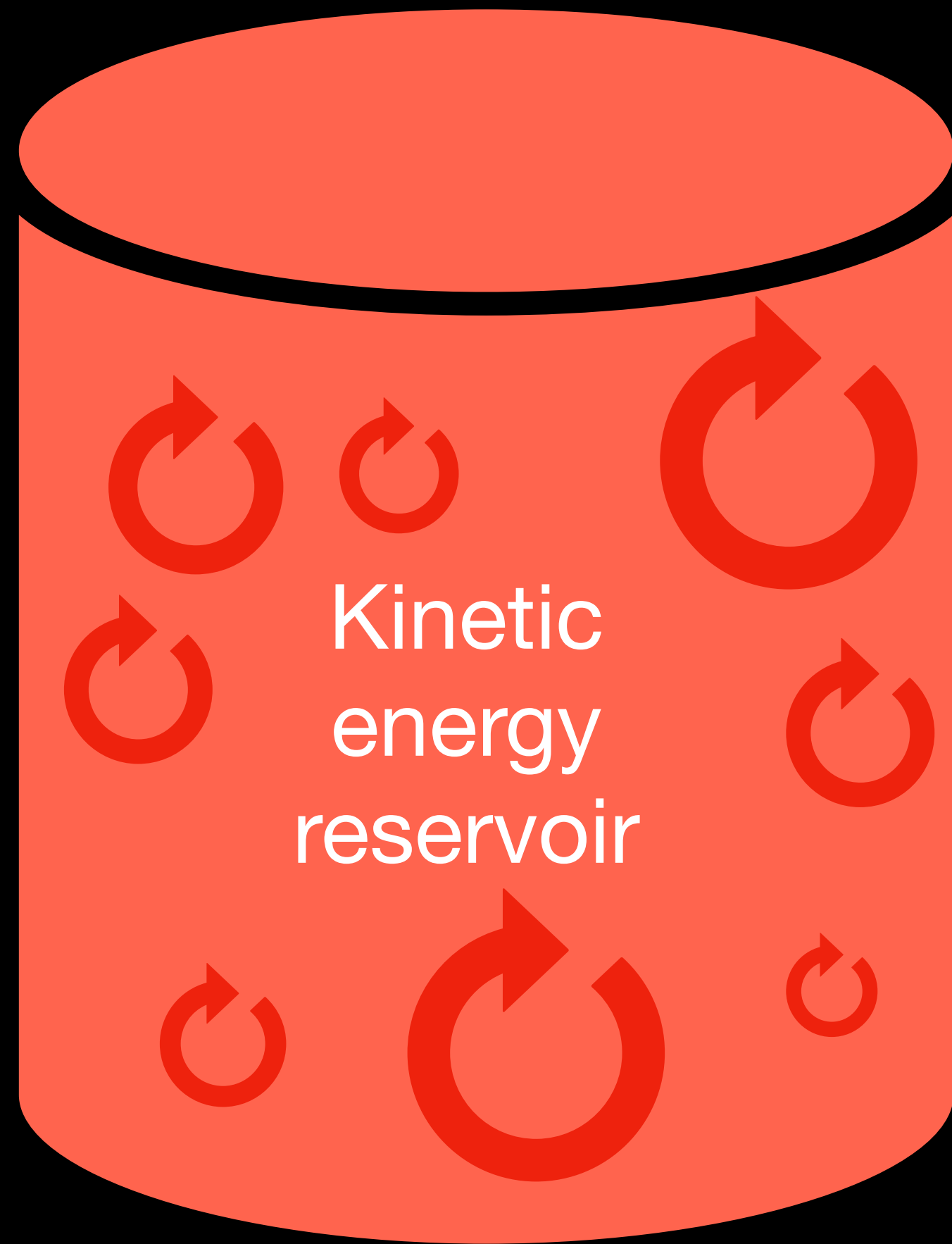
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Kinematic growth

Kriel, Beattie + (2026; PRD)

$$\gamma \sim \text{Re}^{1/2} = \left(\frac{UL}{\nu} \right)^{1/2}$$

$$\sim \frac{\text{advection}}{\text{diffusion}}$$



Kinetic
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reservoir

energy flux in

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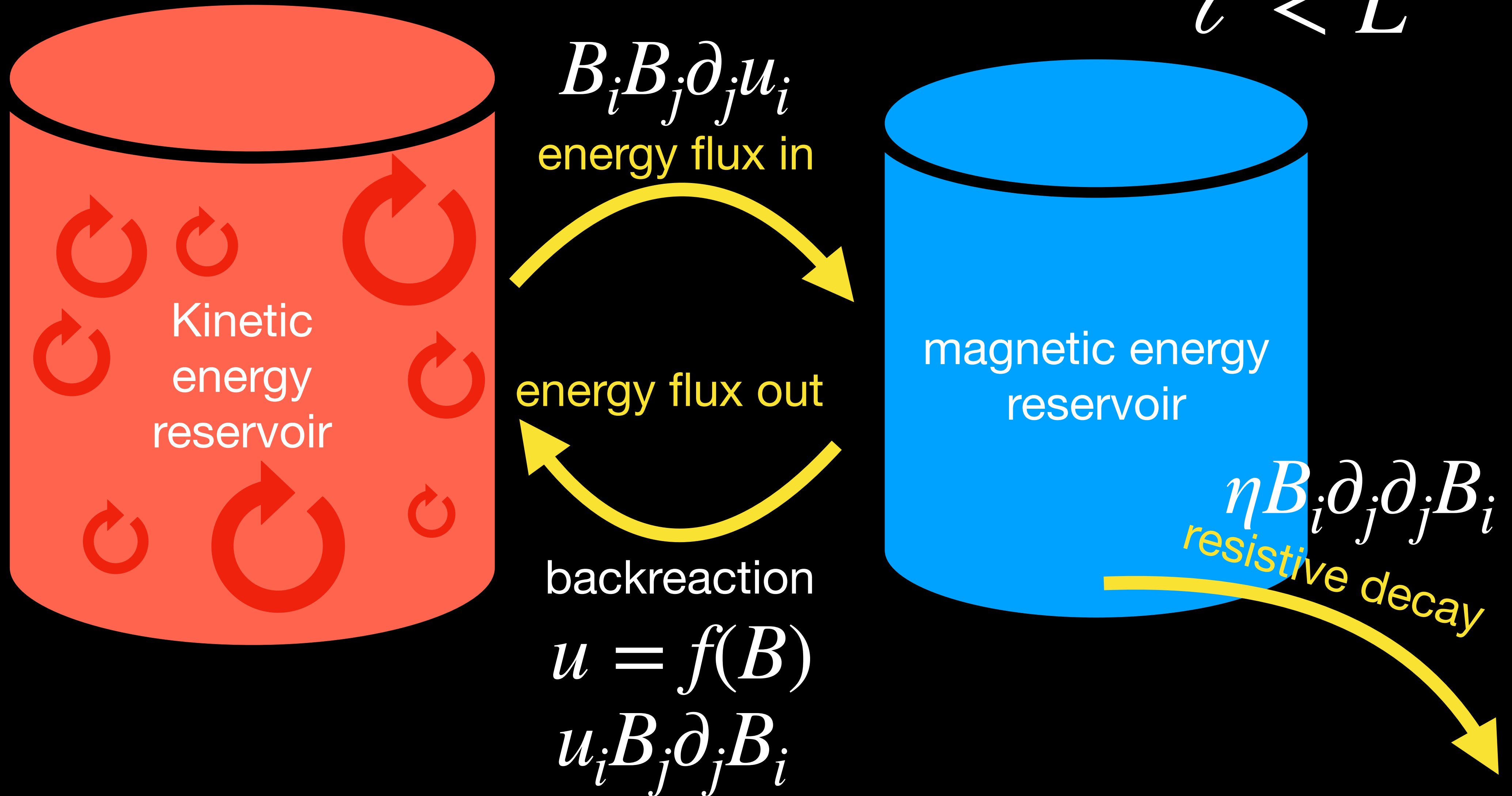
$$U^2 \gg B^2$$

What is a magnetic dynamo?

Nonlinearities and backreaction

$$U_\ell^2 \sim B^2$$

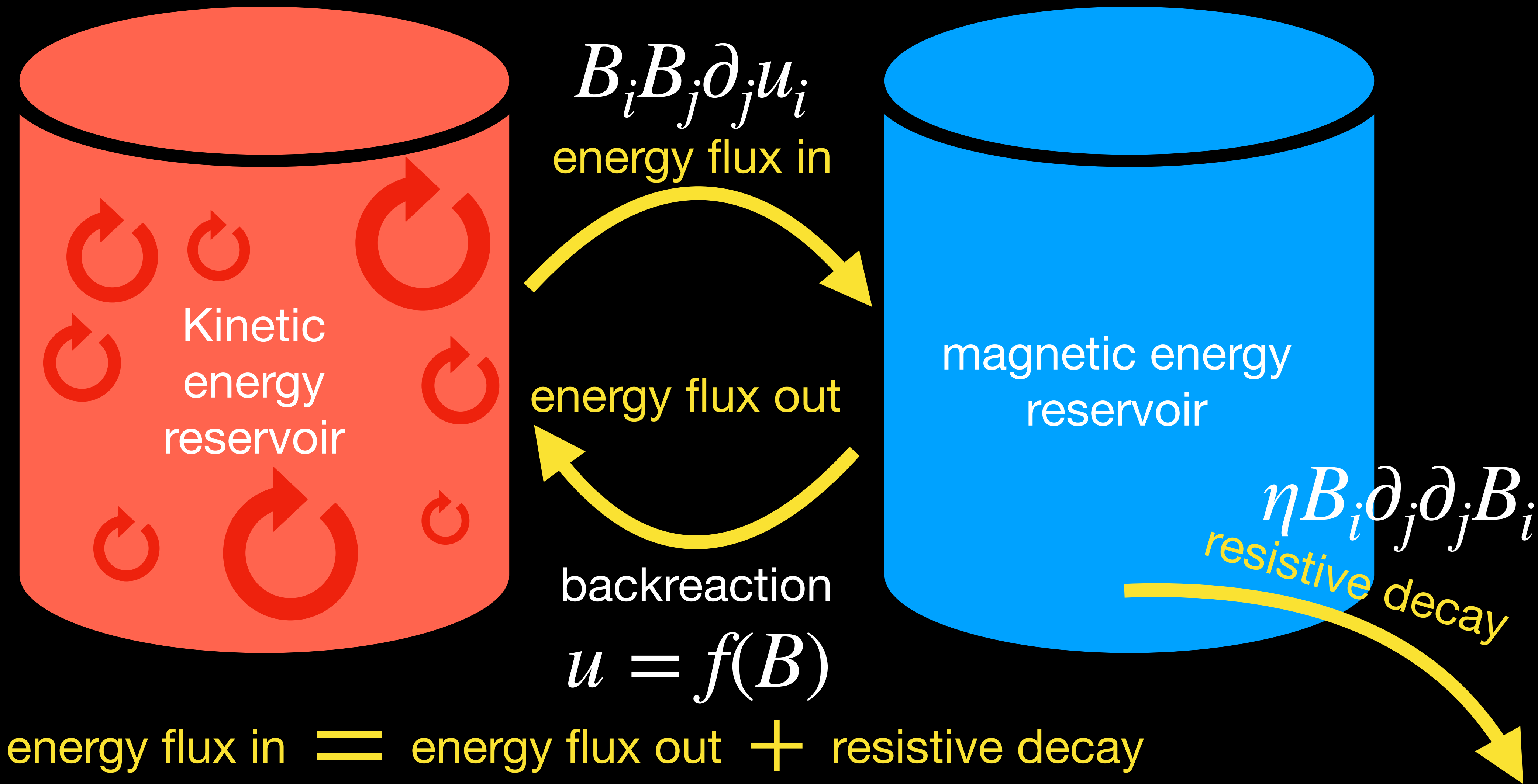
$$\ell < L$$



What is a magnetic dynamo?

Saturation

$$U^2 \sim B^2$$



Examples of dynamos

Milky Way-type galaxies in cosmological sims

exponential growth



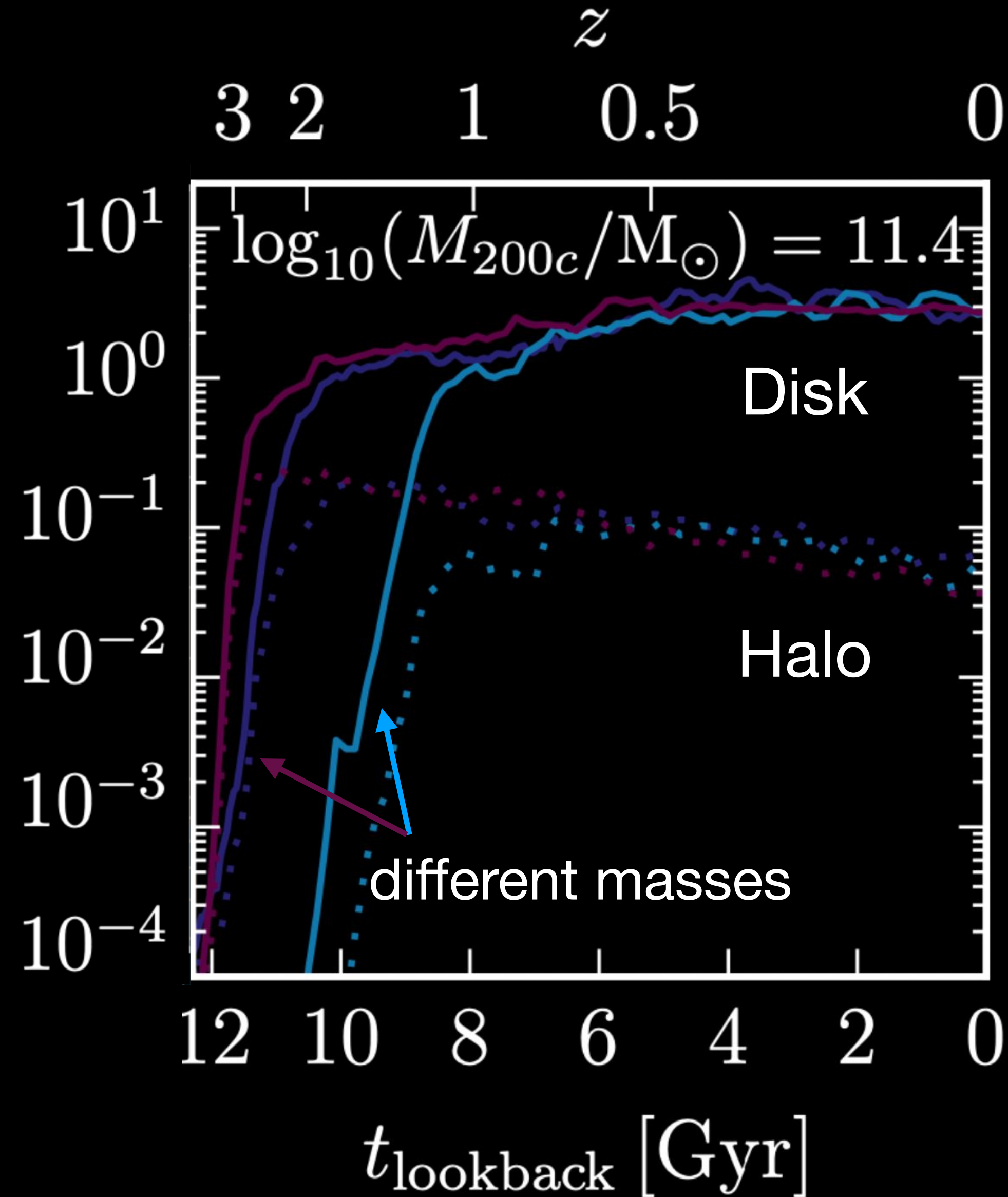
linear growth



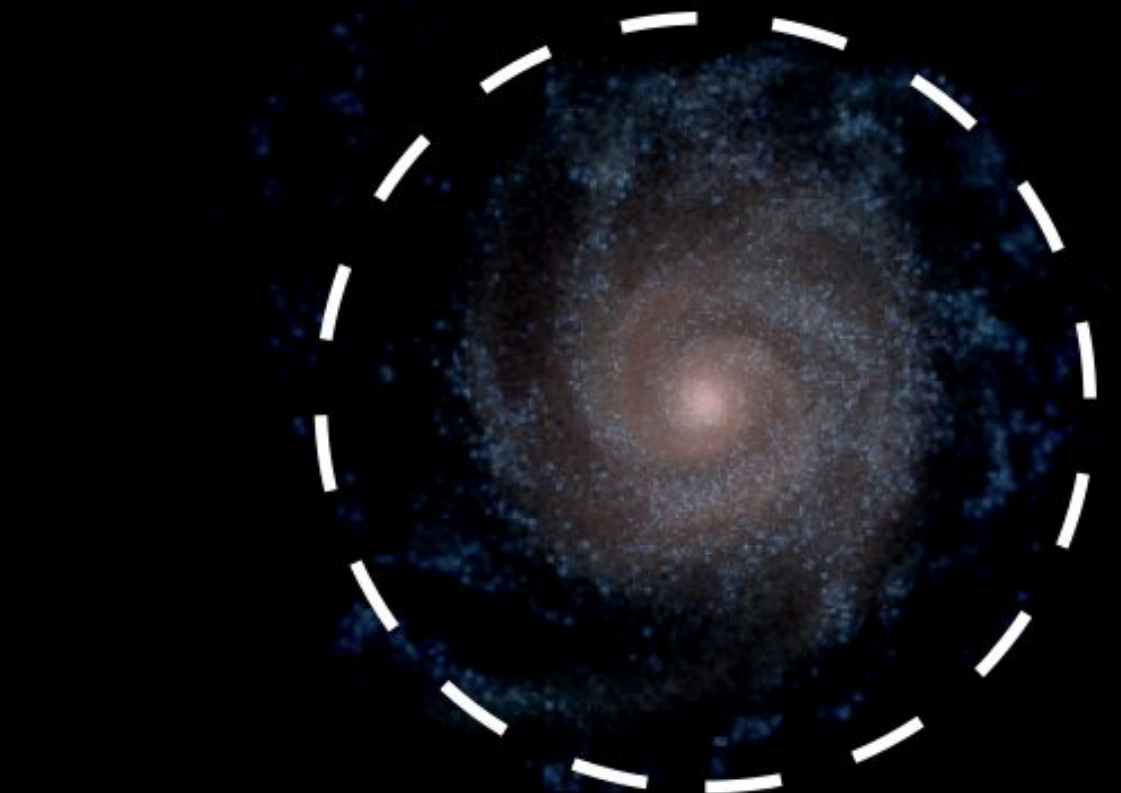
saturation

$$\mathcal{E}_{\text{kin}} \sim \mathcal{E}_{\text{mag}}$$

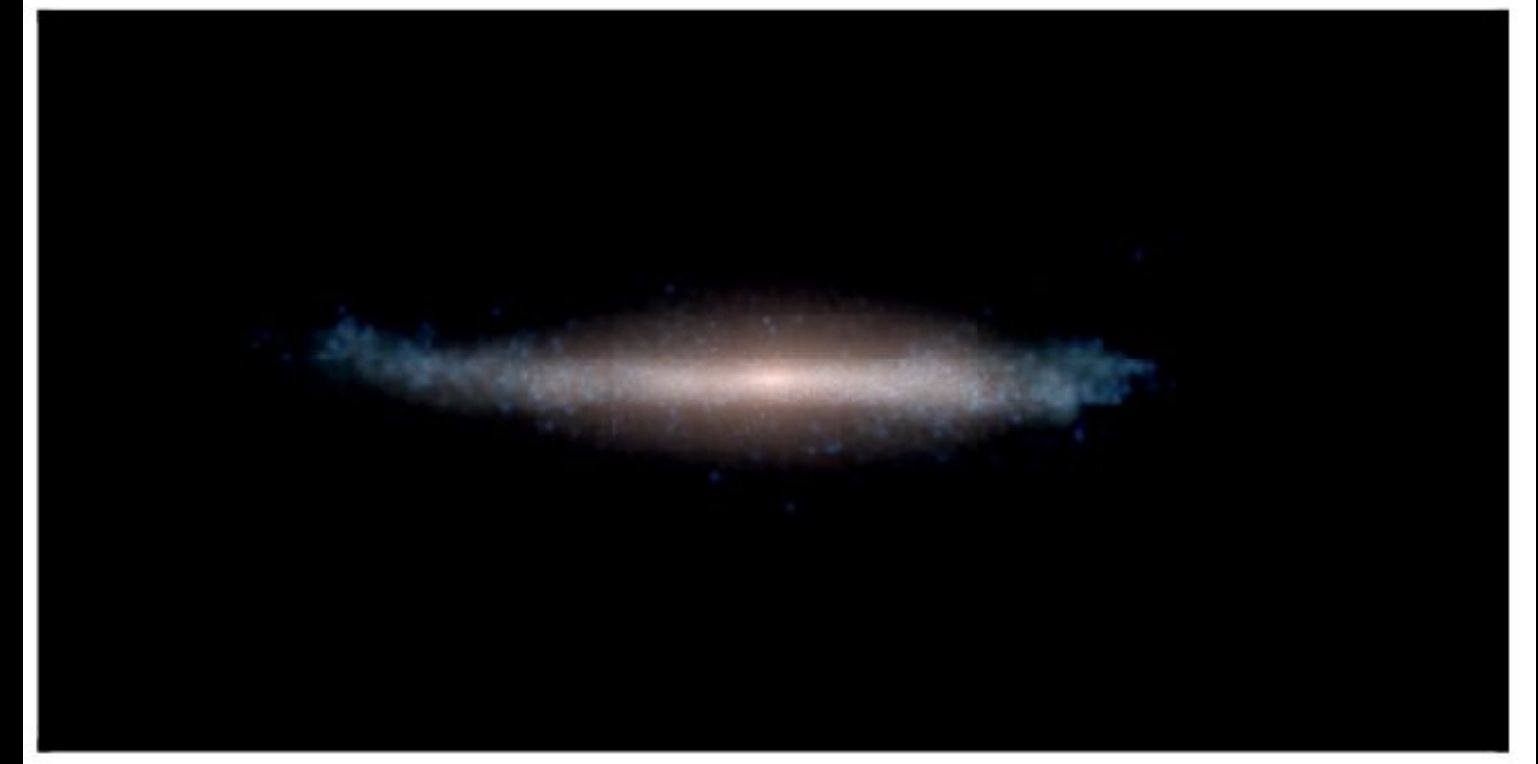
B [μG]



$$\log_{10}(M_{200c}/M_{\odot}) = 11.4$$



5kpc



Pakmor+ (2024) *AREPO*

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Milky Way-type galaxies in cosmological sims

exponential growth



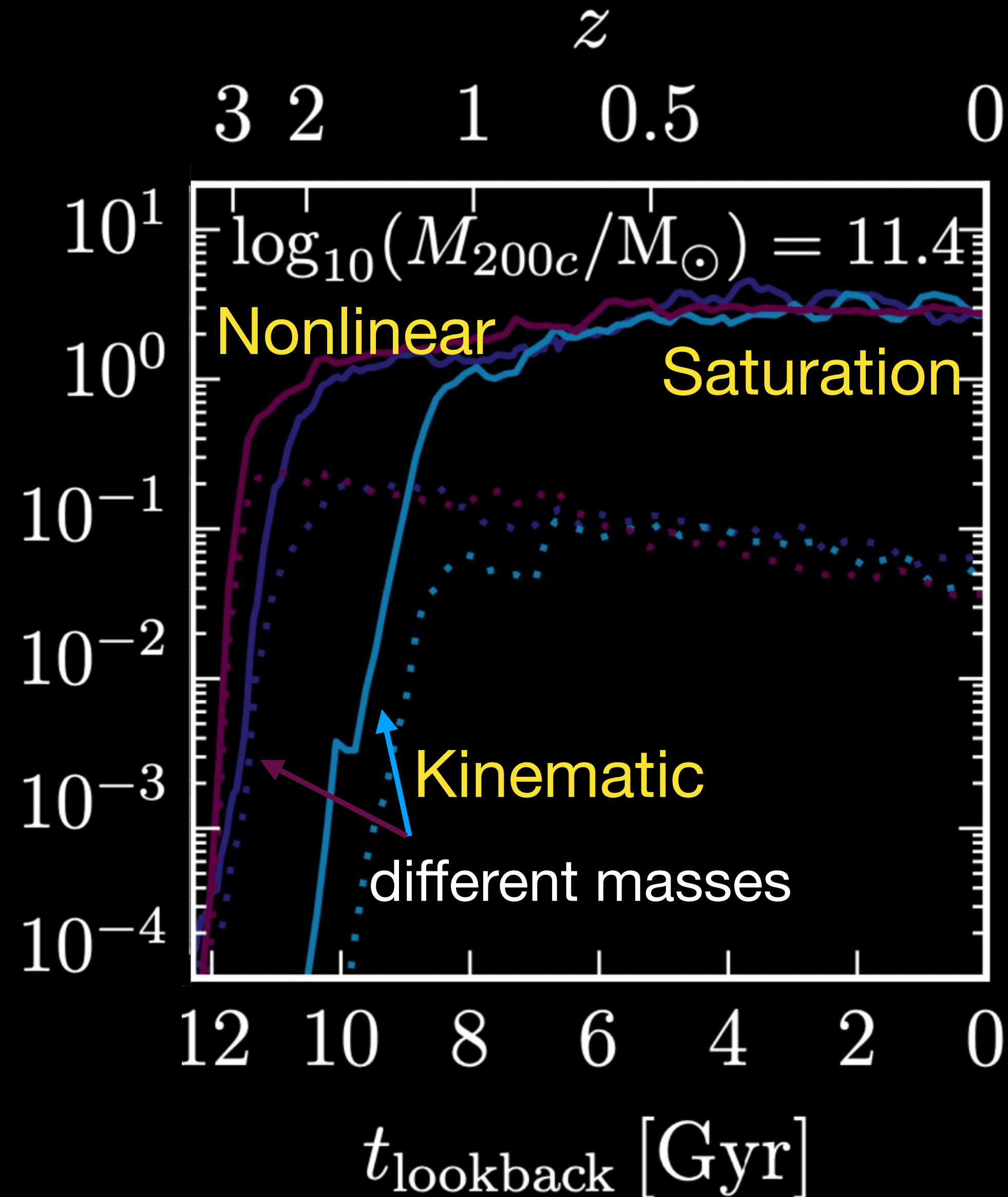
linear growth



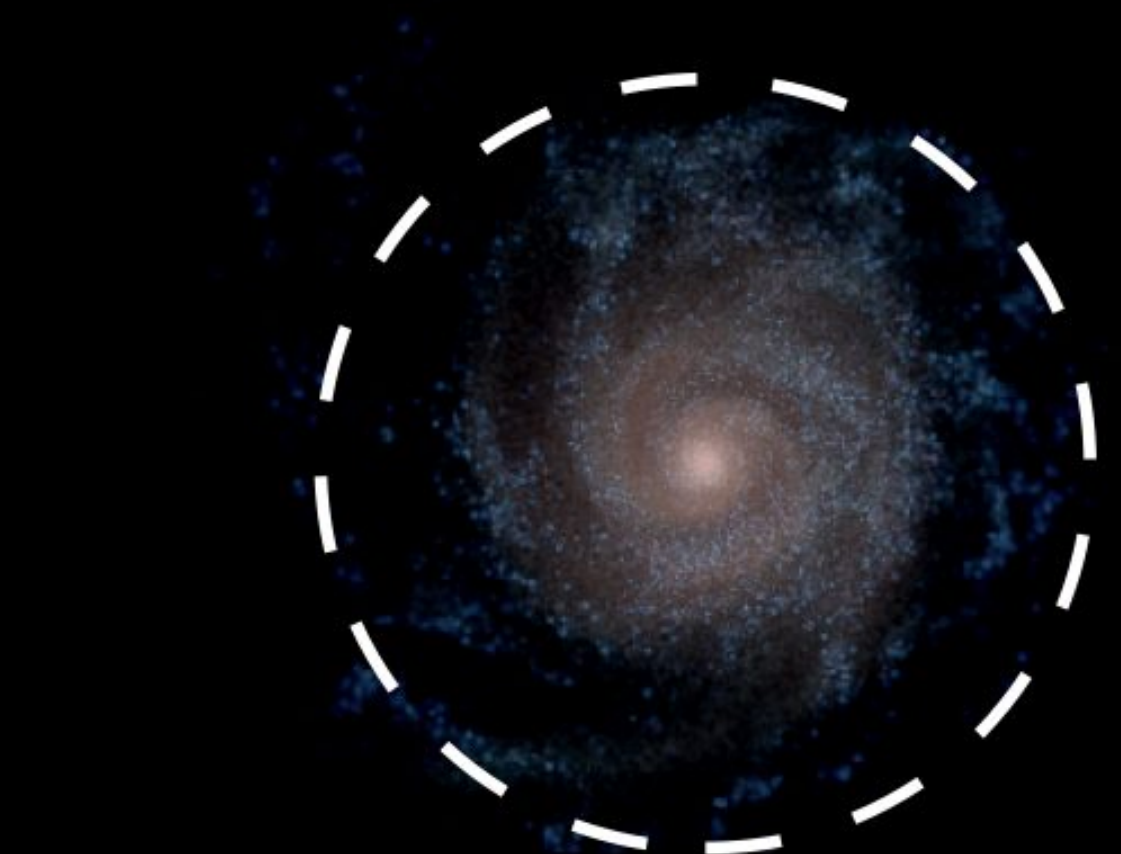
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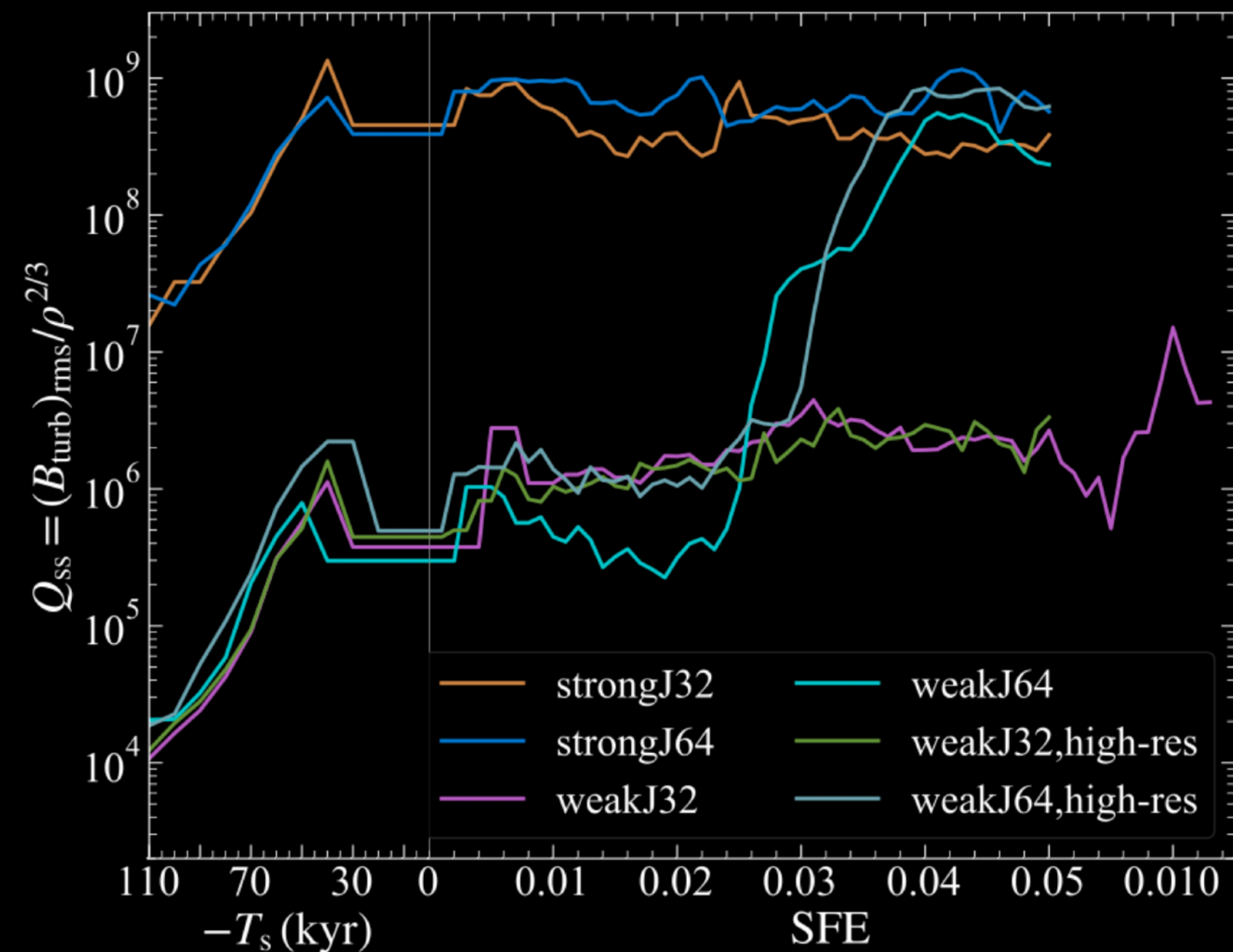


Pakmor+ (2024) *AREPO*

Examples of dynamos

There are many, across all scales

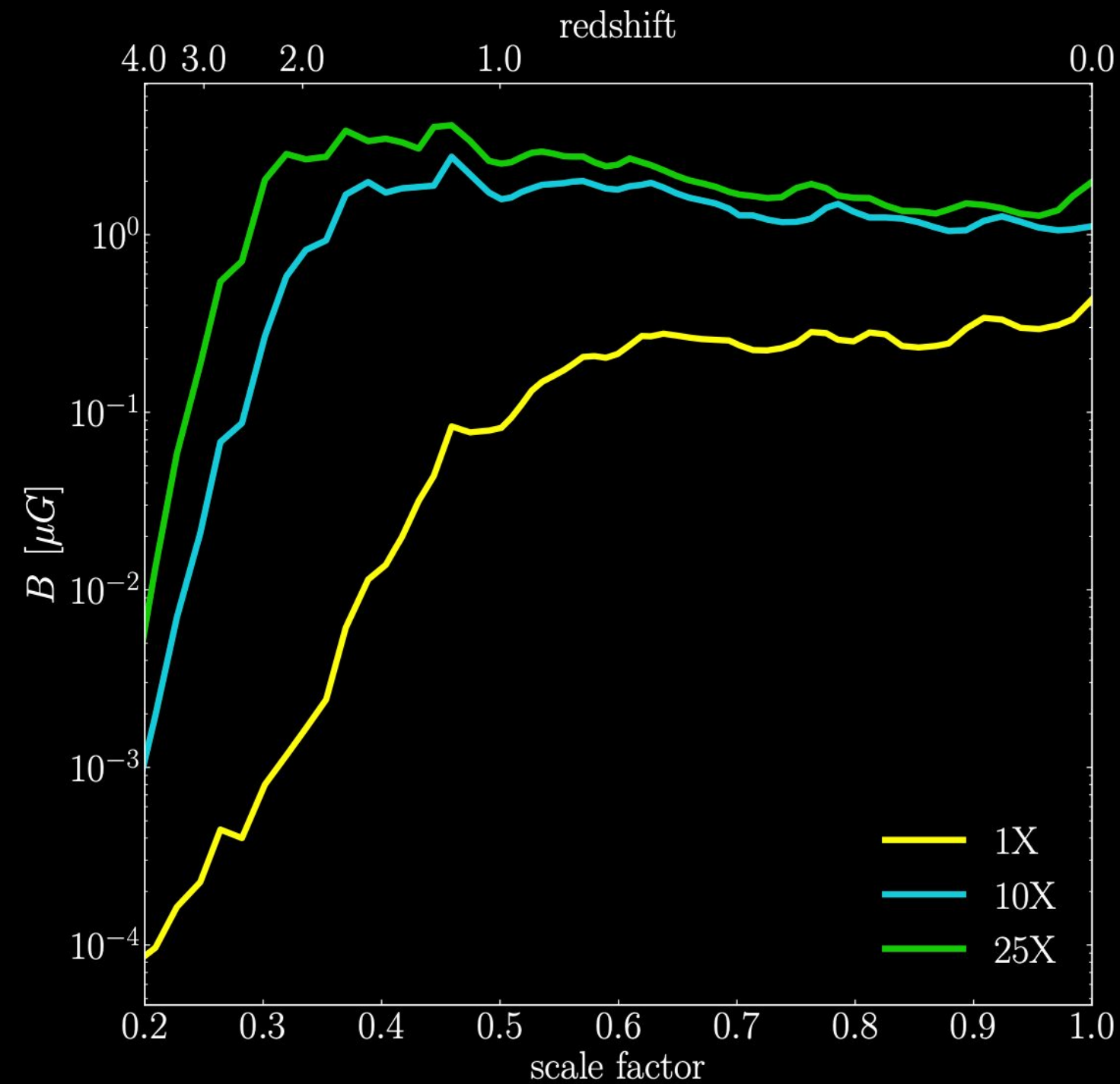
Sharda+2021



Molecular clouds in first generation stars

FLASH

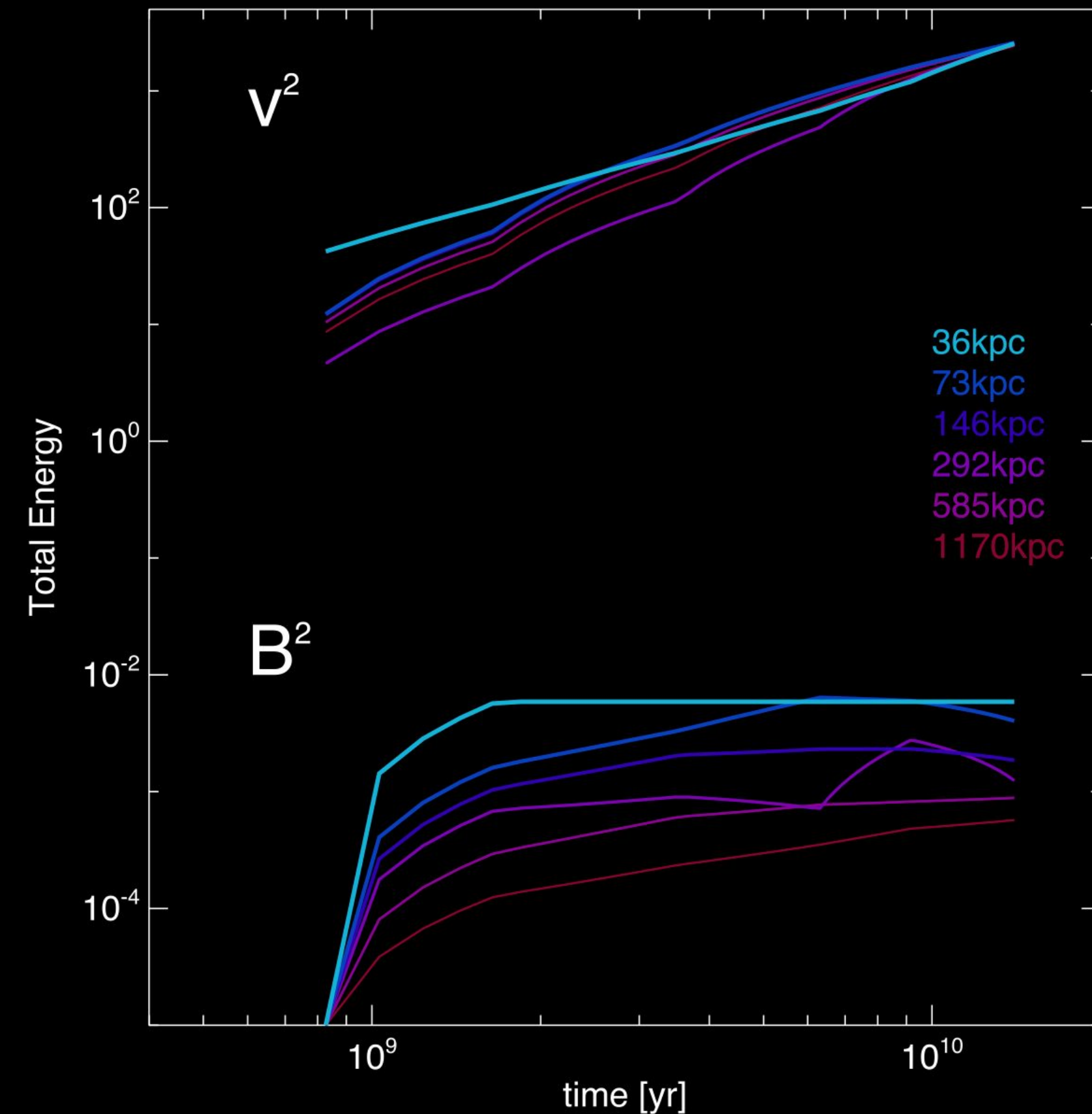
Steinwandel+2021



Intracluster medium

RAMSES

Vazza+2014



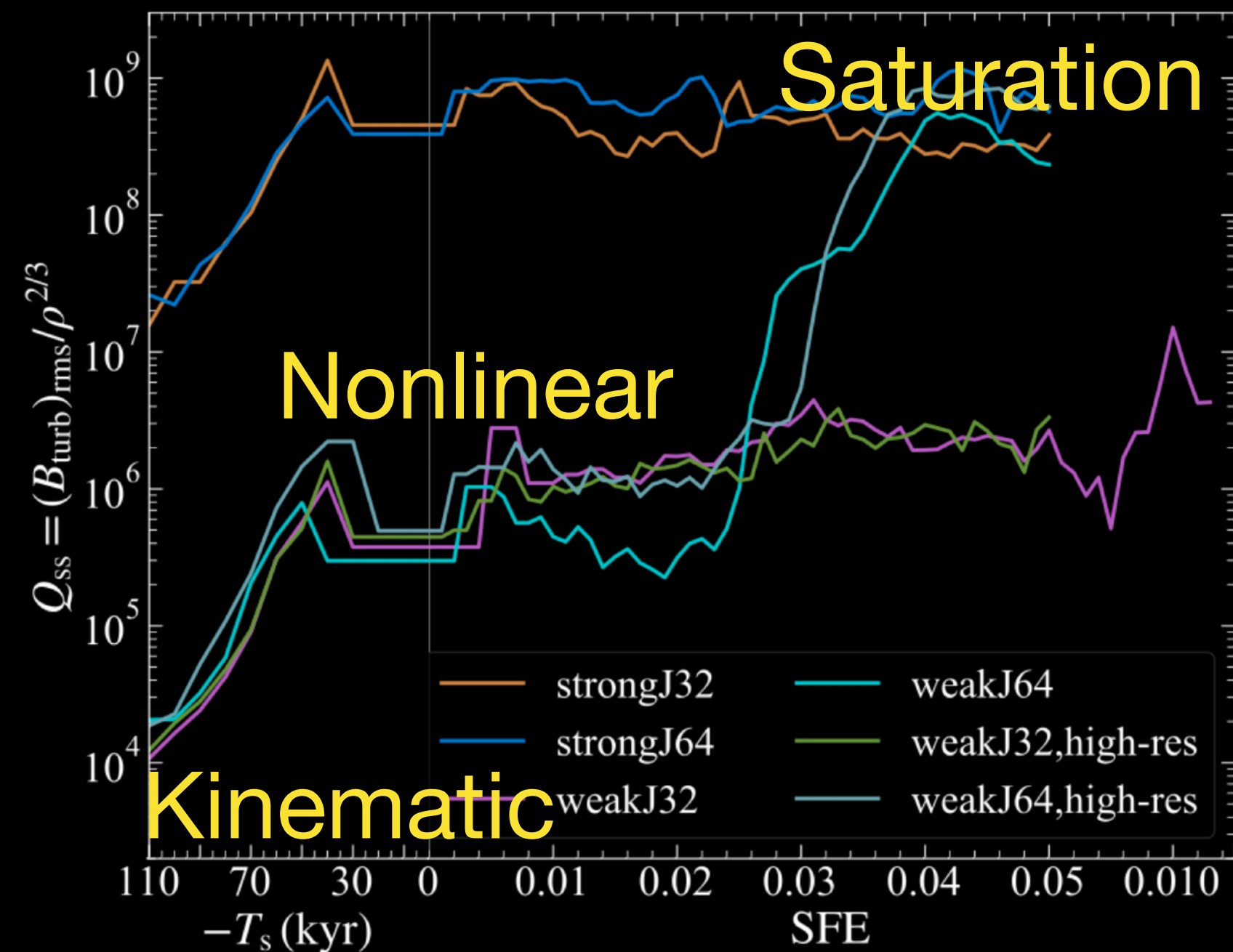
Cosmic filaments

ENZO

Examples of dynamos

There are many, across all scales

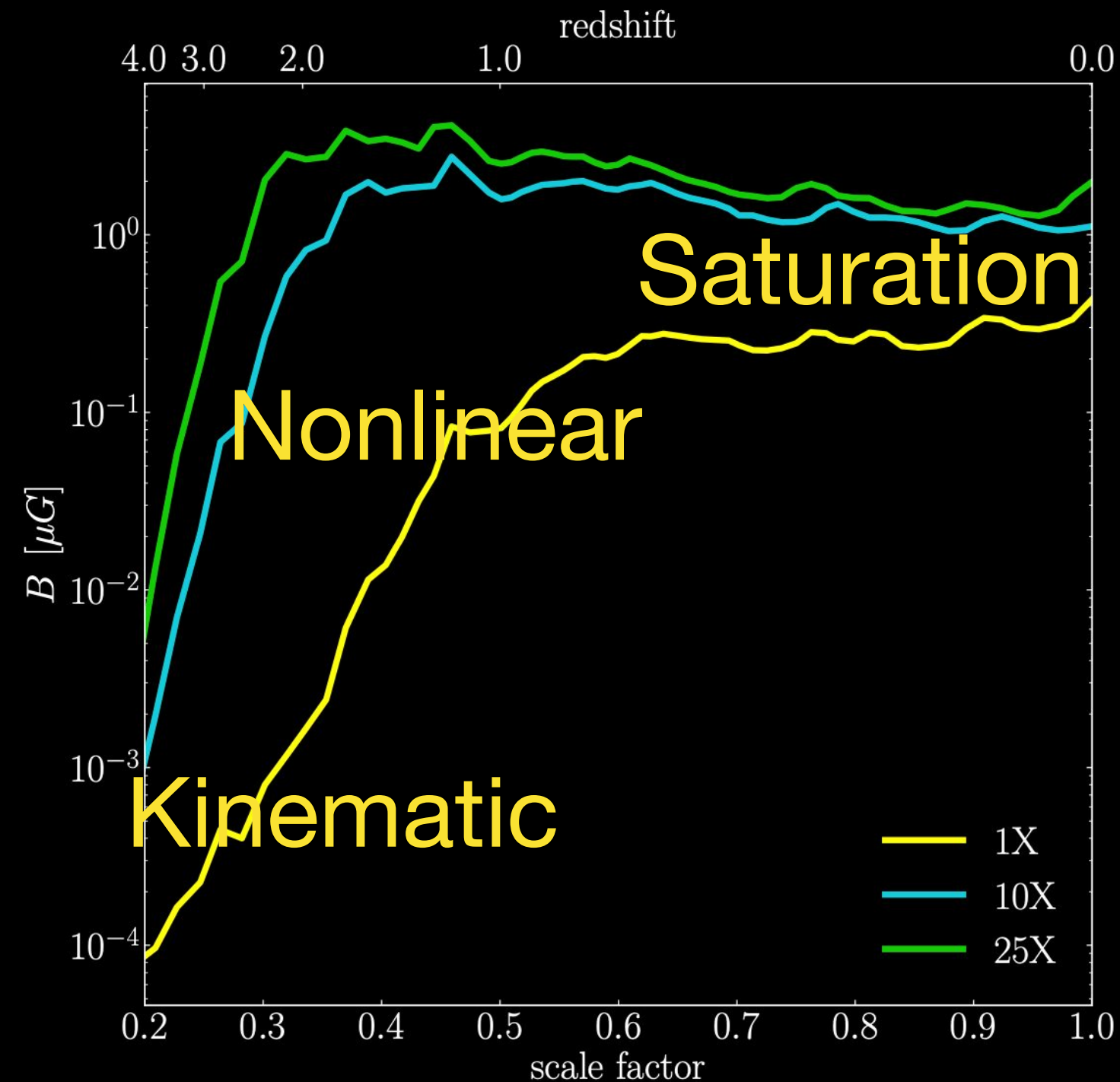
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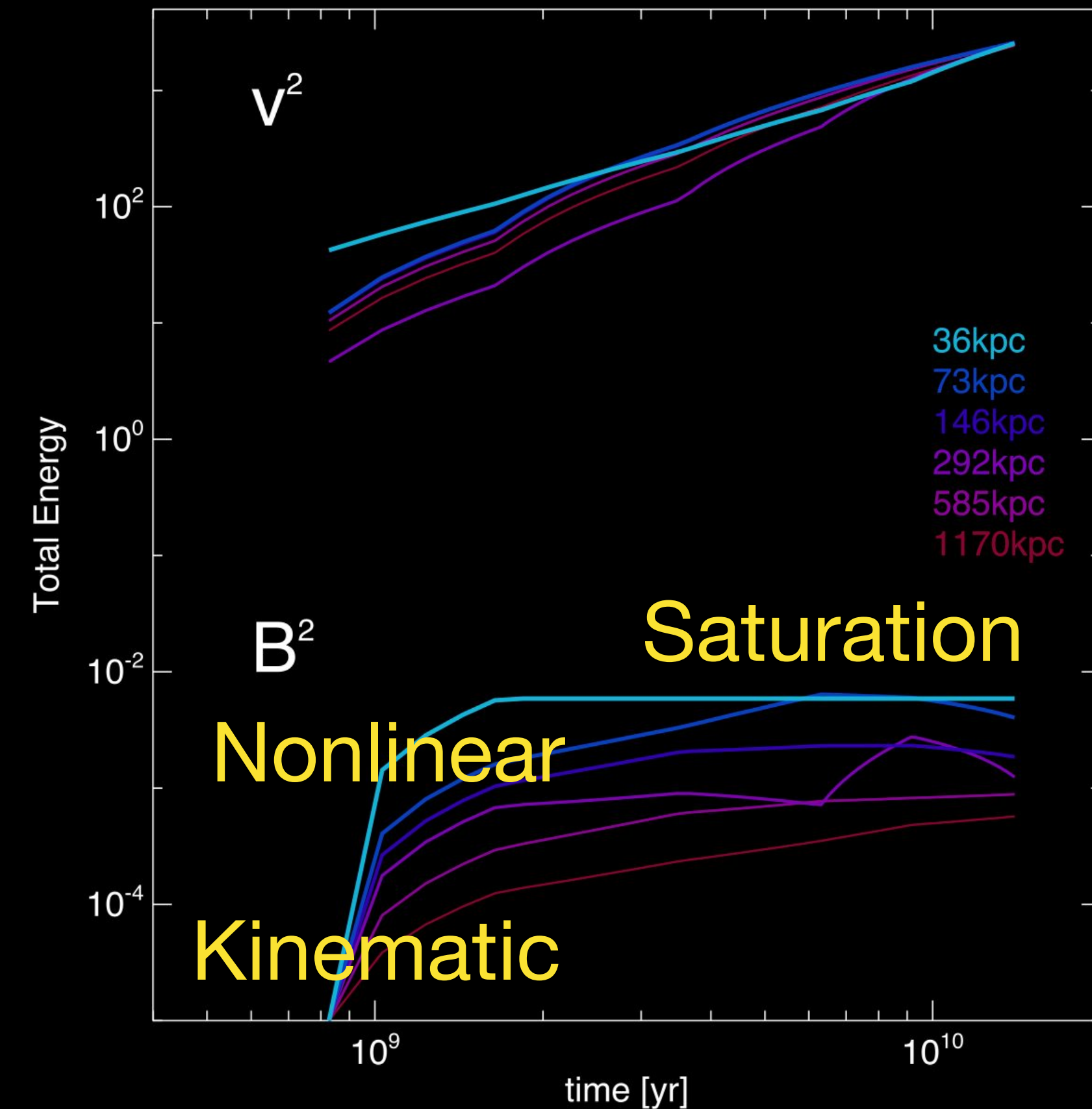
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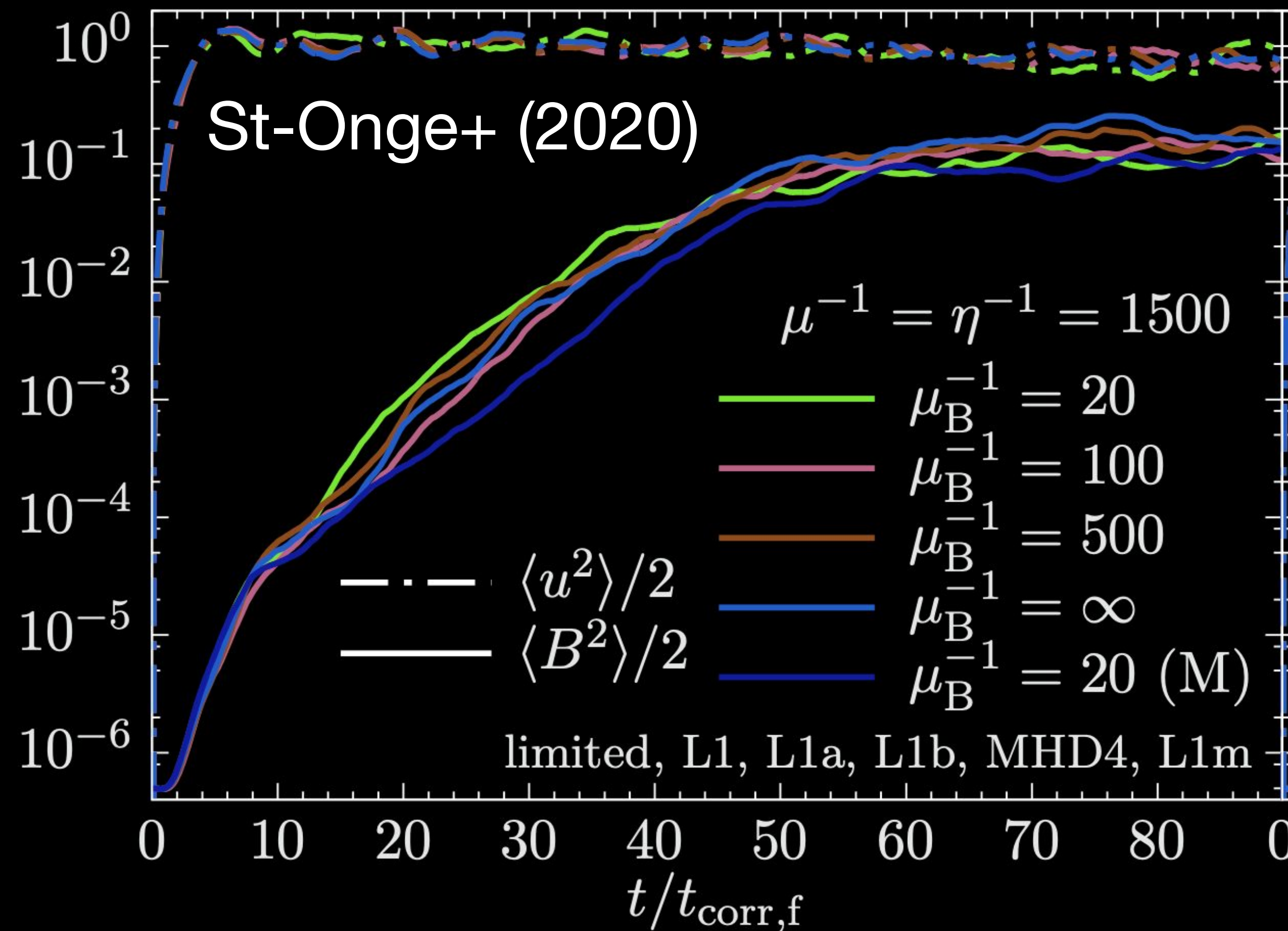
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Examples of dynamos and plasma regimes

ICM

Weakly collisional Braginskii MHD



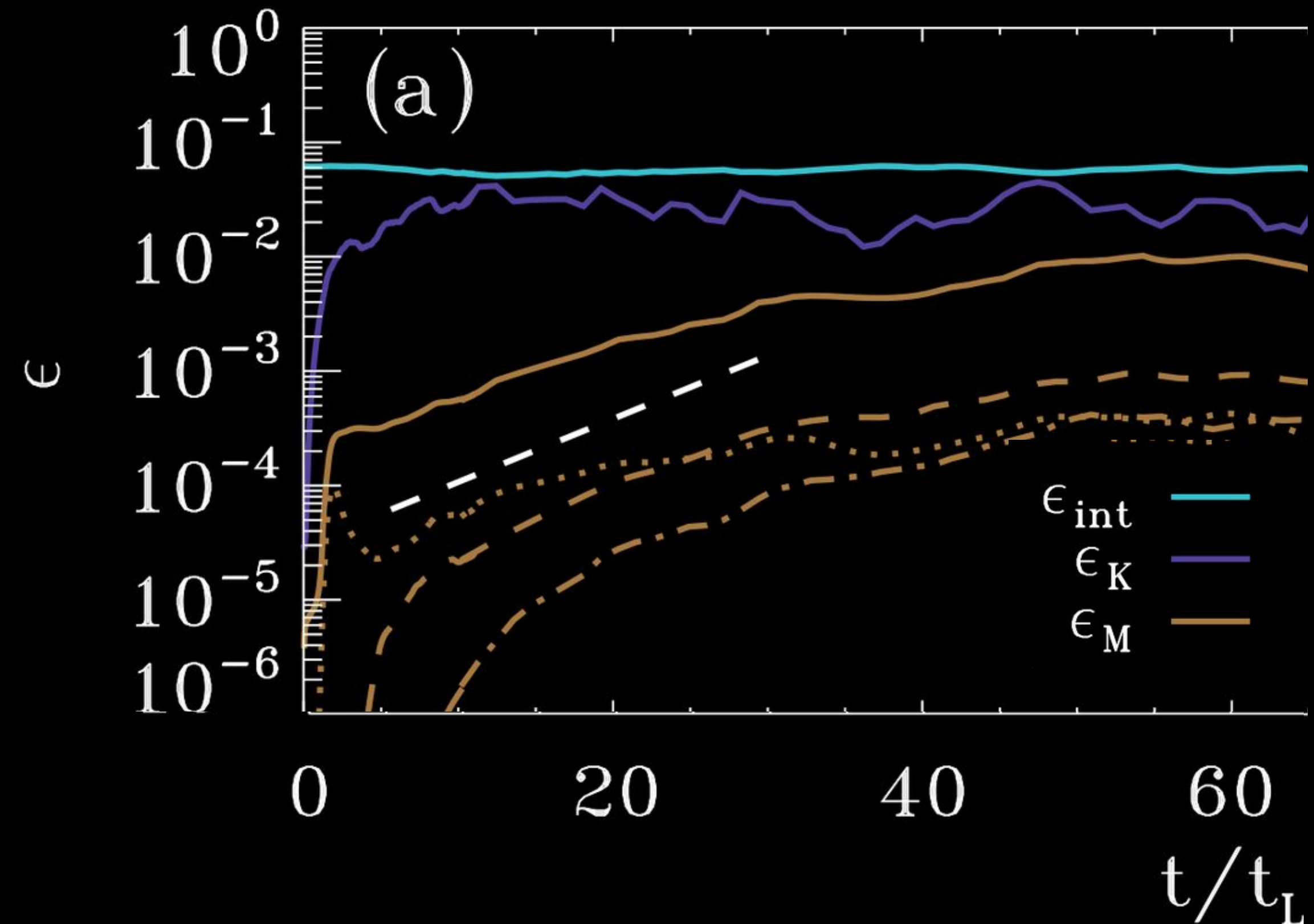
(added anisotropic viscous Braginskii stress term into MHD)

$$\nabla \cdot (\hat{\mathbf{b}} \otimes \hat{\mathbf{b}} (\hat{\mathbf{b}} \otimes \hat{\mathbf{b}} : \nabla \mathbf{v})) \quad \text{Snoopy}$$

Cosmological voids

Collisionless plasma

magnetogenesis coupled to dynamo



Sironi+2023

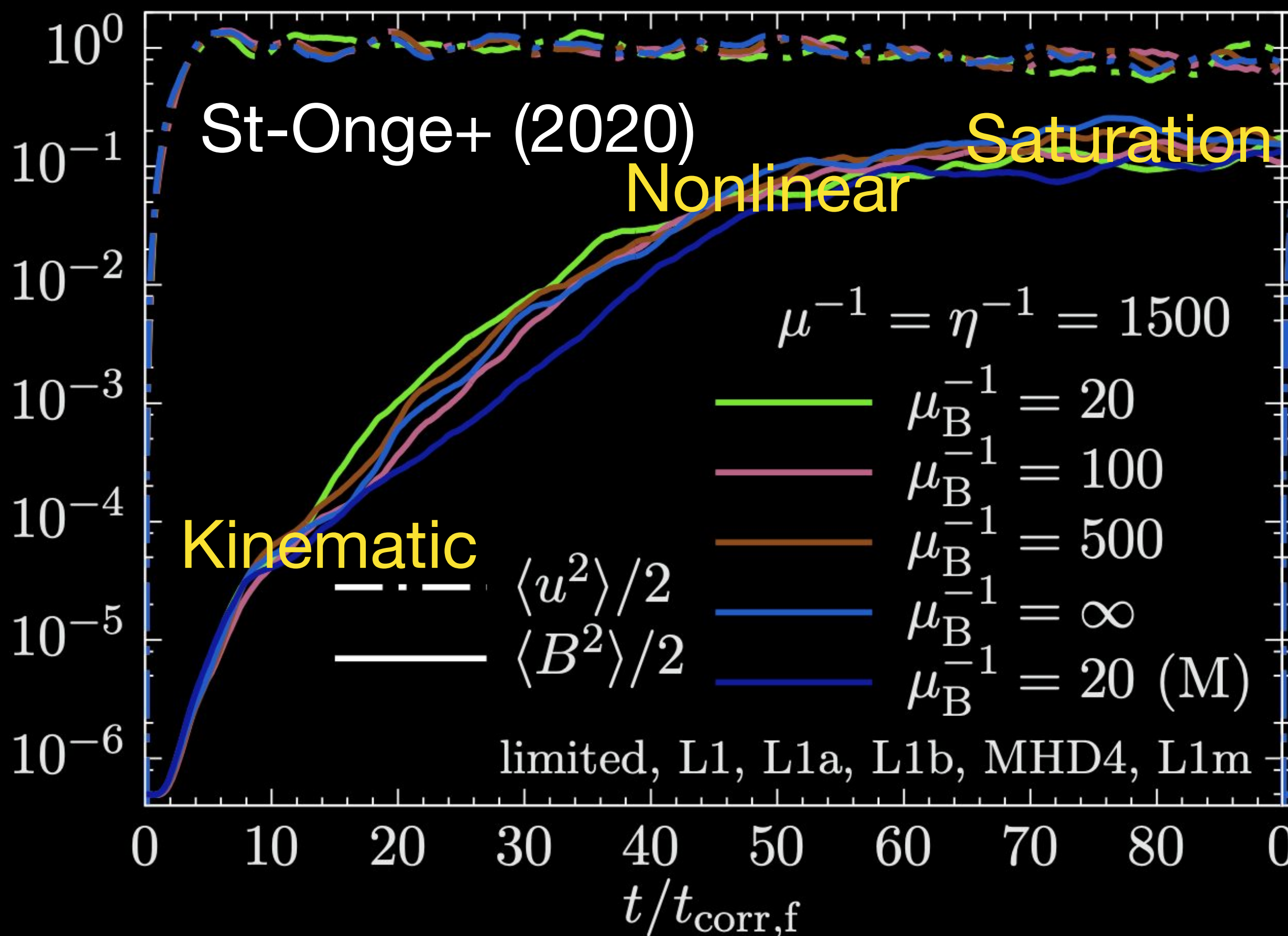
(PIC: pair plasma)

TRISTAN-MP

Examples of dynamos and plasma regimes

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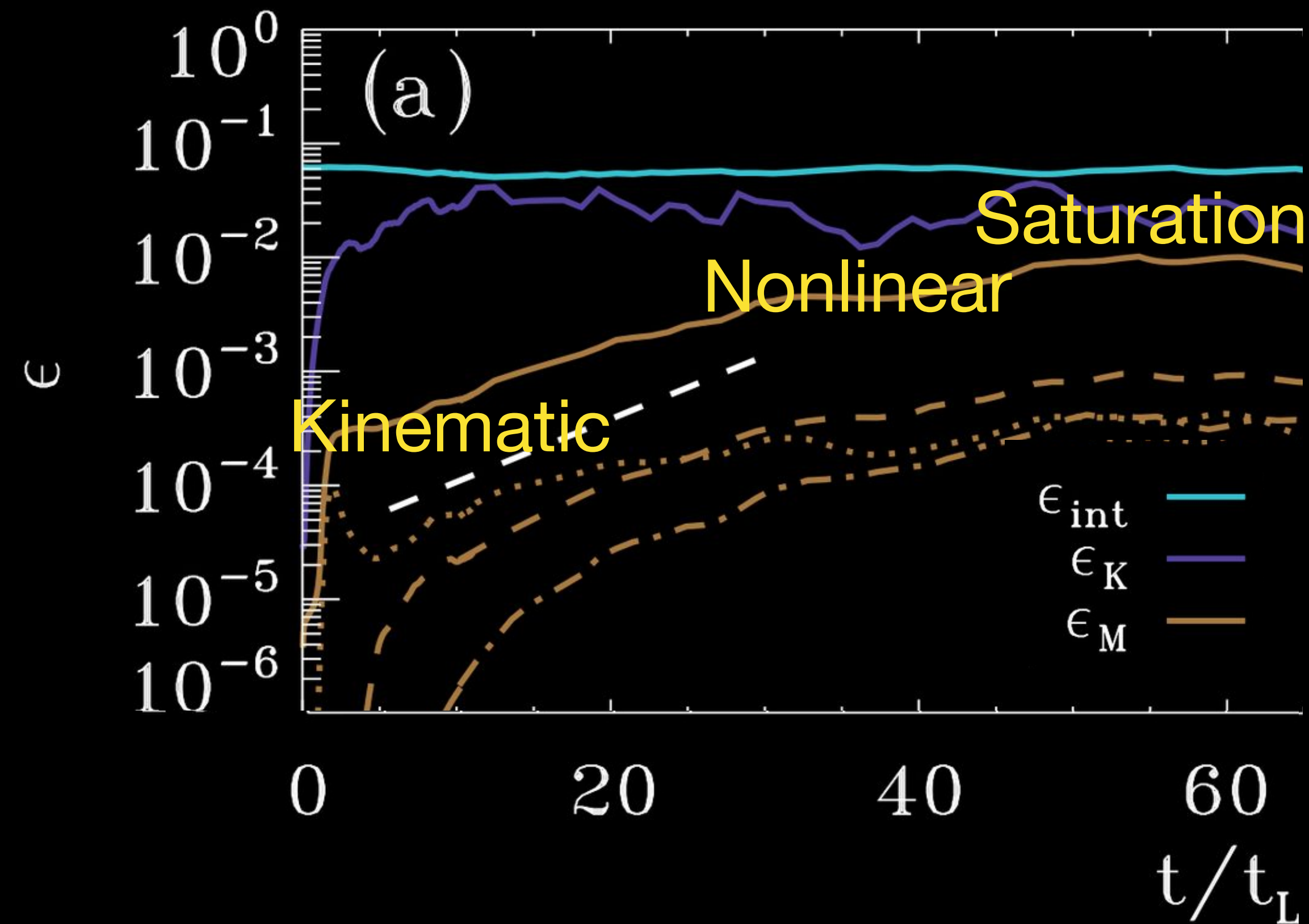
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Collisionless plasma

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Sironi+2023

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TRISTAN-MP

Dynamos

Ubiquitous across multiple phenomena / regimes

Extremely simple ingredients

1. $\nabla \mathbf{u}$ (the Universe is full of these!)

2. $R_m = \frac{U_0 L}{\eta} = \frac{\text{induction}}{\text{diffusion}} \gtrsim 100$

(have to be reasonably conductive)

One of the most extreme magnetic dynamos

Merging Binary Neutron Stars: Chronology

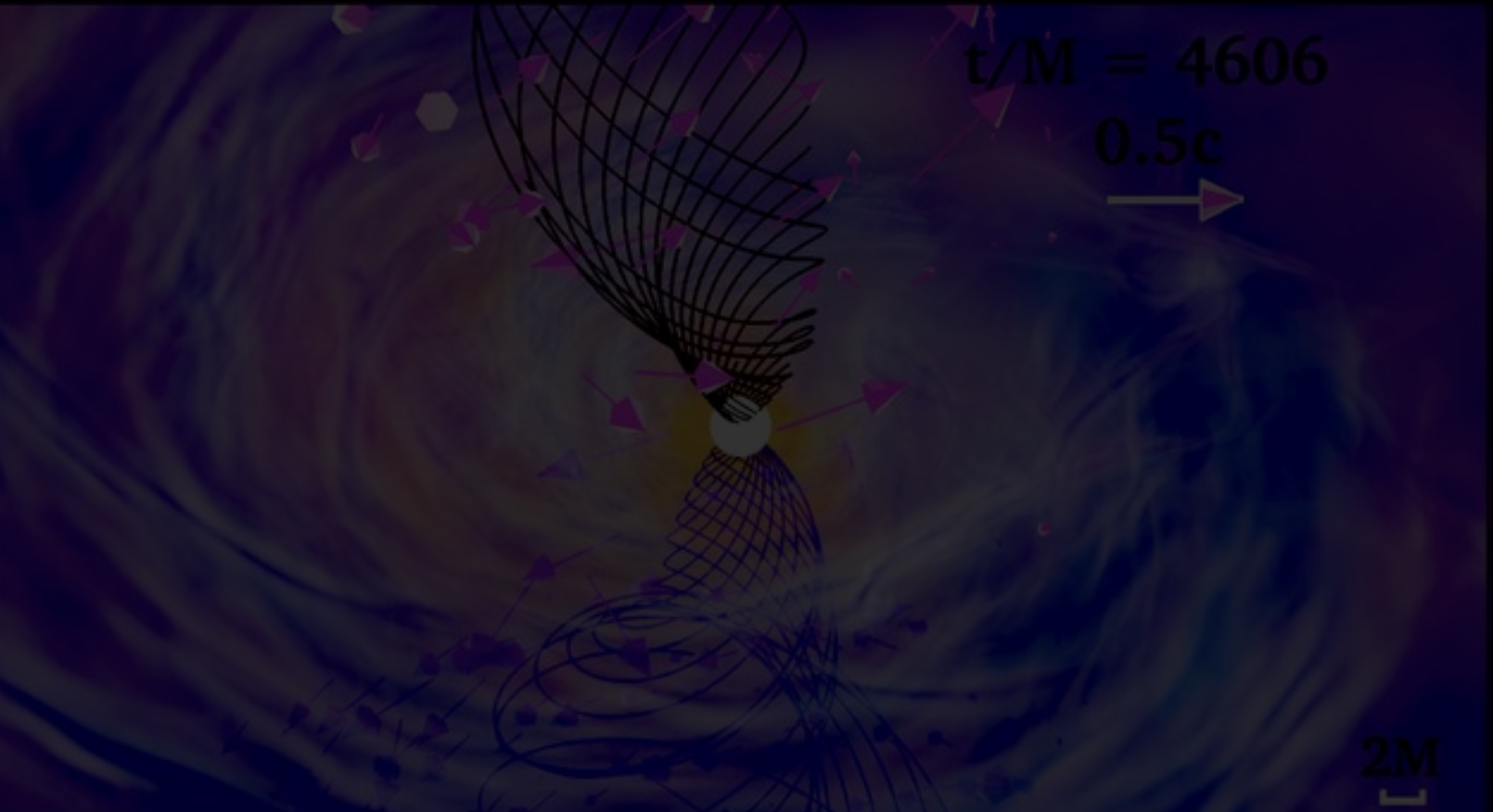
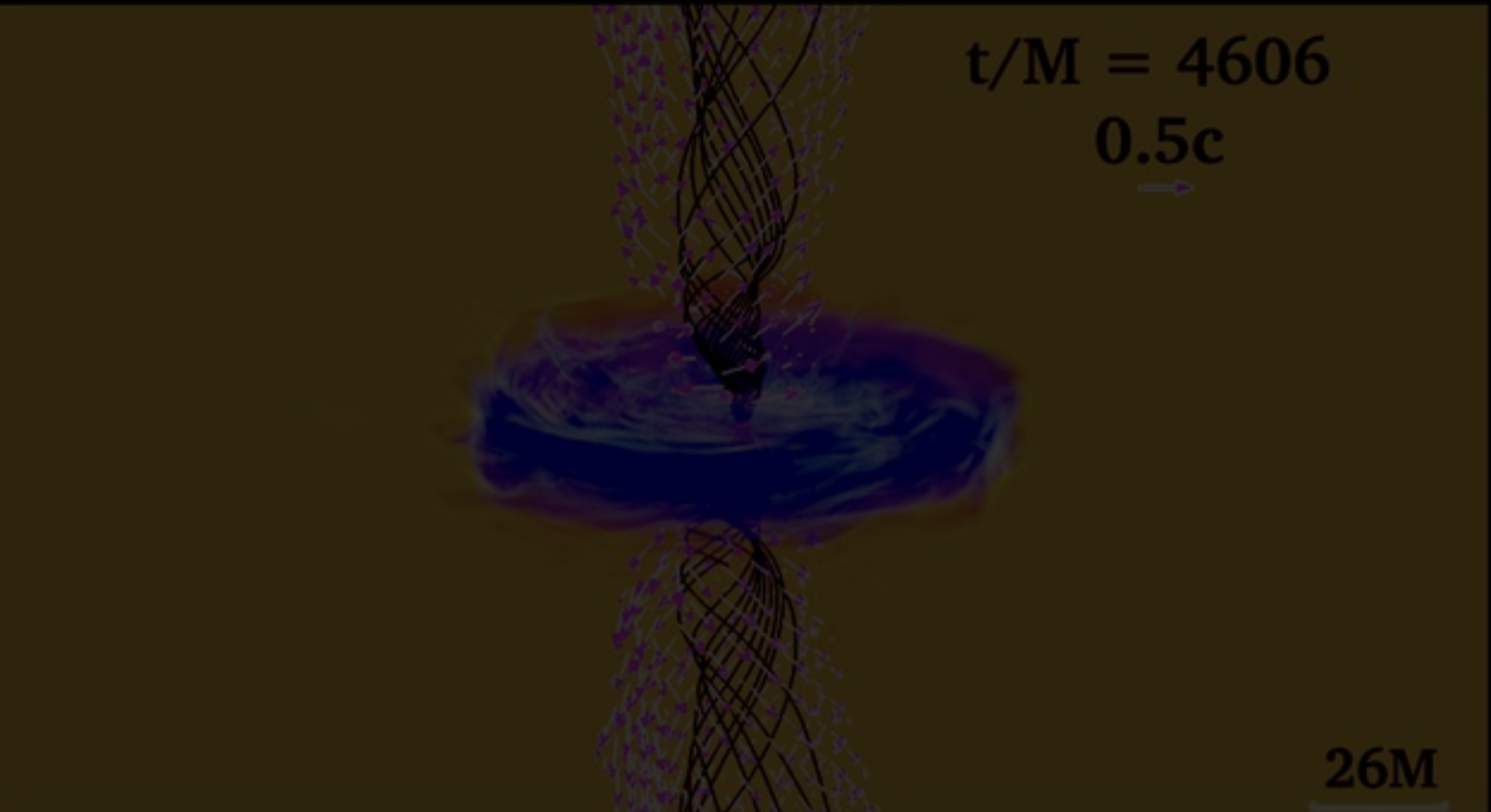
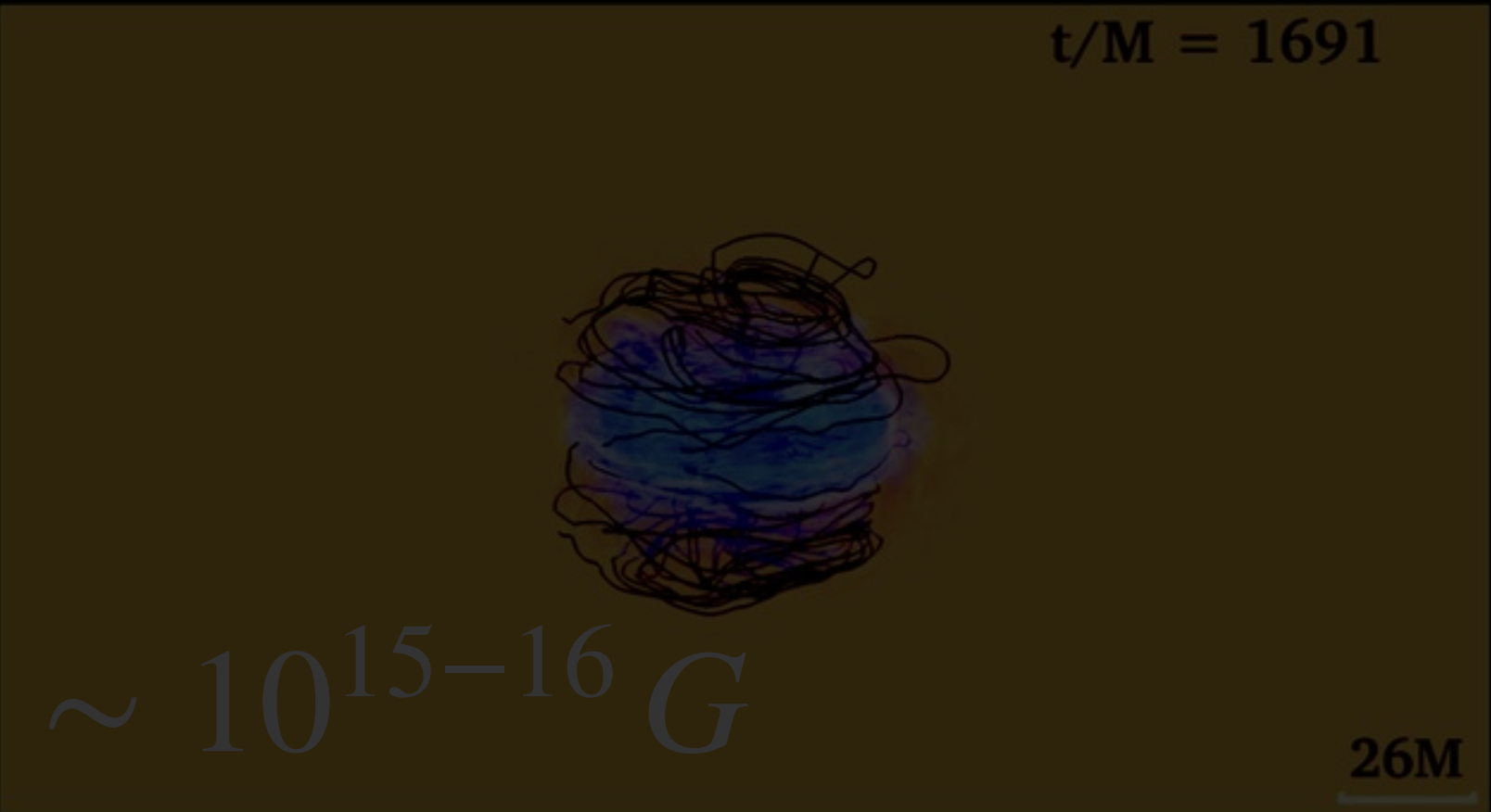
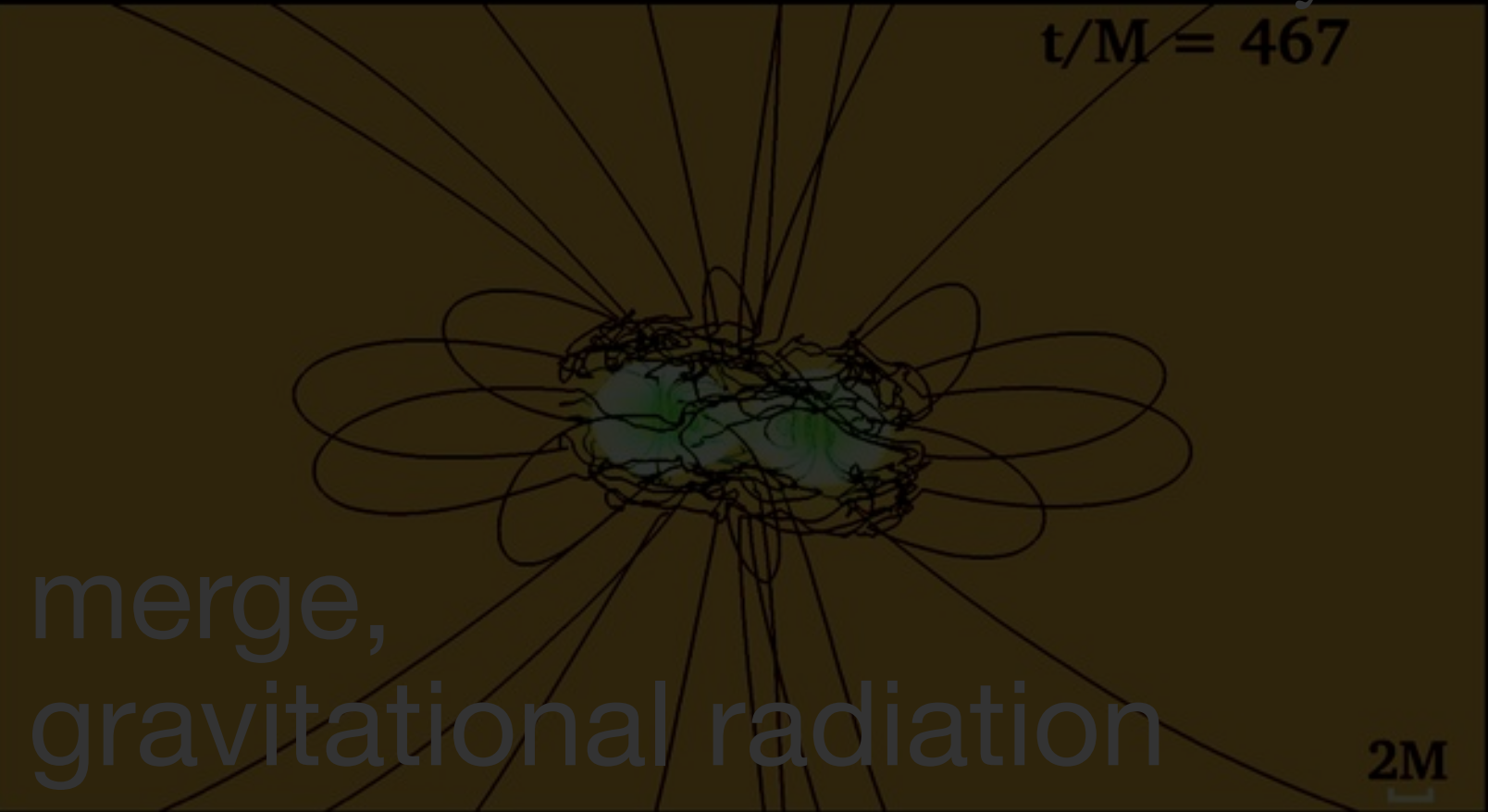
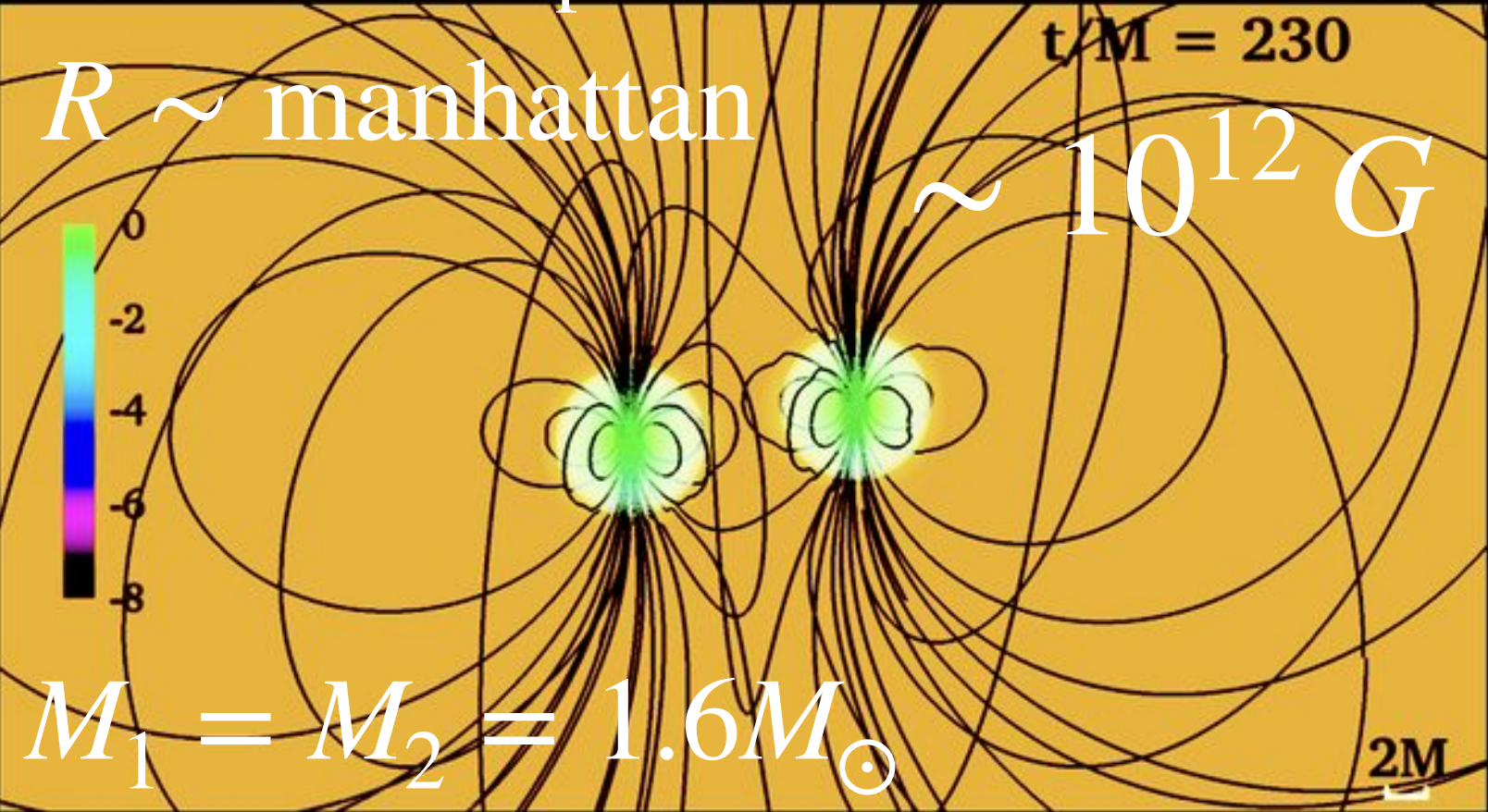
Ruiz+(2018)

inspiraling stars

$t_{\text{inspiral}} \gtrsim 10^8 \text{ yrs}$

something happens

$t_{\text{dynamical}} \sim 1 \text{ ms}$



Significant restructuring of
B-field

Blandford-Zanjek-type jet
need vertical field...

GRB
electromagnetic counterpart

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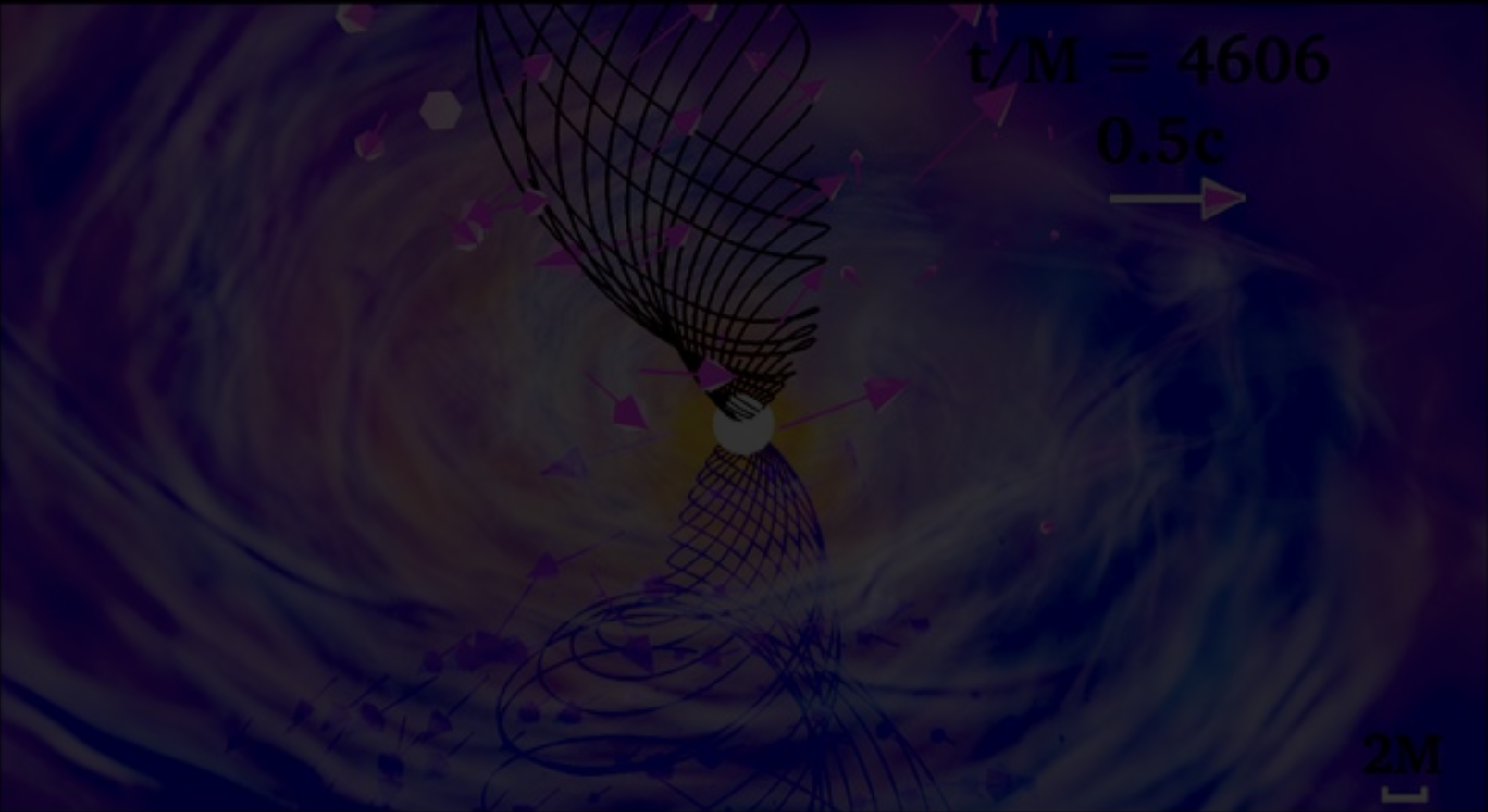
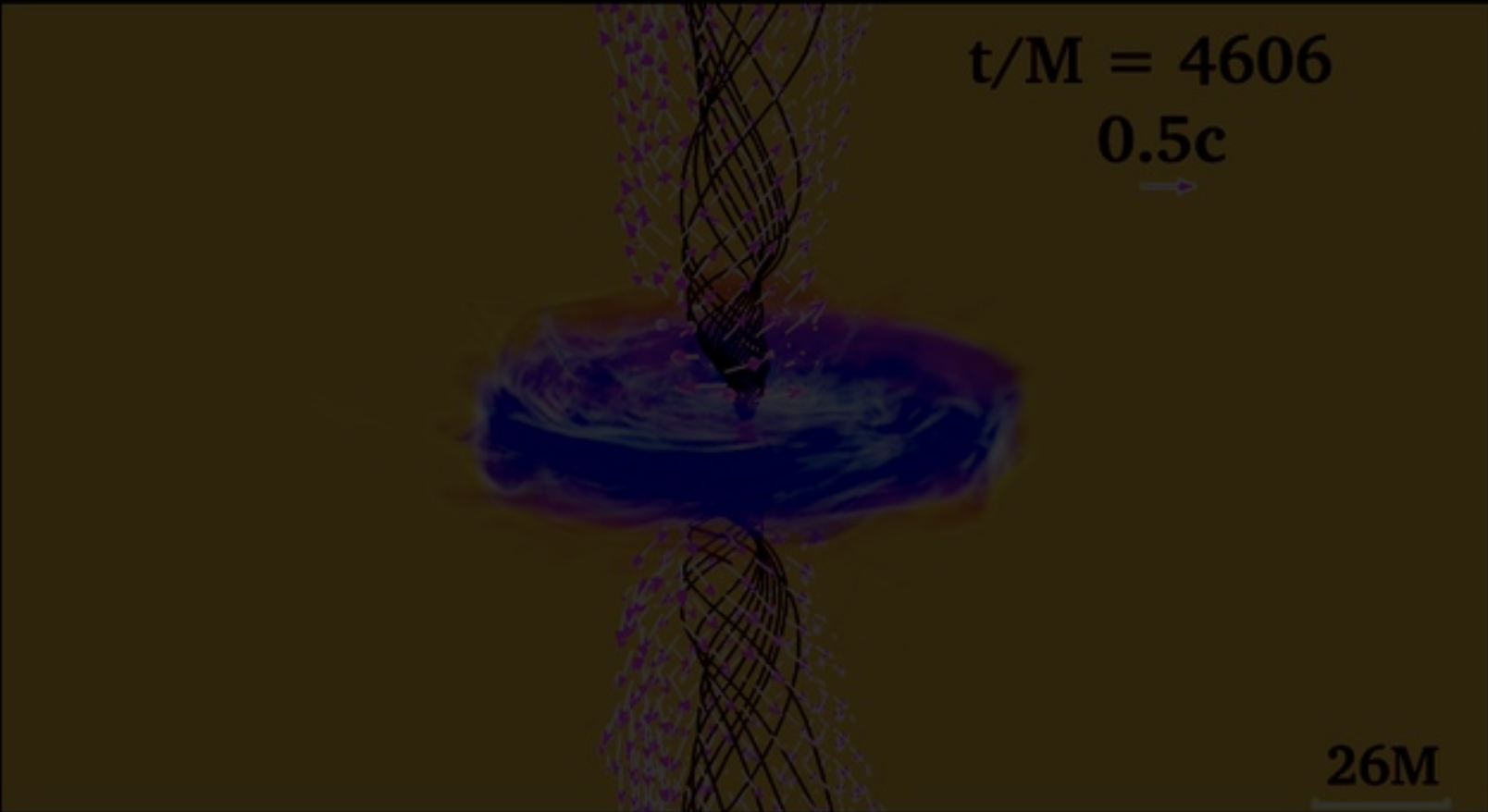
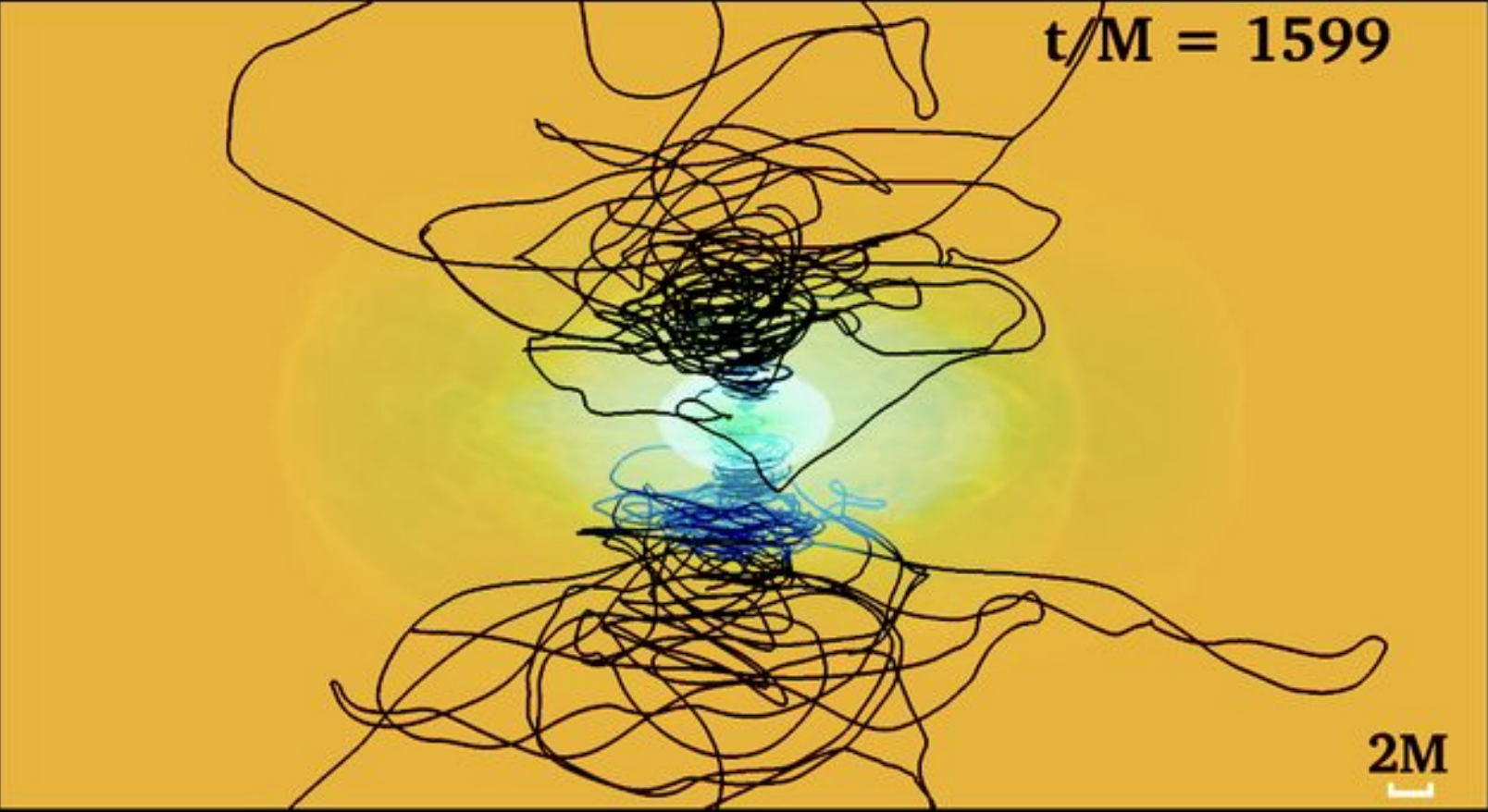
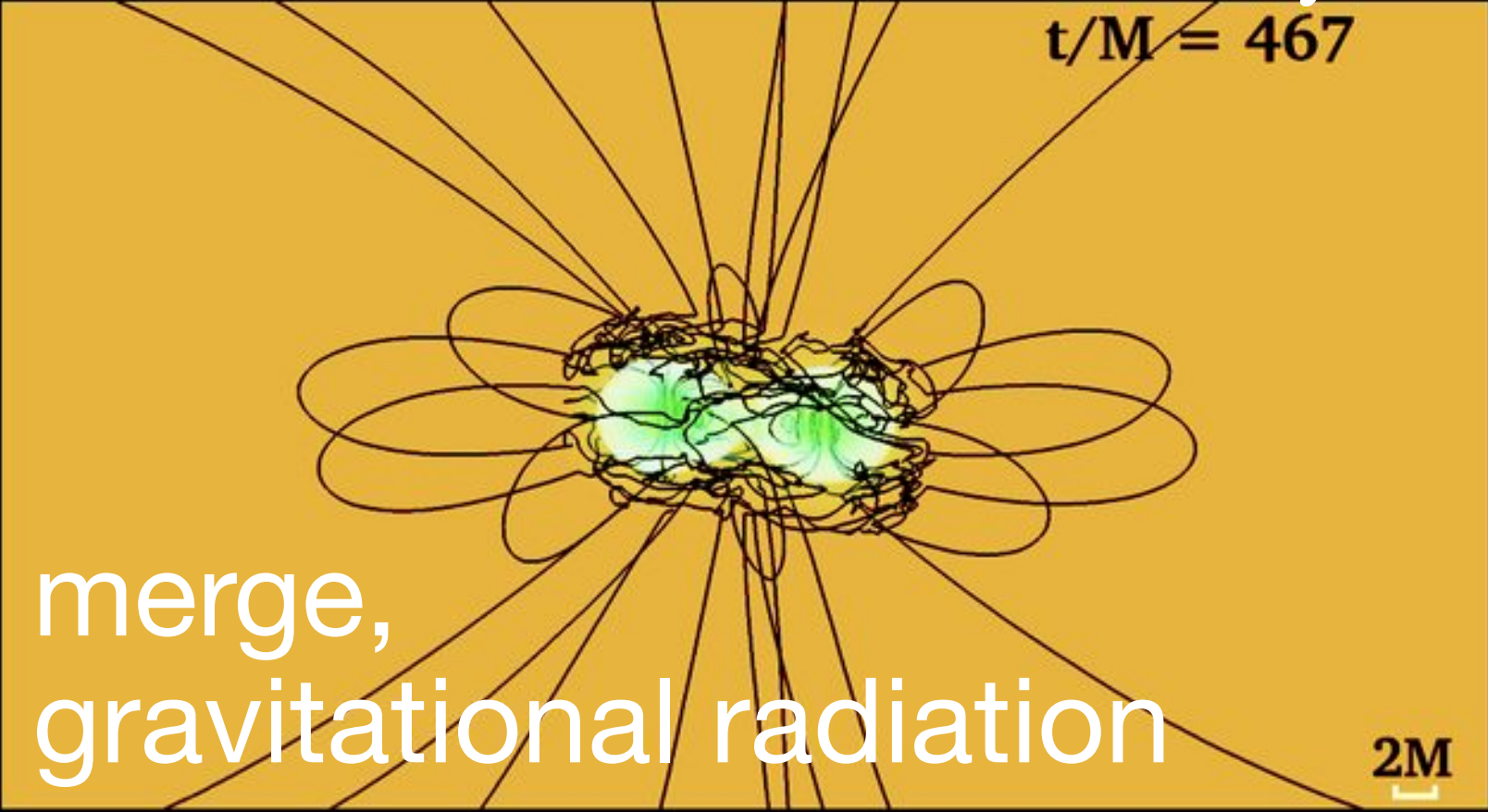
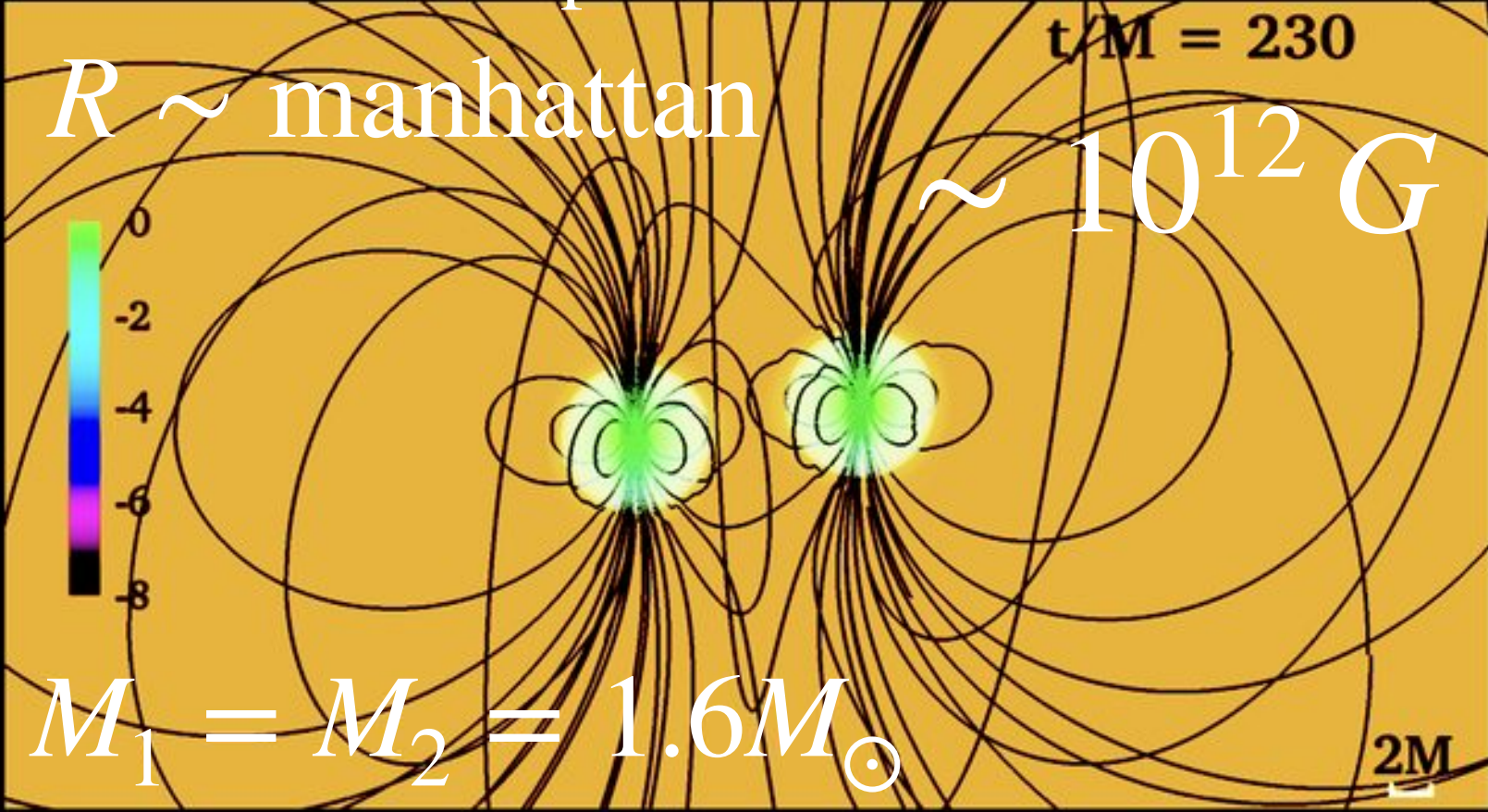
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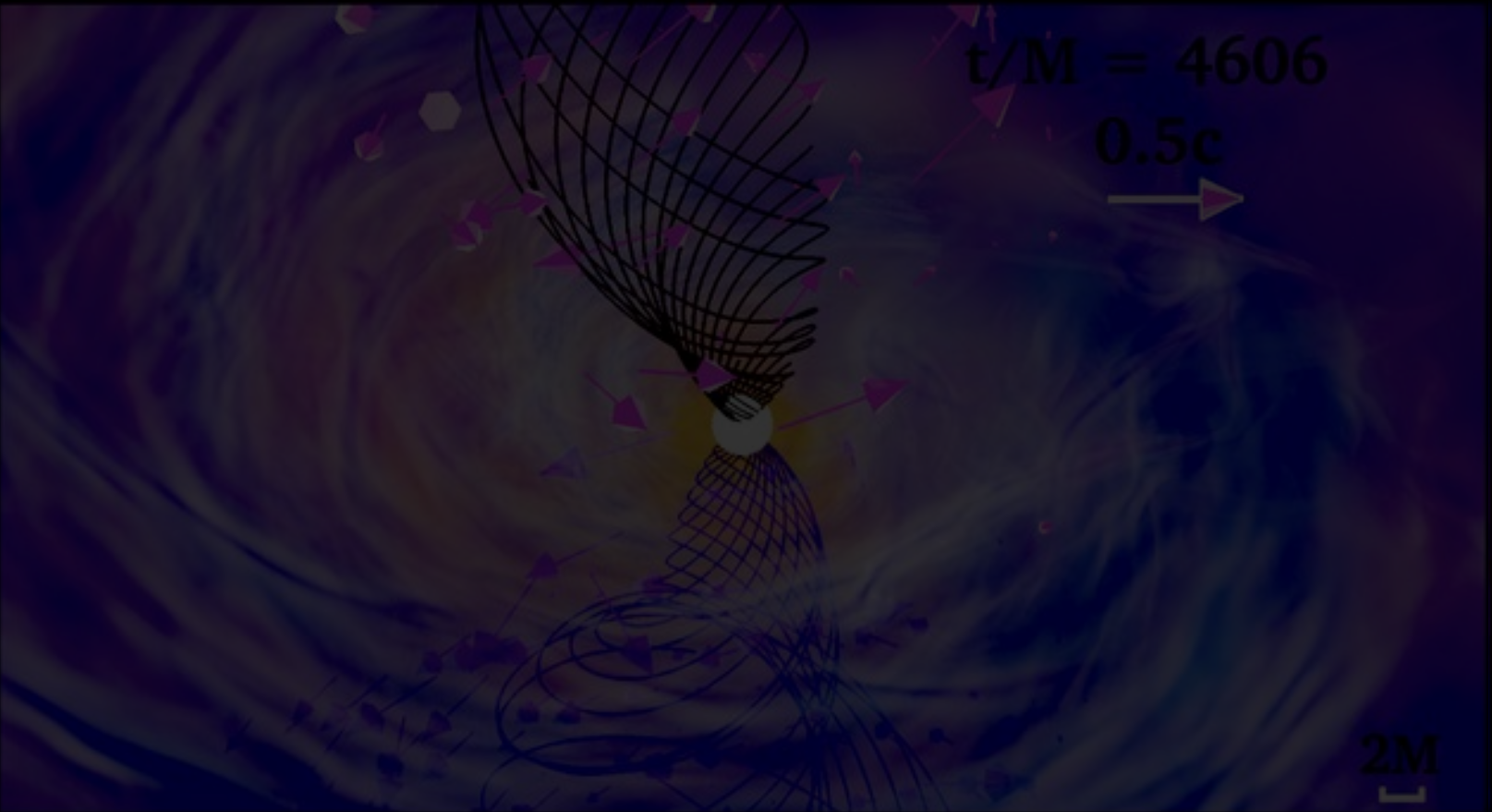
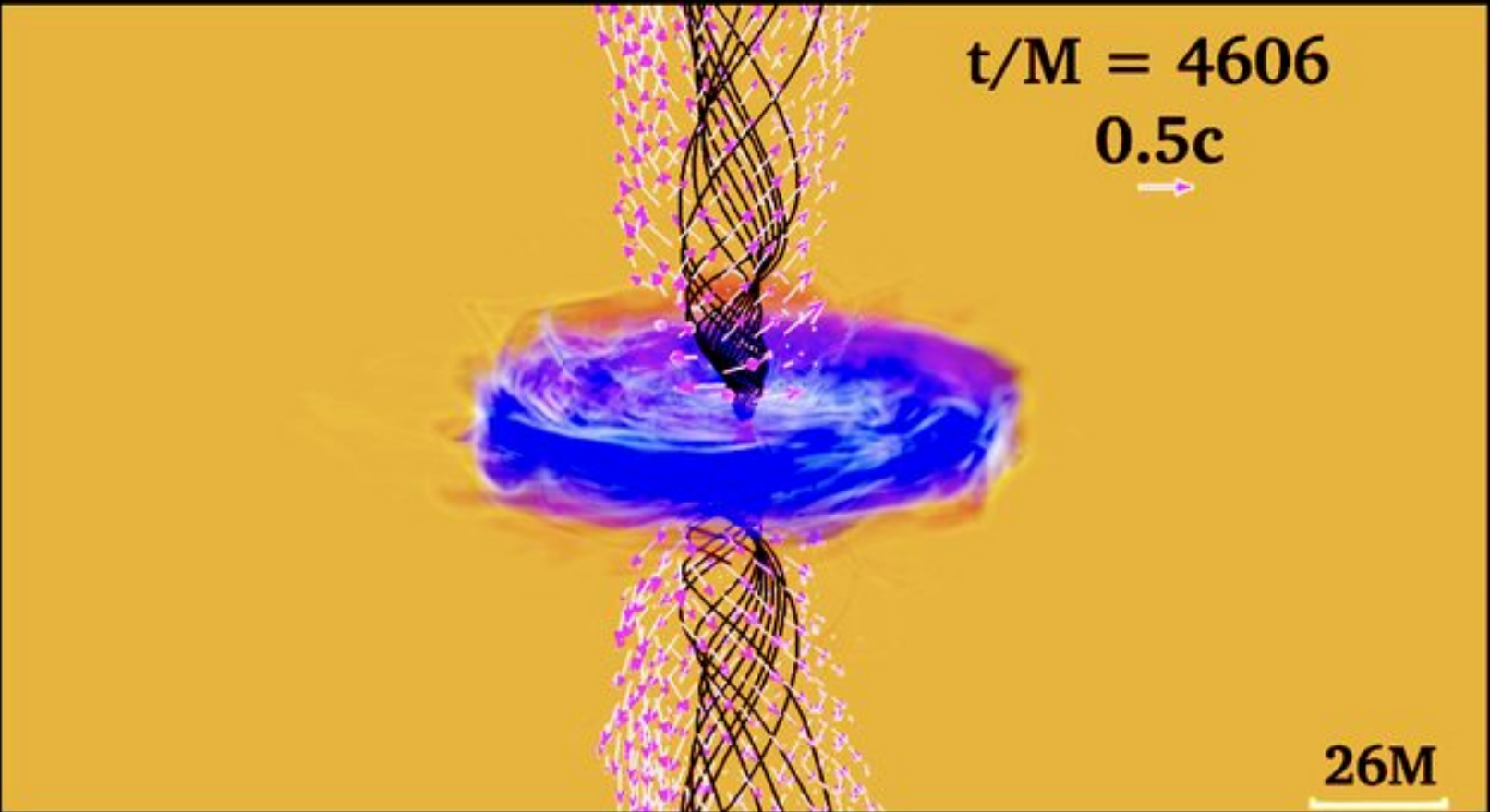
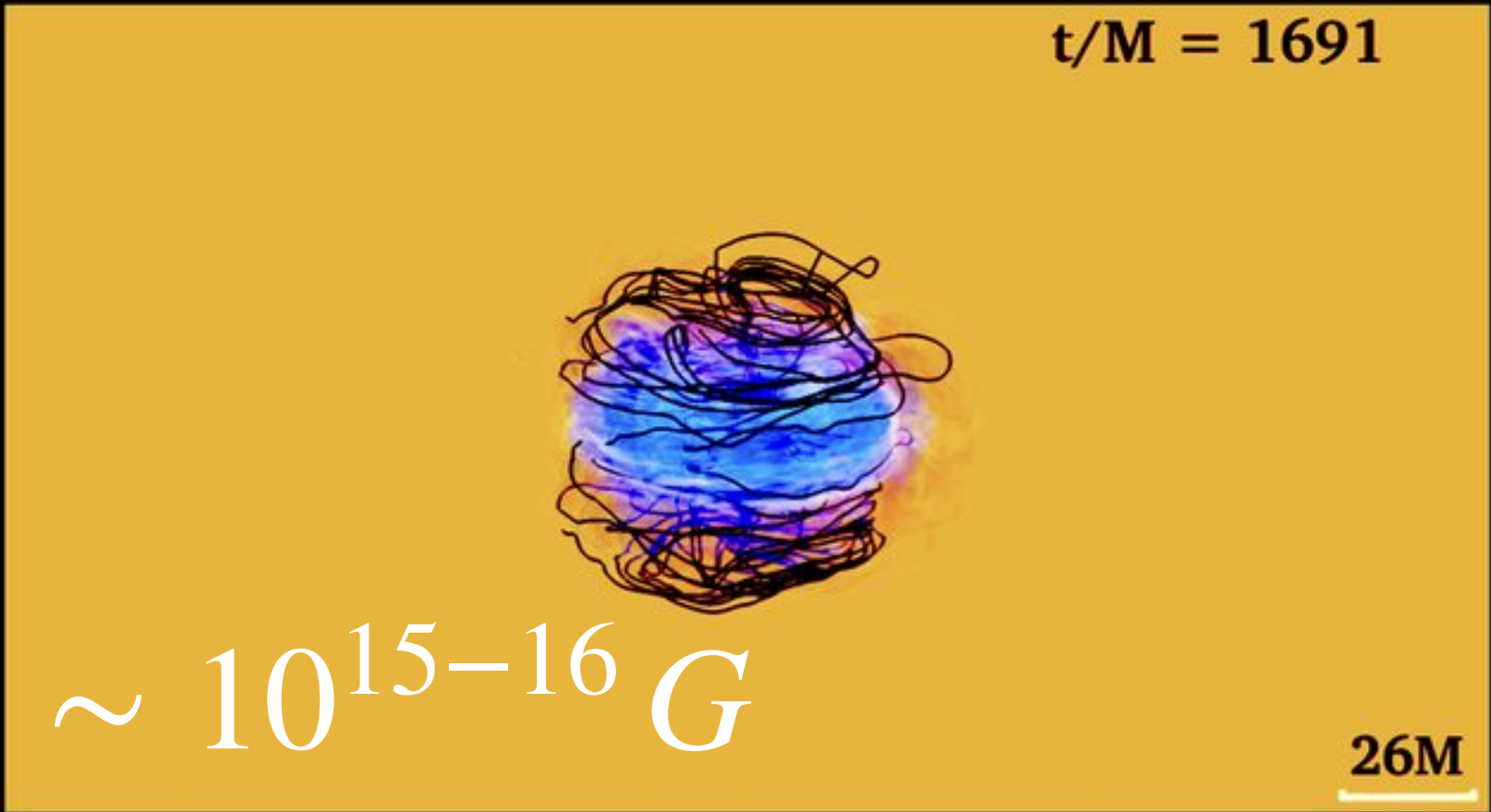
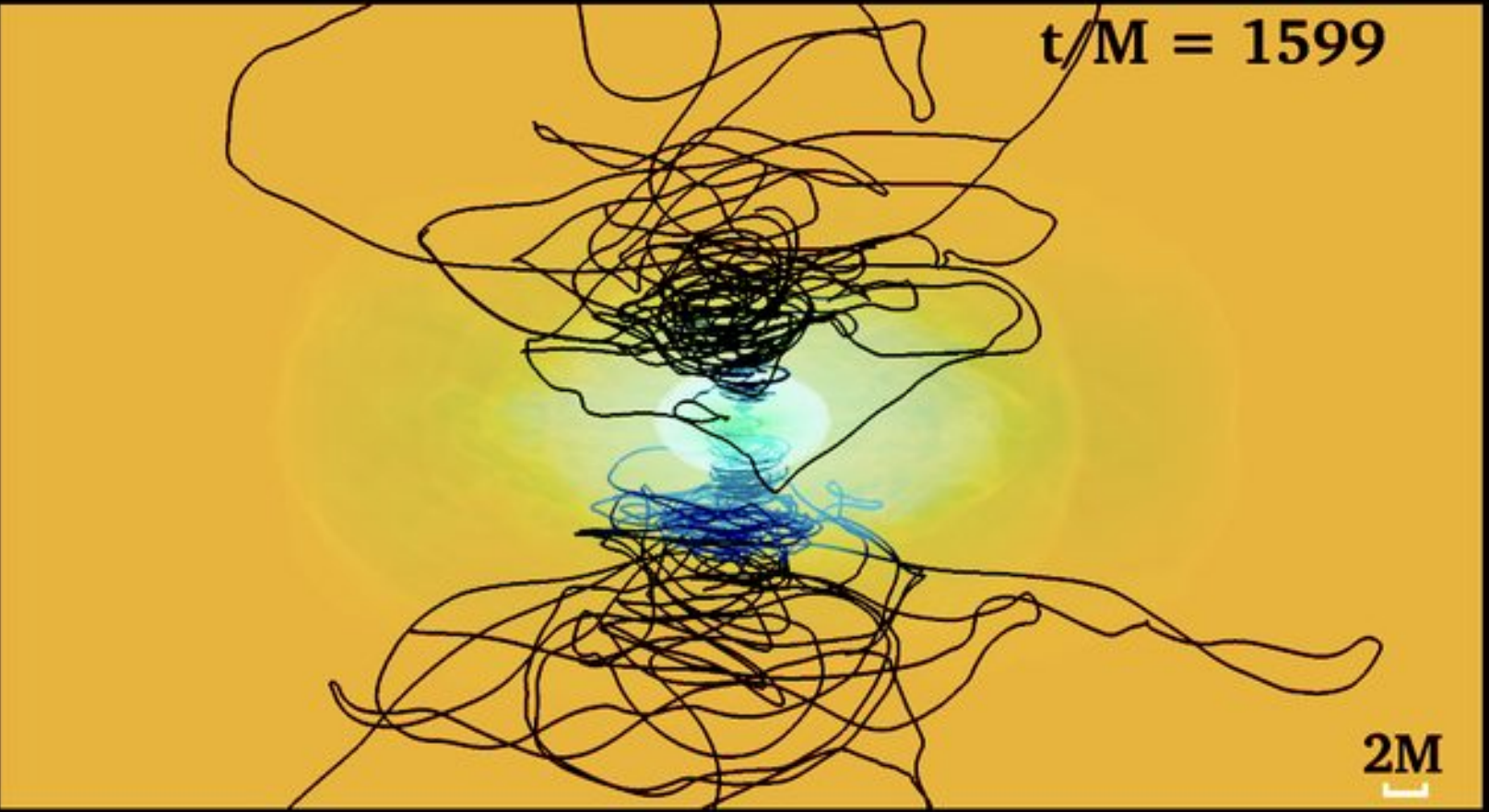
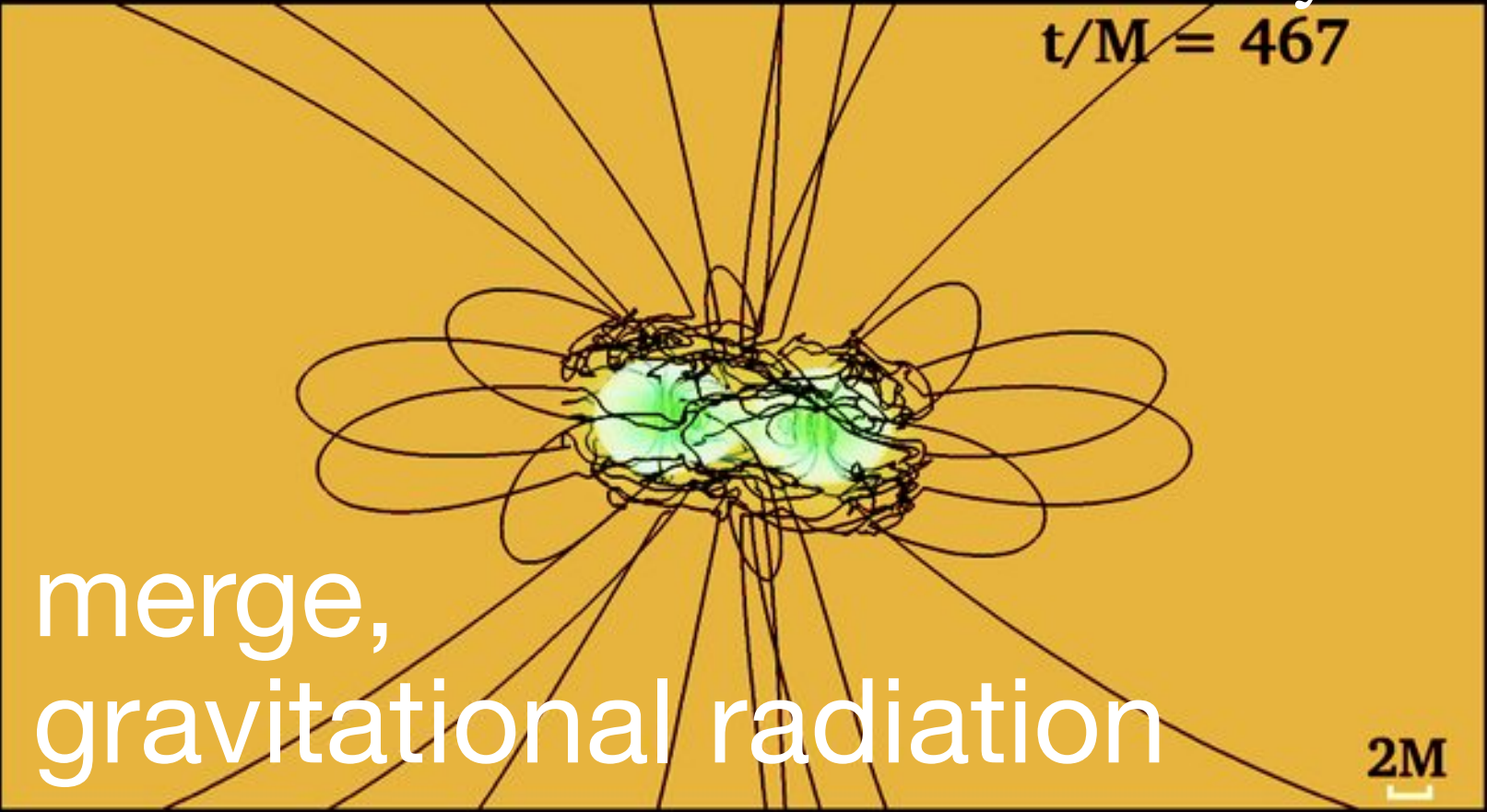
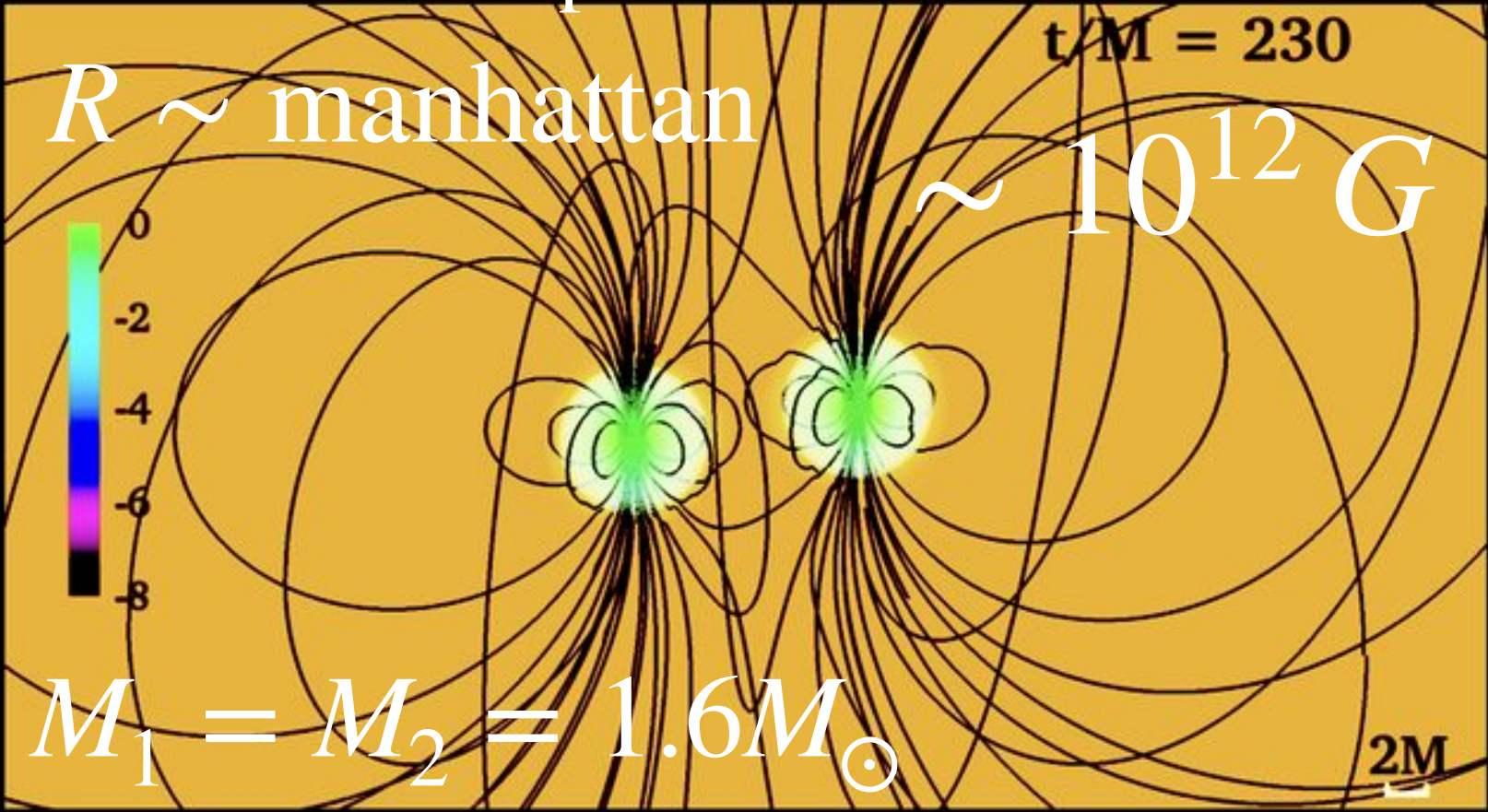
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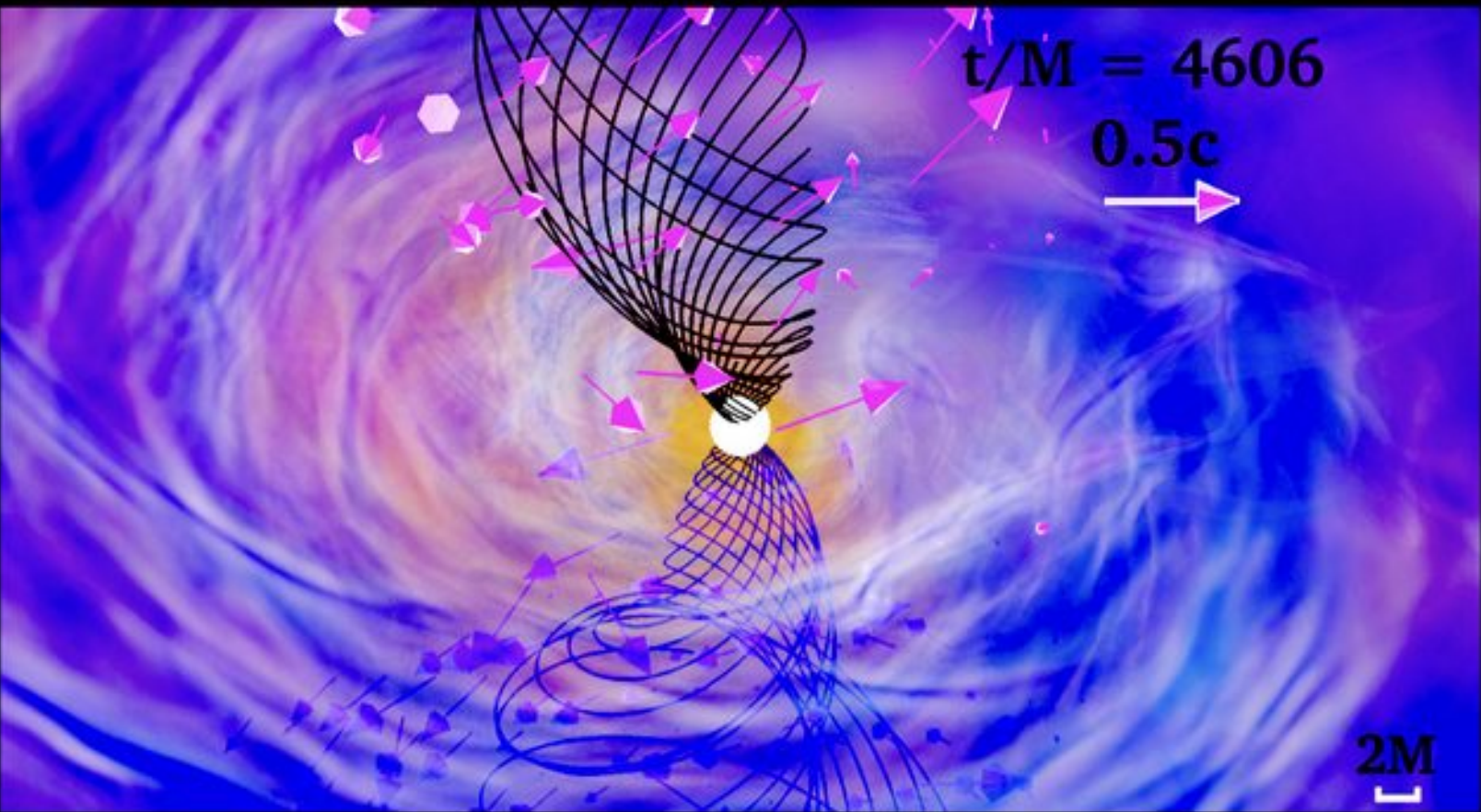
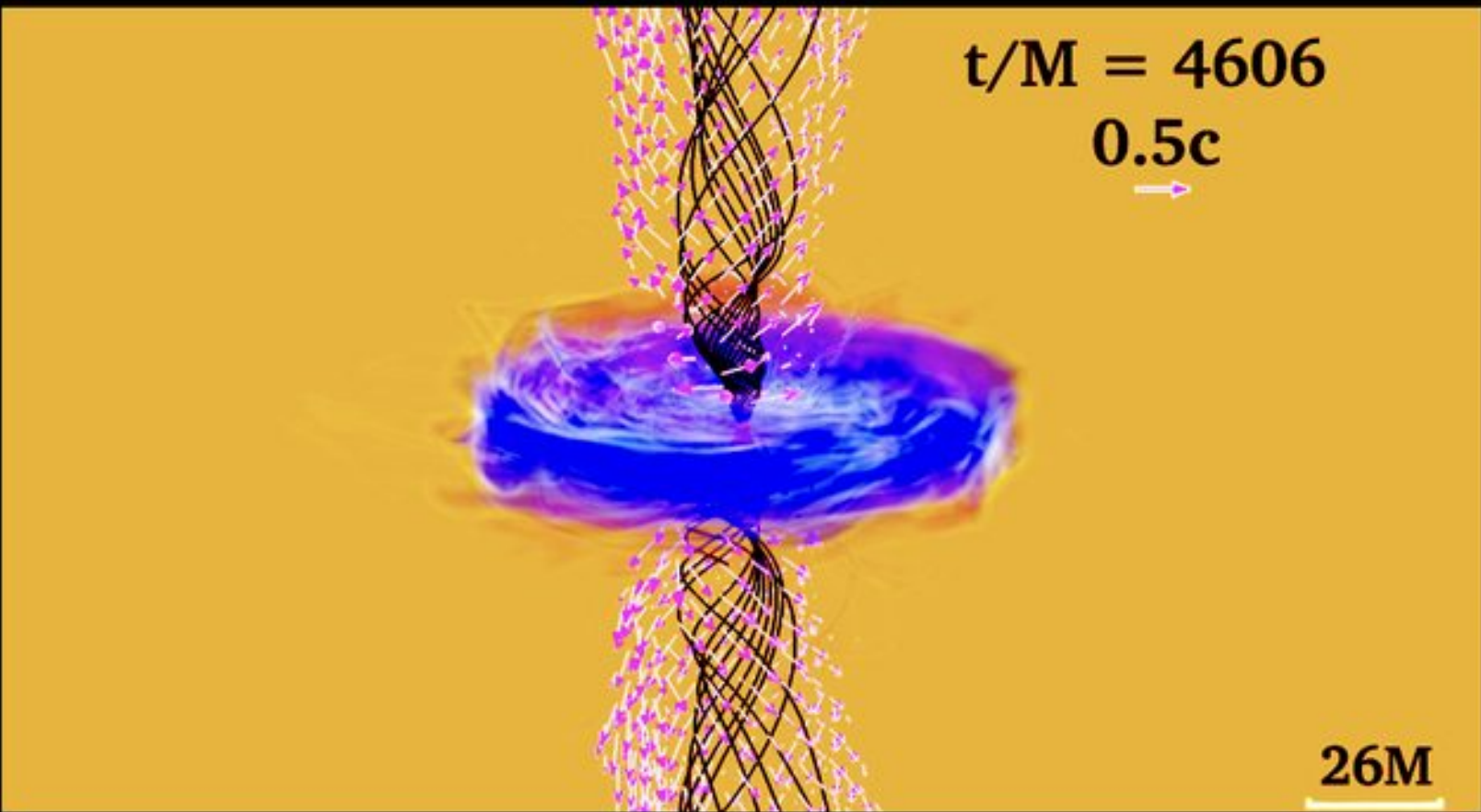
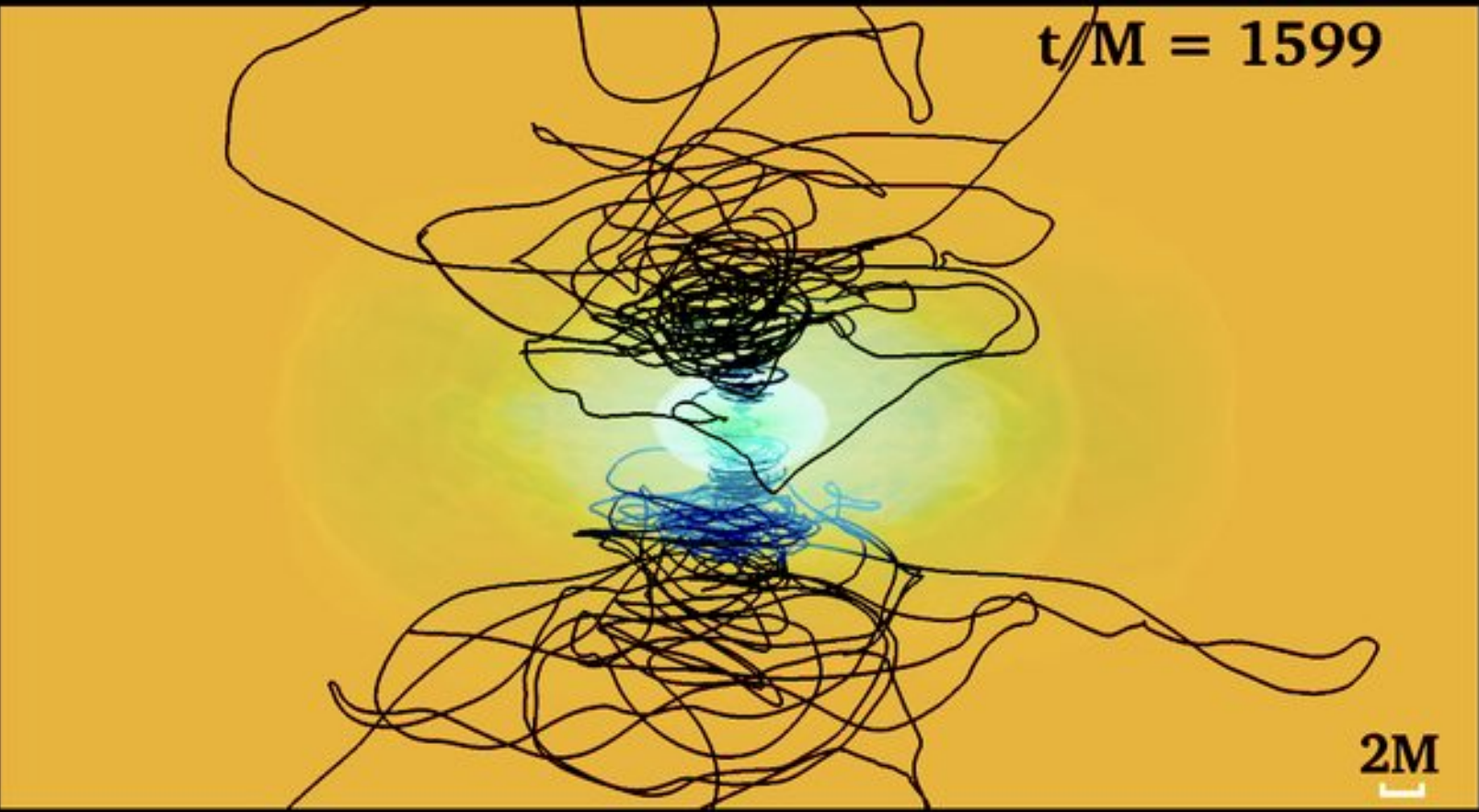
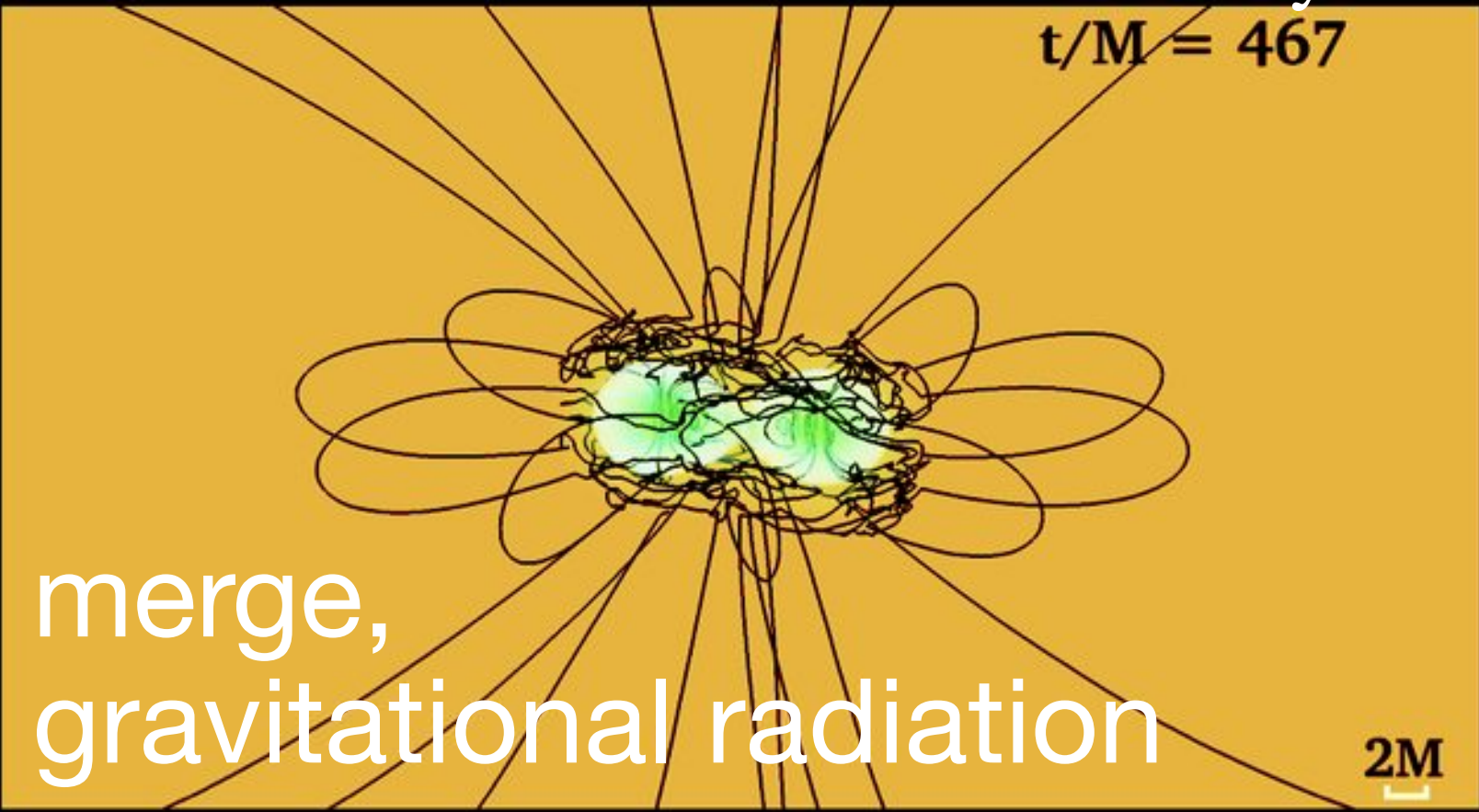
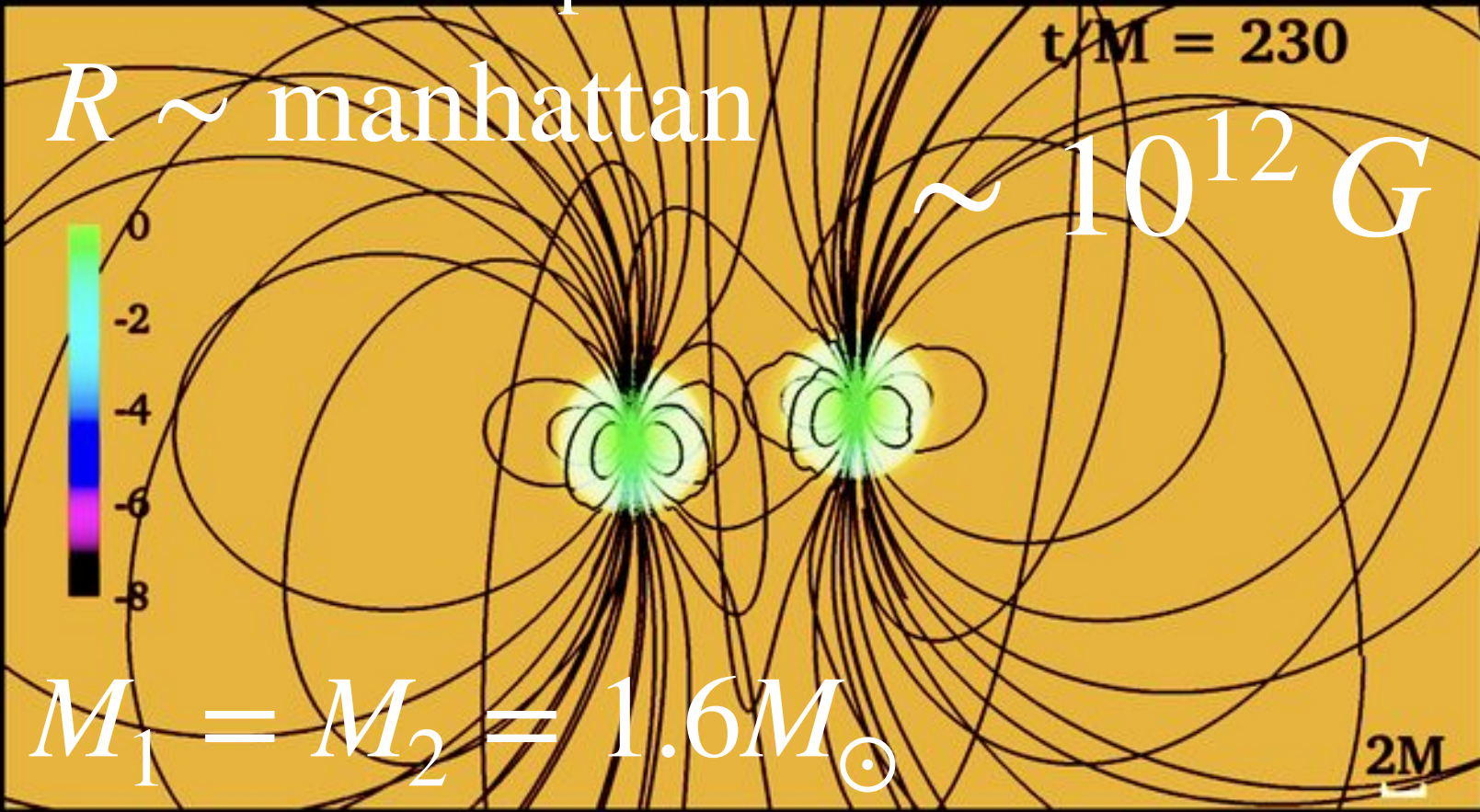
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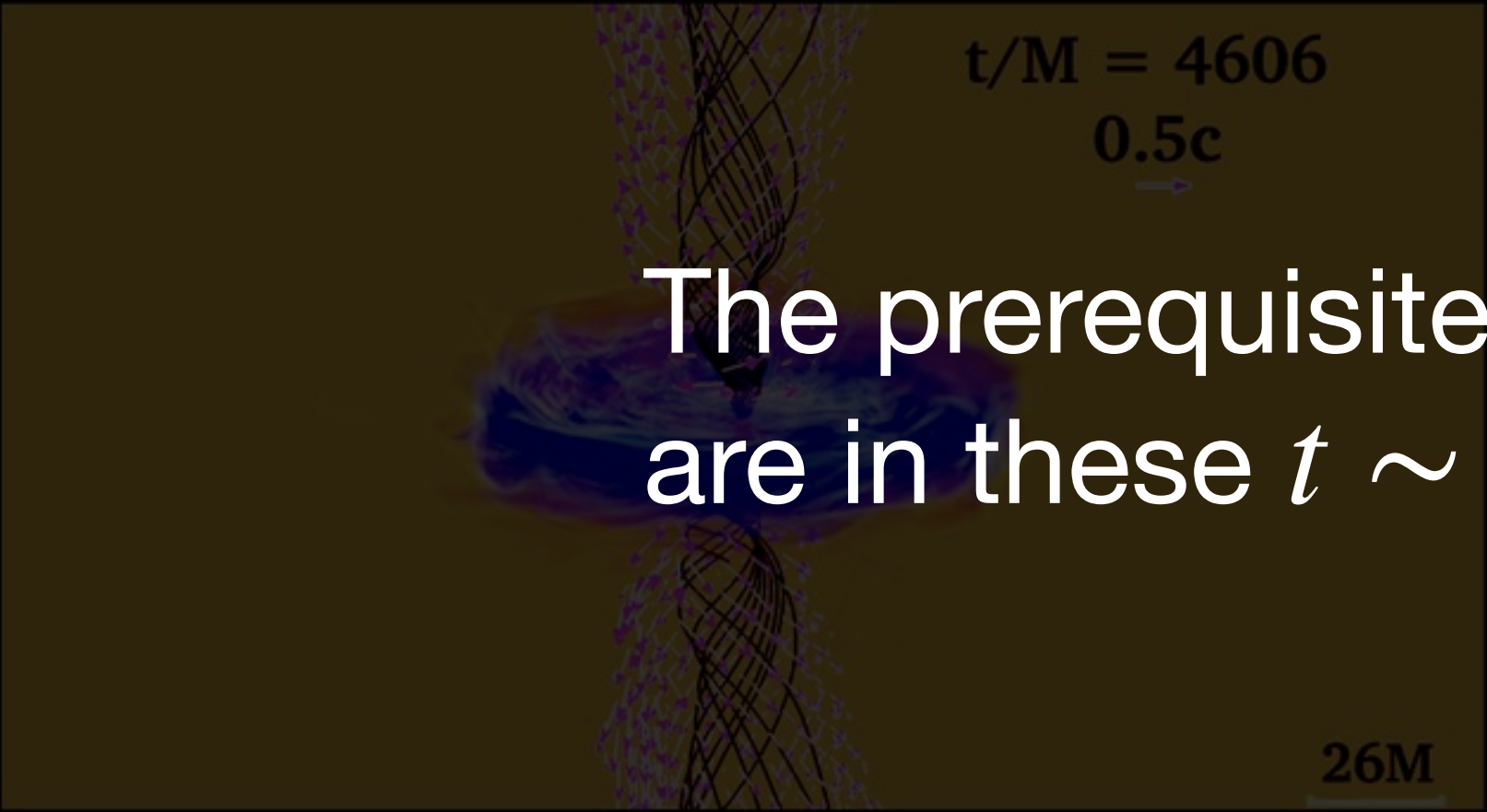
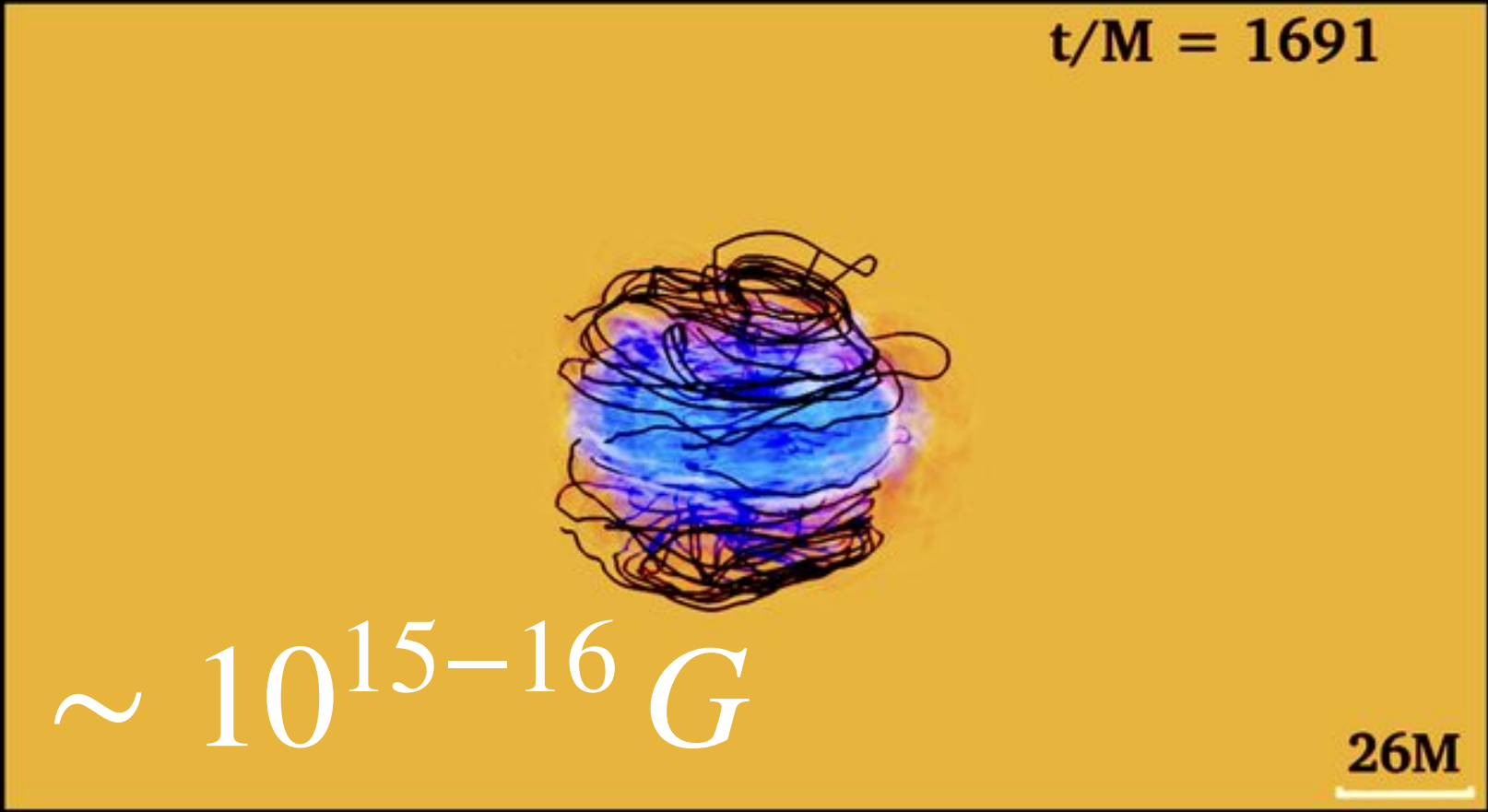
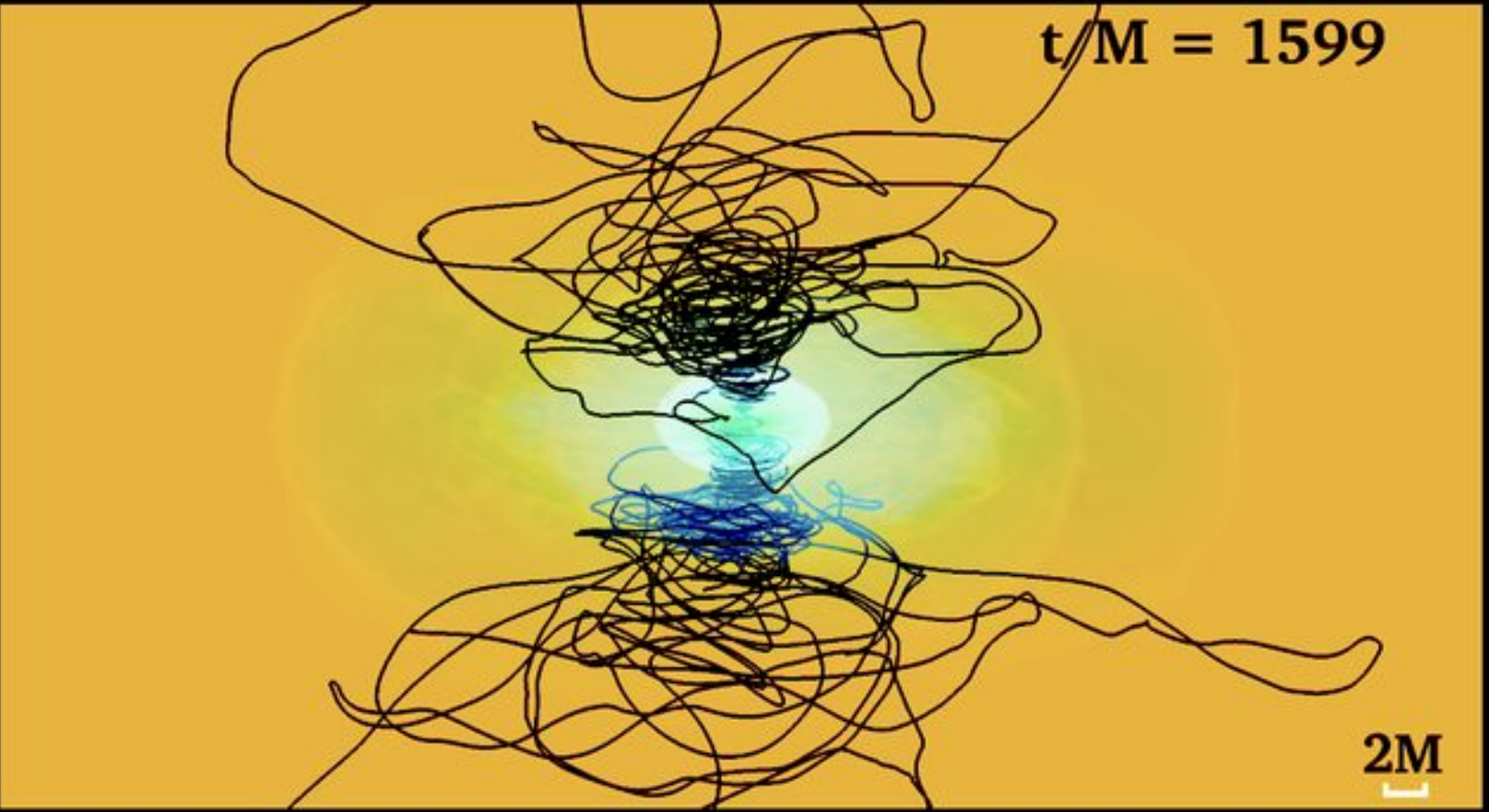
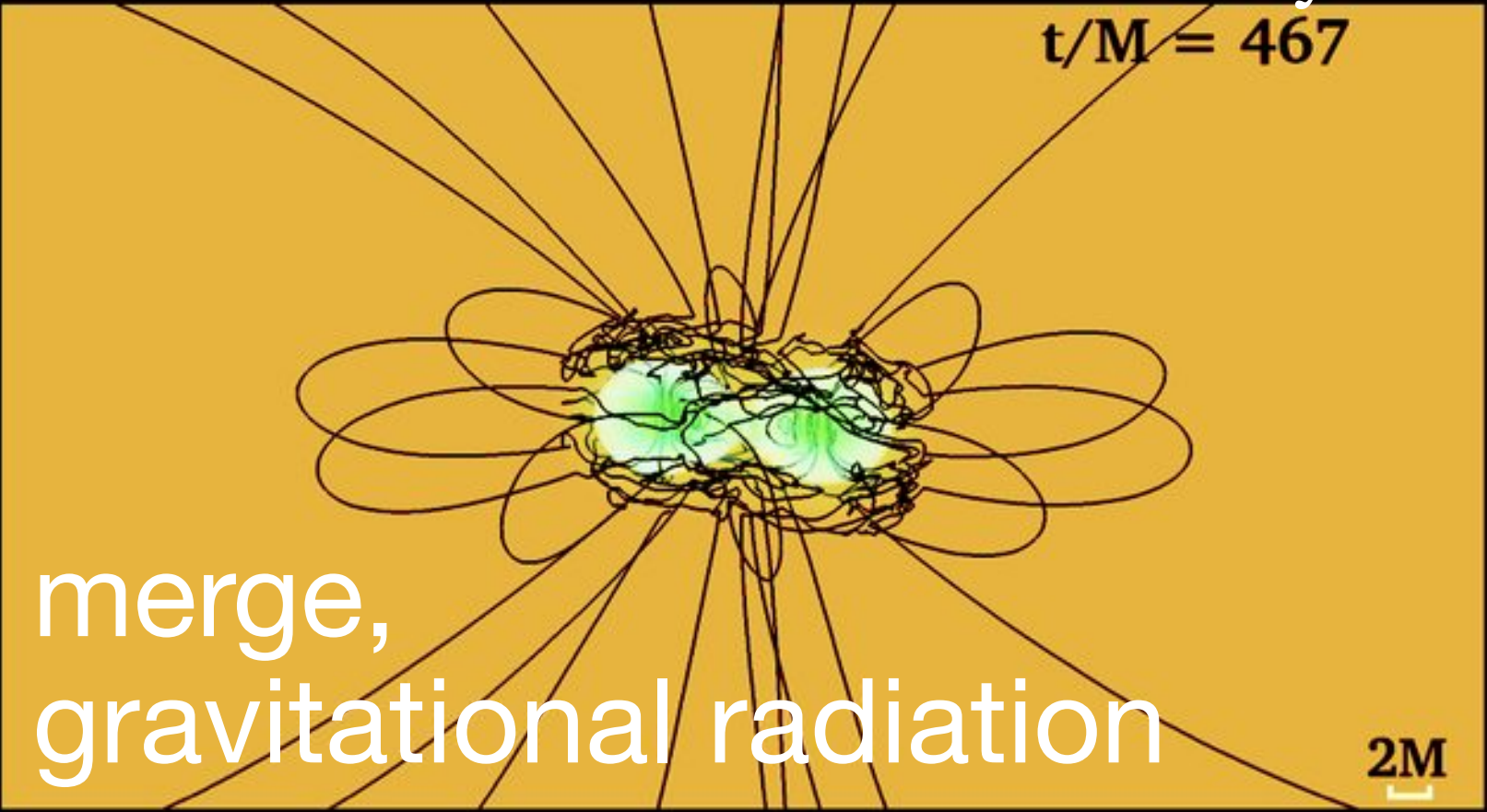
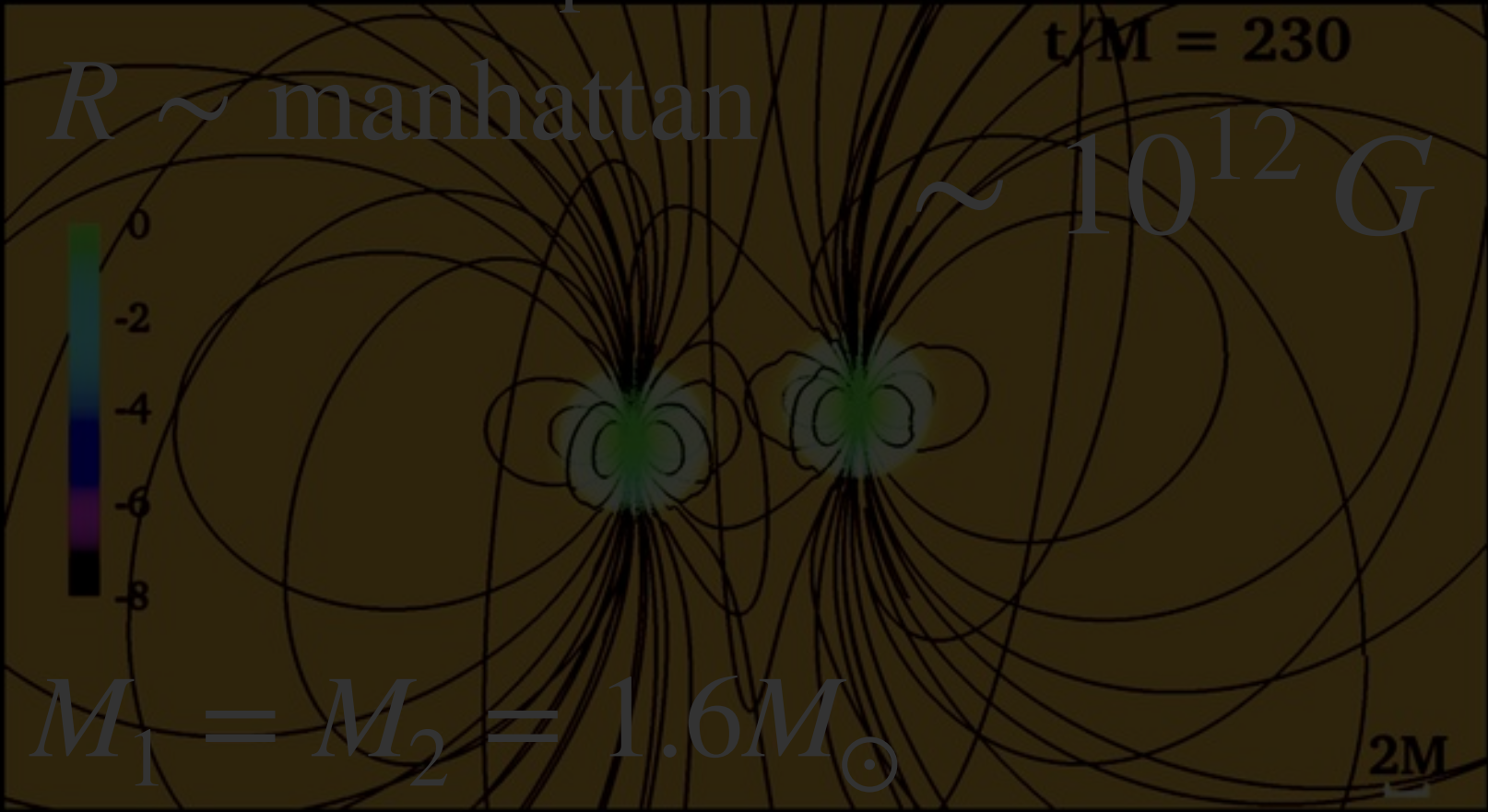
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The prerequisites for EM counterpart are in these $t \sim \text{ms}$ stages!

Significant restructuring of B-field

Blandford-Zanjek-type jet need vertical field...

GRB electromagnetic counterpart

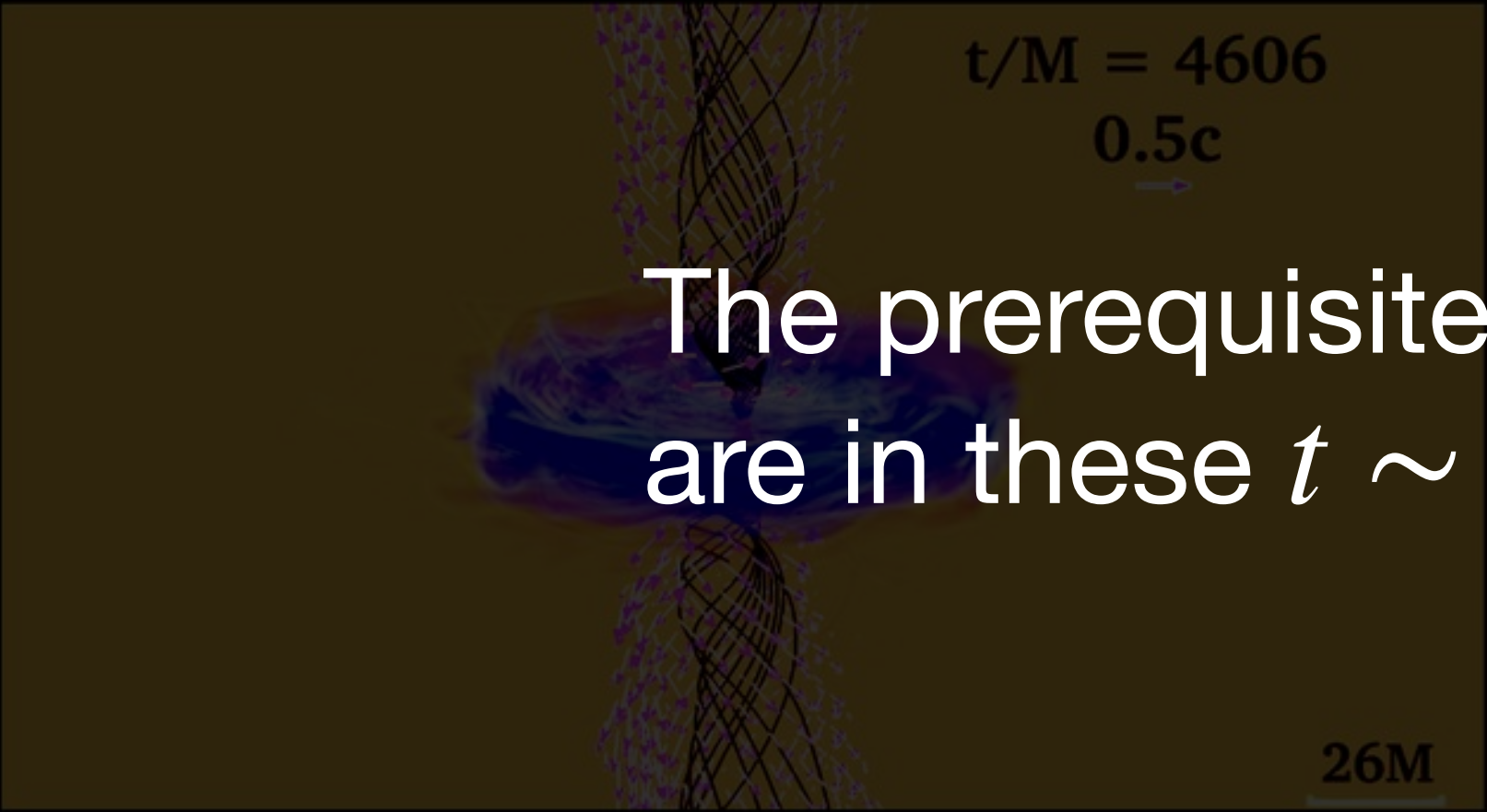
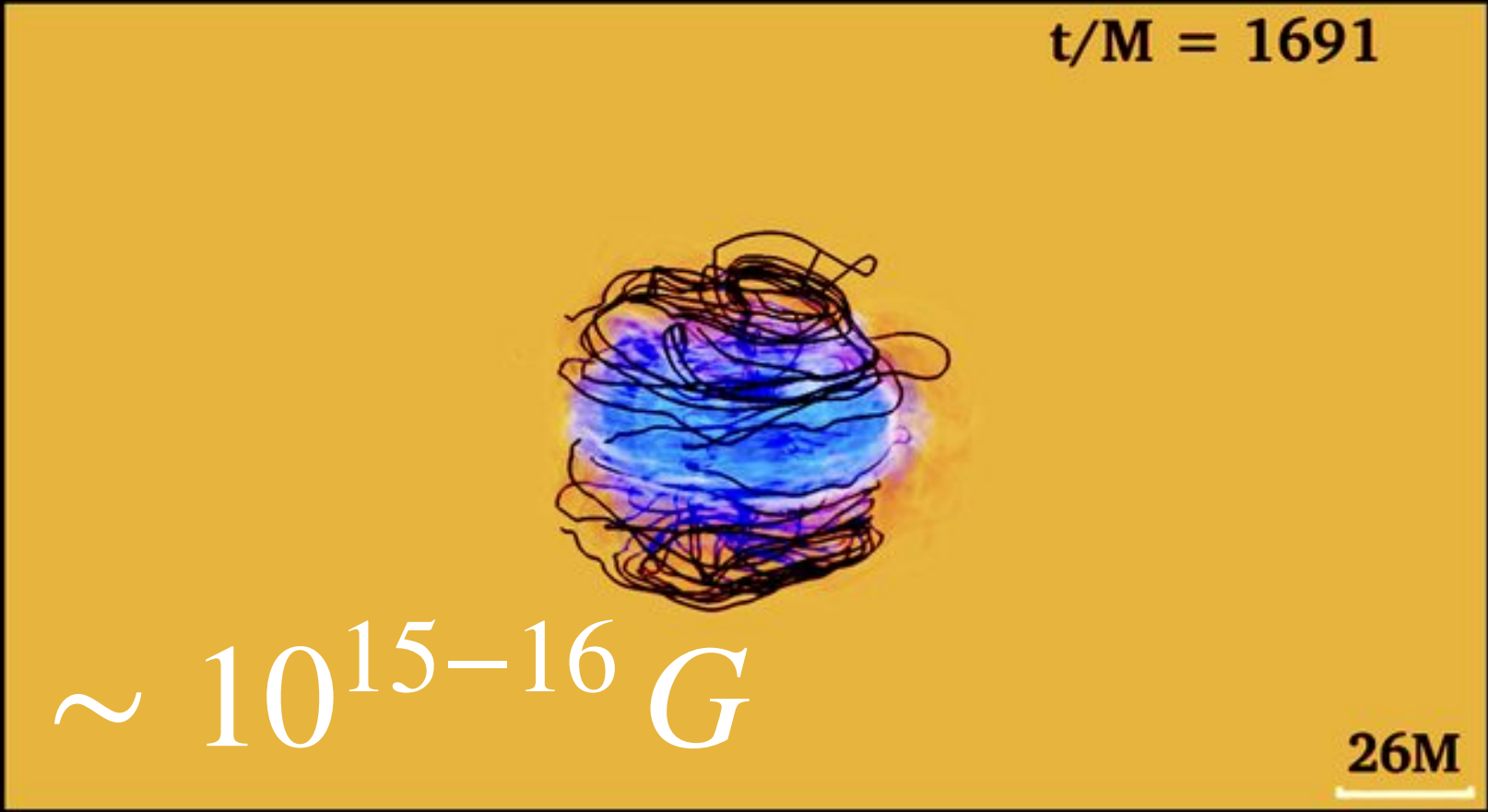
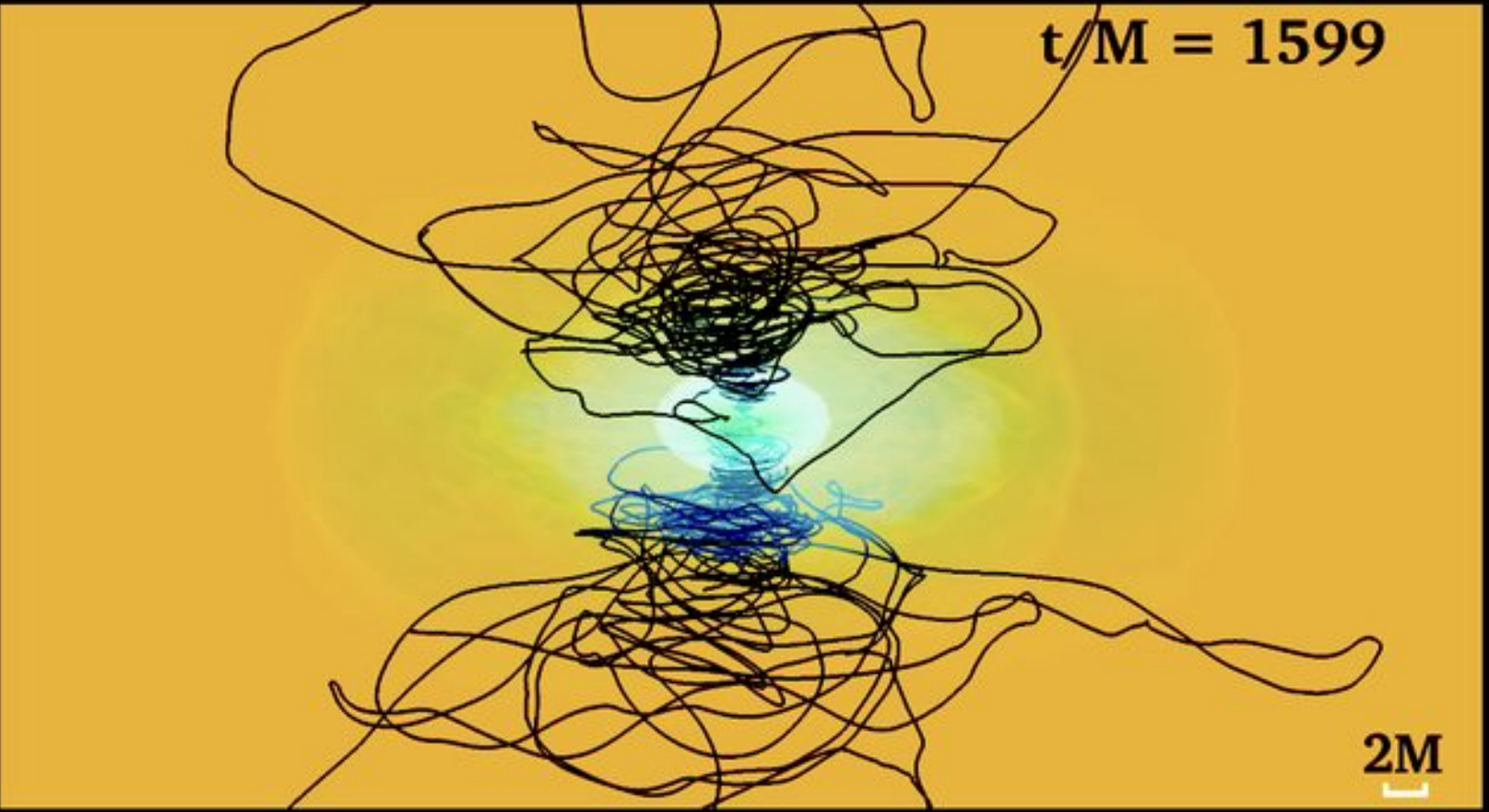
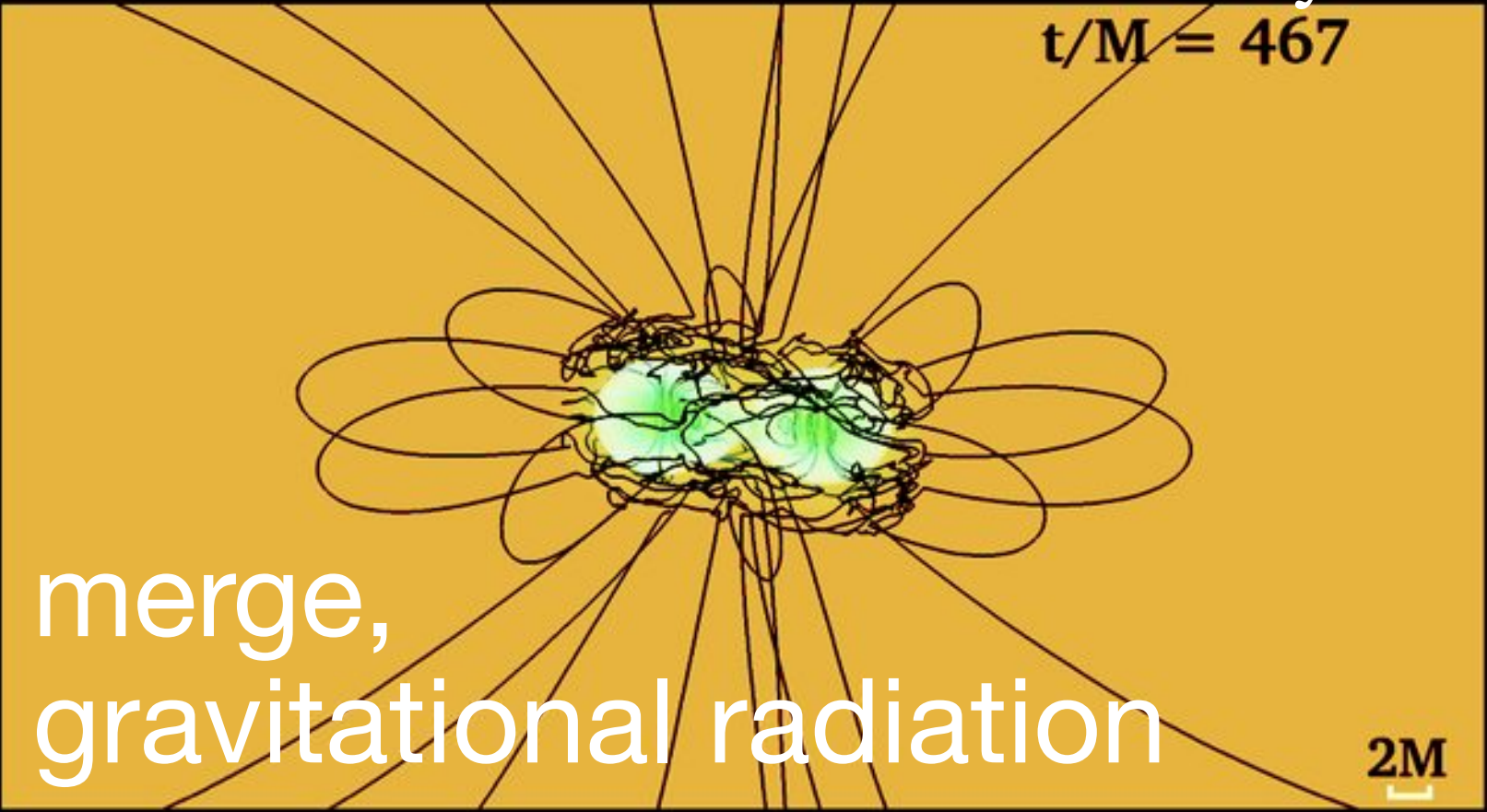
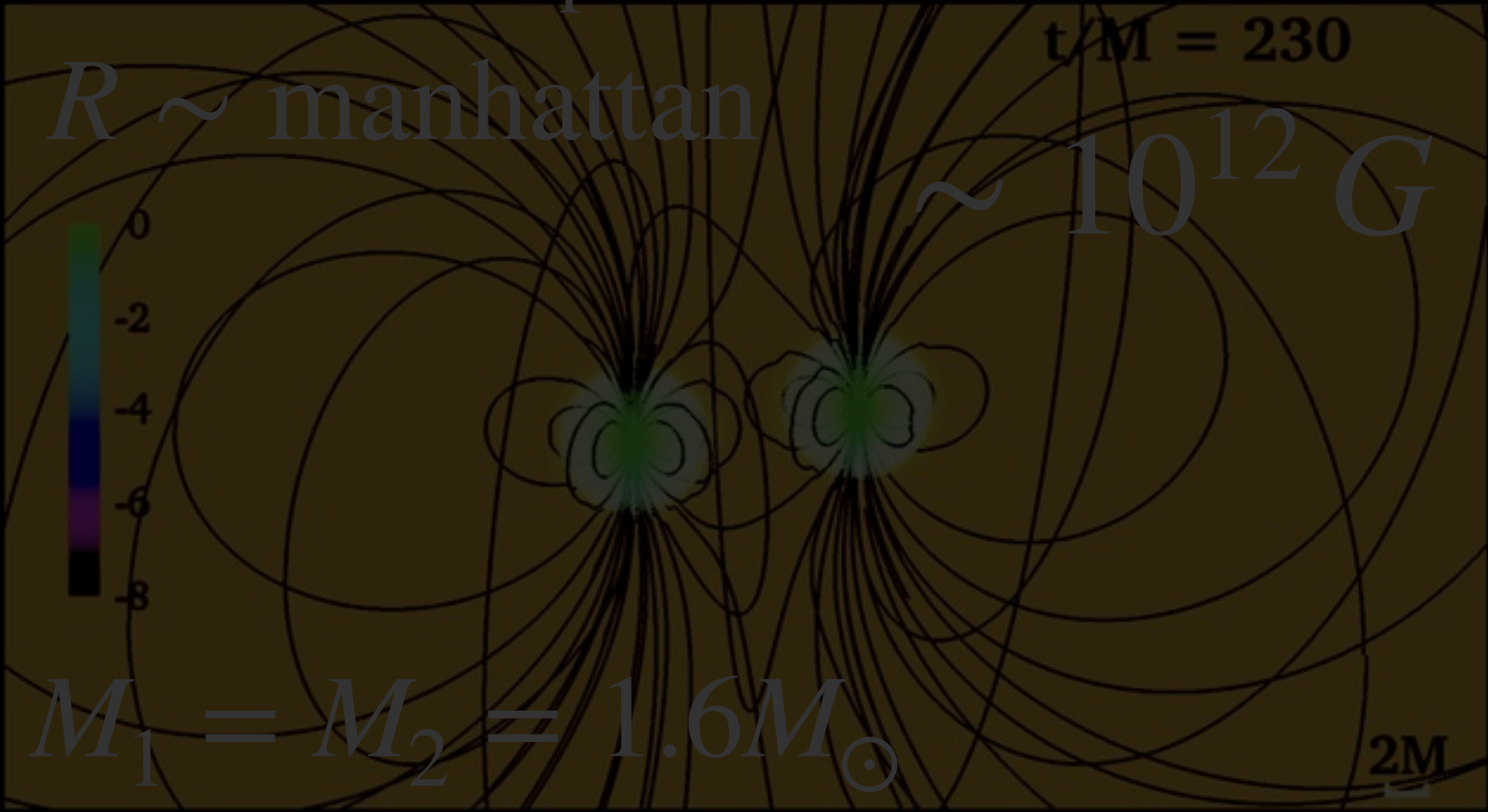
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Merging Binary Neutron Stars: our focus

Ruiz+(2018)

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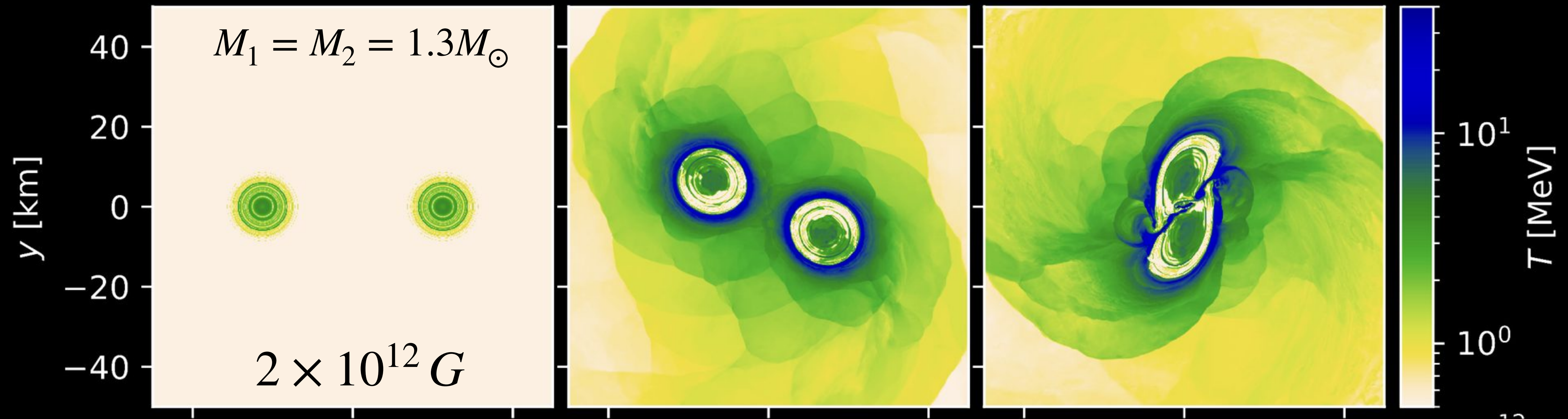
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Taking a closer look at “something happens”

Merging Binary Neutron Stars

Gutiérrez + (2026)

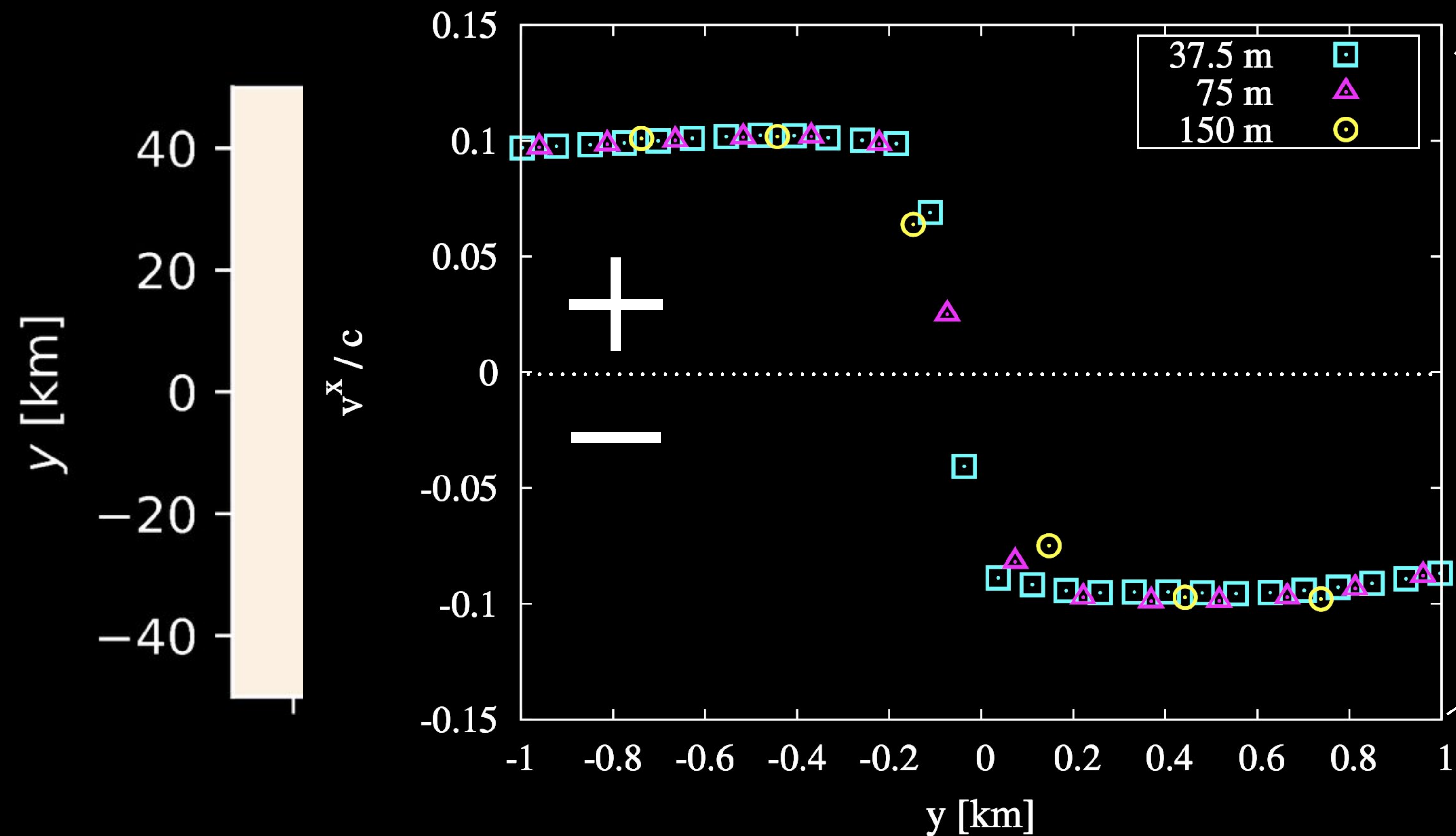
GRMHD (AthenaK)



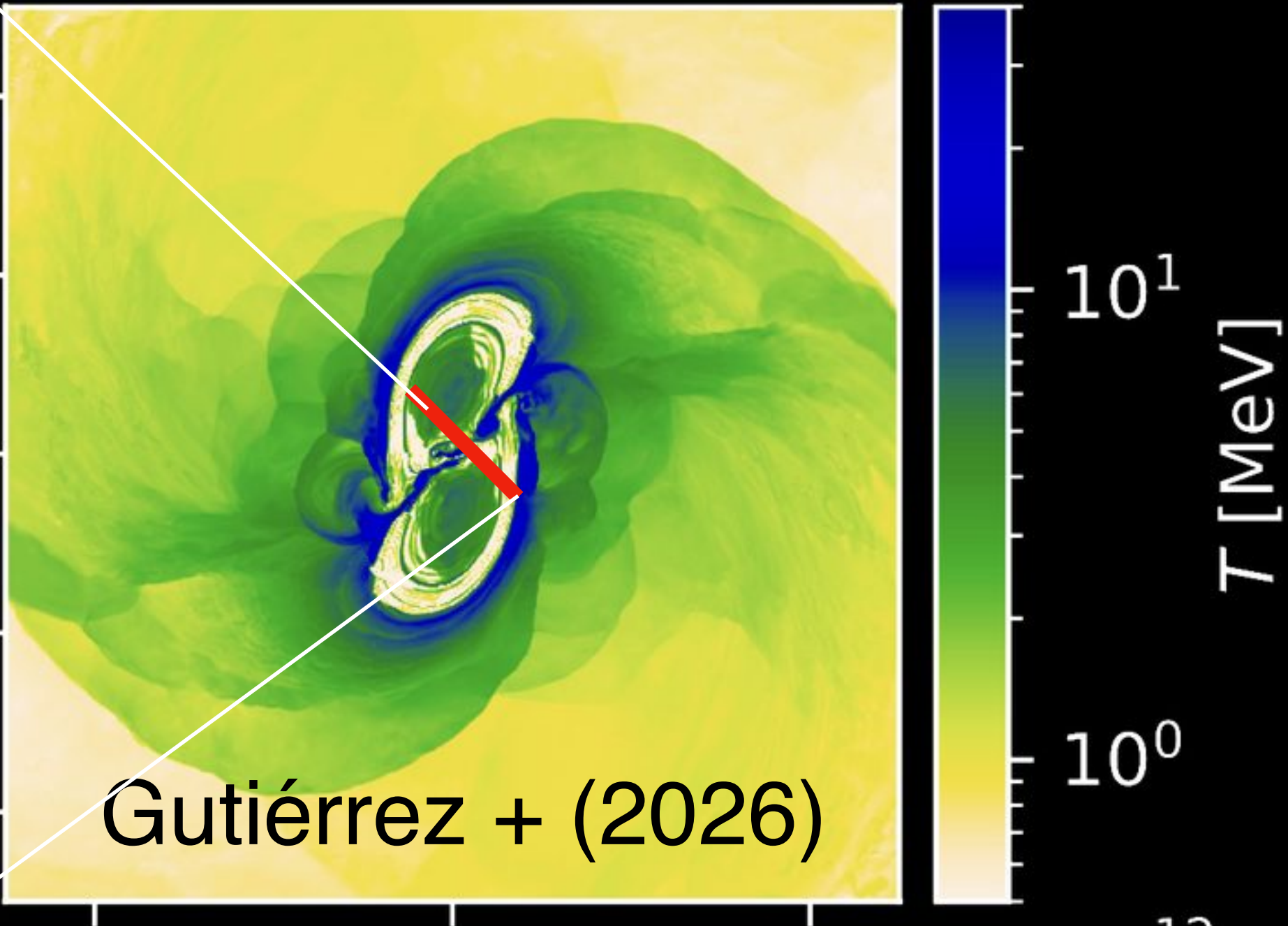
A shear layer!!!

Merging Binary Neutron Stars

Kiuchi+(2018)



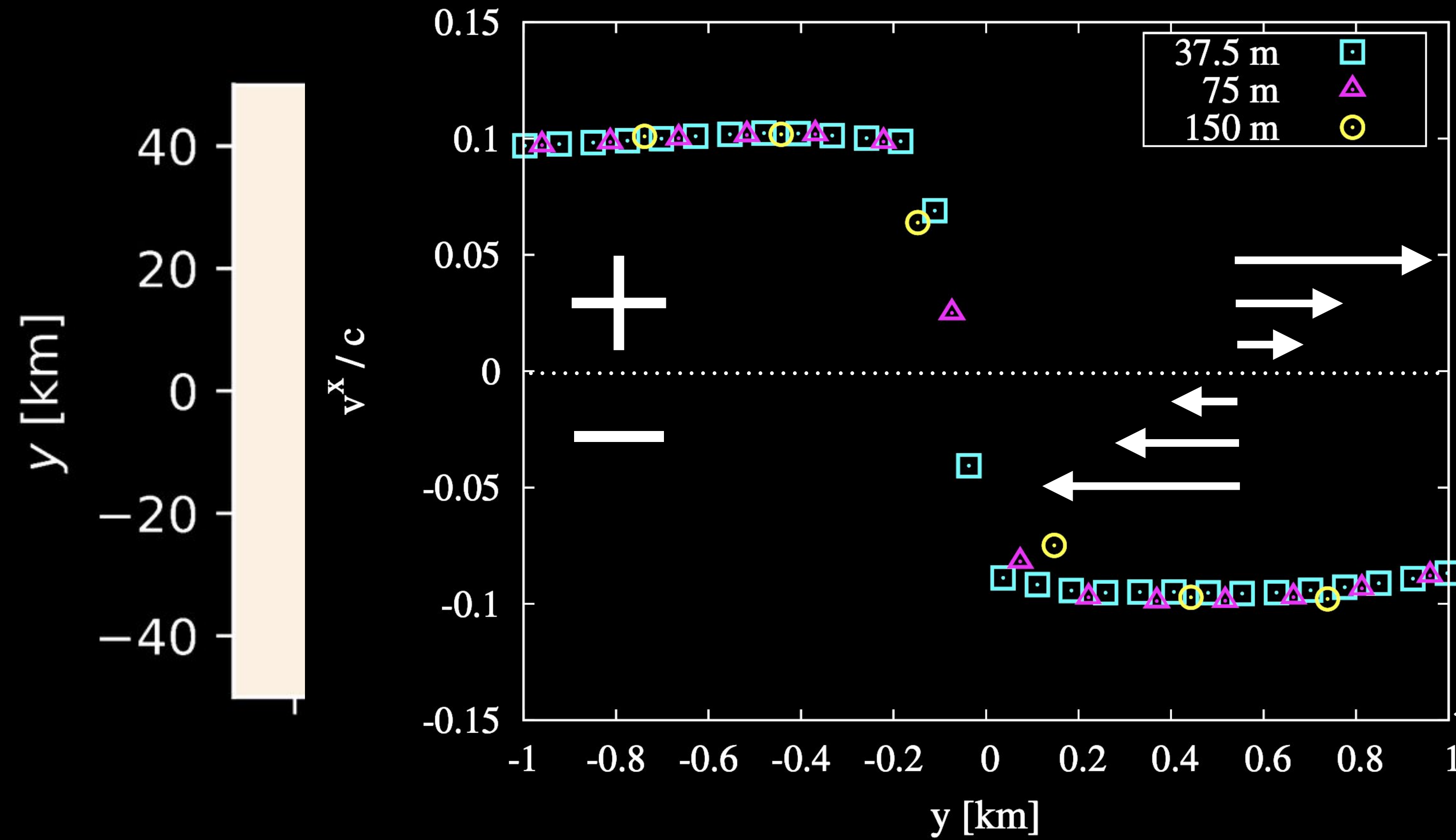
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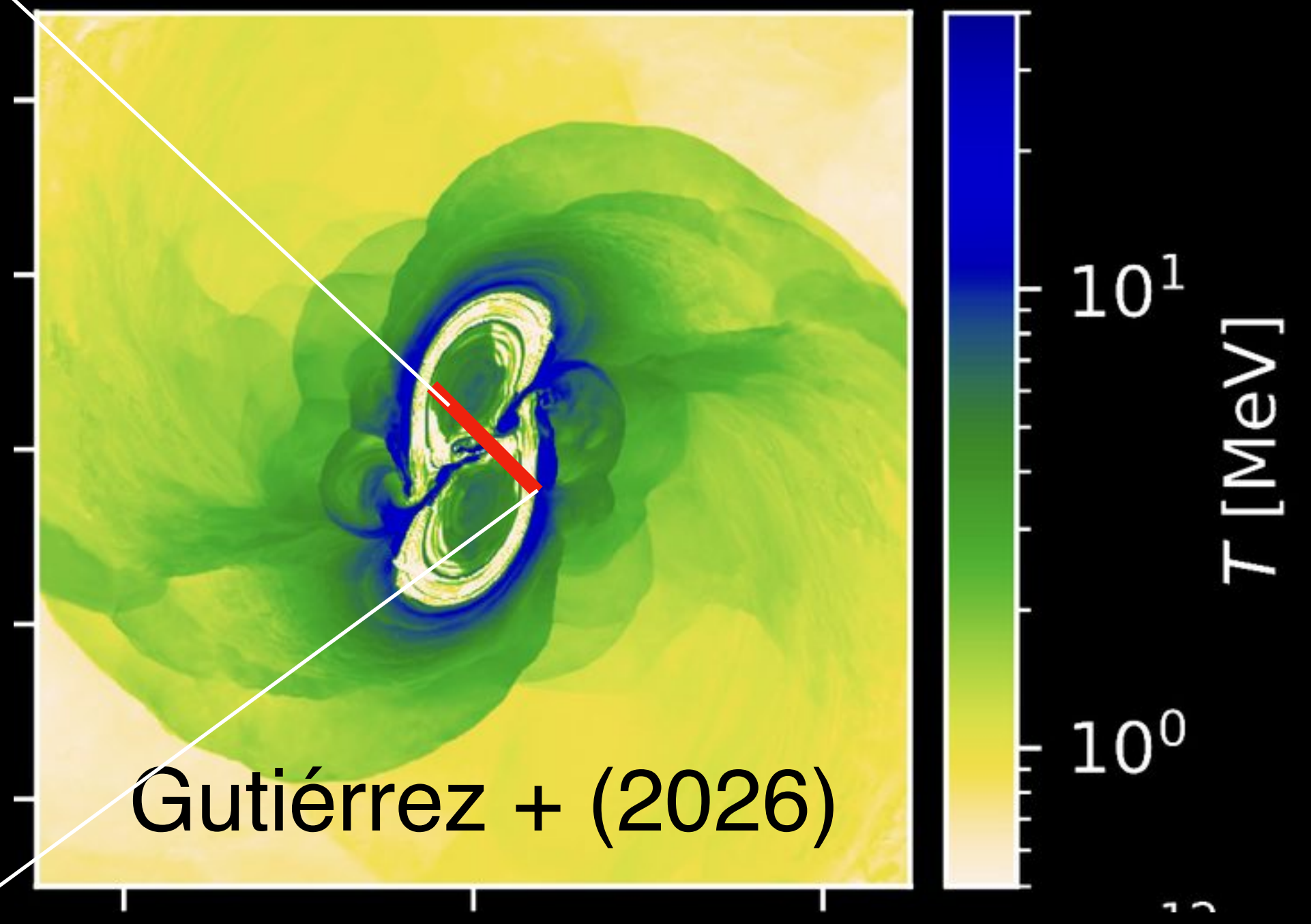
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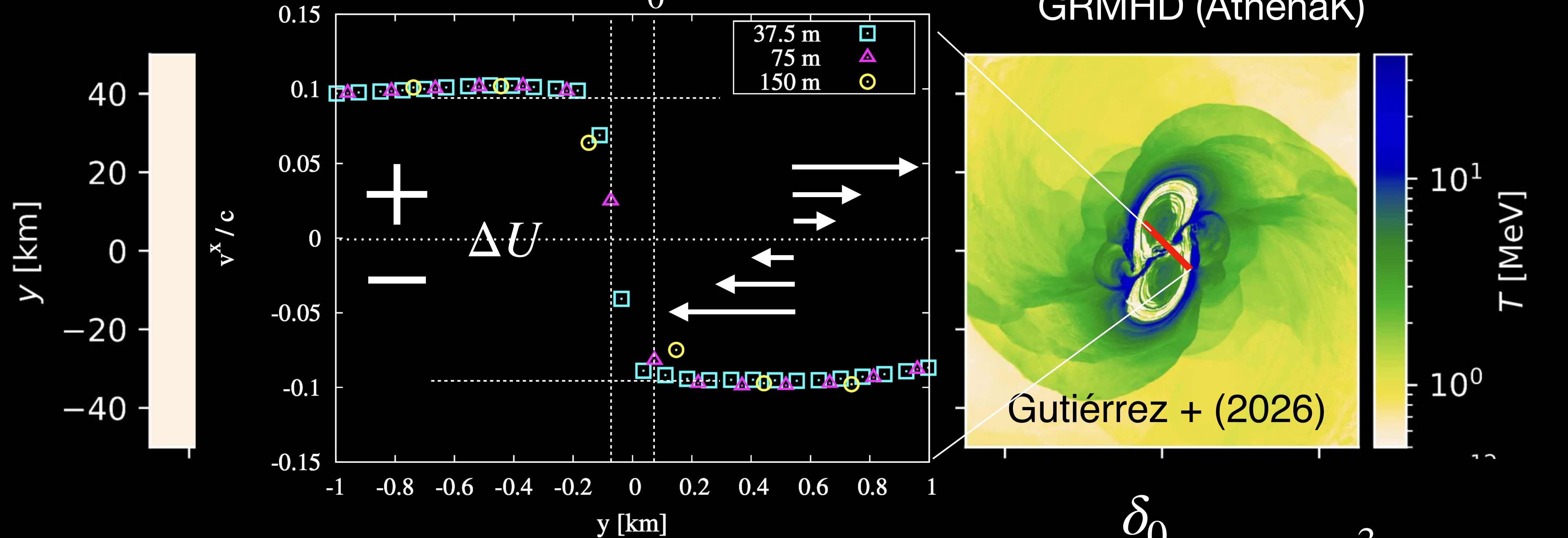
$$\Delta U \sim 0.2c$$

$$\delta_0 \sim 50 \text{ m}$$

Kiuchi+(2018)

$$2\delta_0$$

GRMHD (AthenaK)



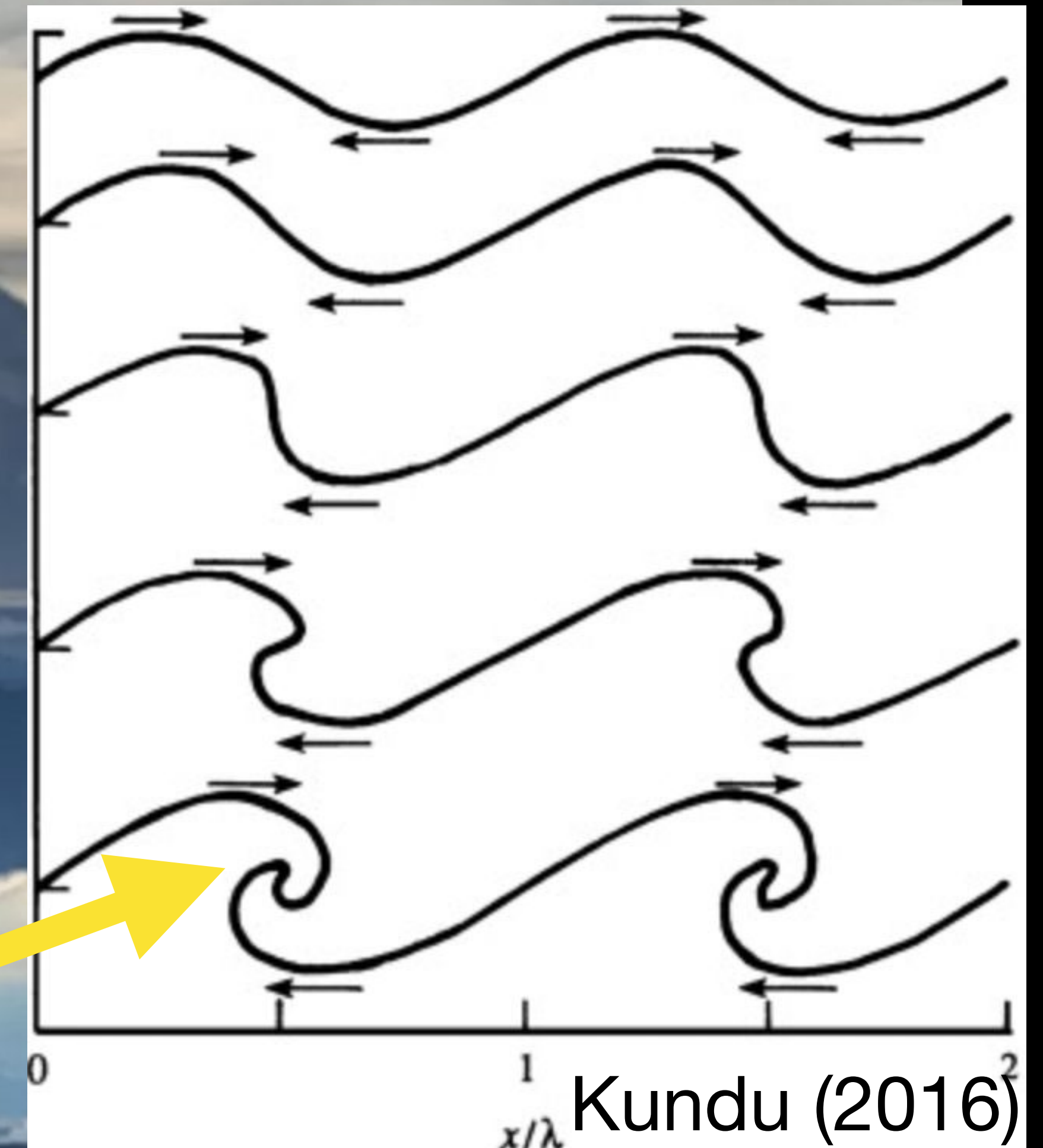
$$t_{\Delta} = \frac{\delta_0}{\Delta U} \sim 10^{-3} t_{\text{dynamical}}$$

Shear layers are linearly unstable to Kelvin-Helmholtz instability!

linear theory $\sim t_{\Delta}$

$$t_{\Delta} = \frac{\delta_0}{\Delta U} \sim 10^{-3} t_{\text{dynamical}}$$

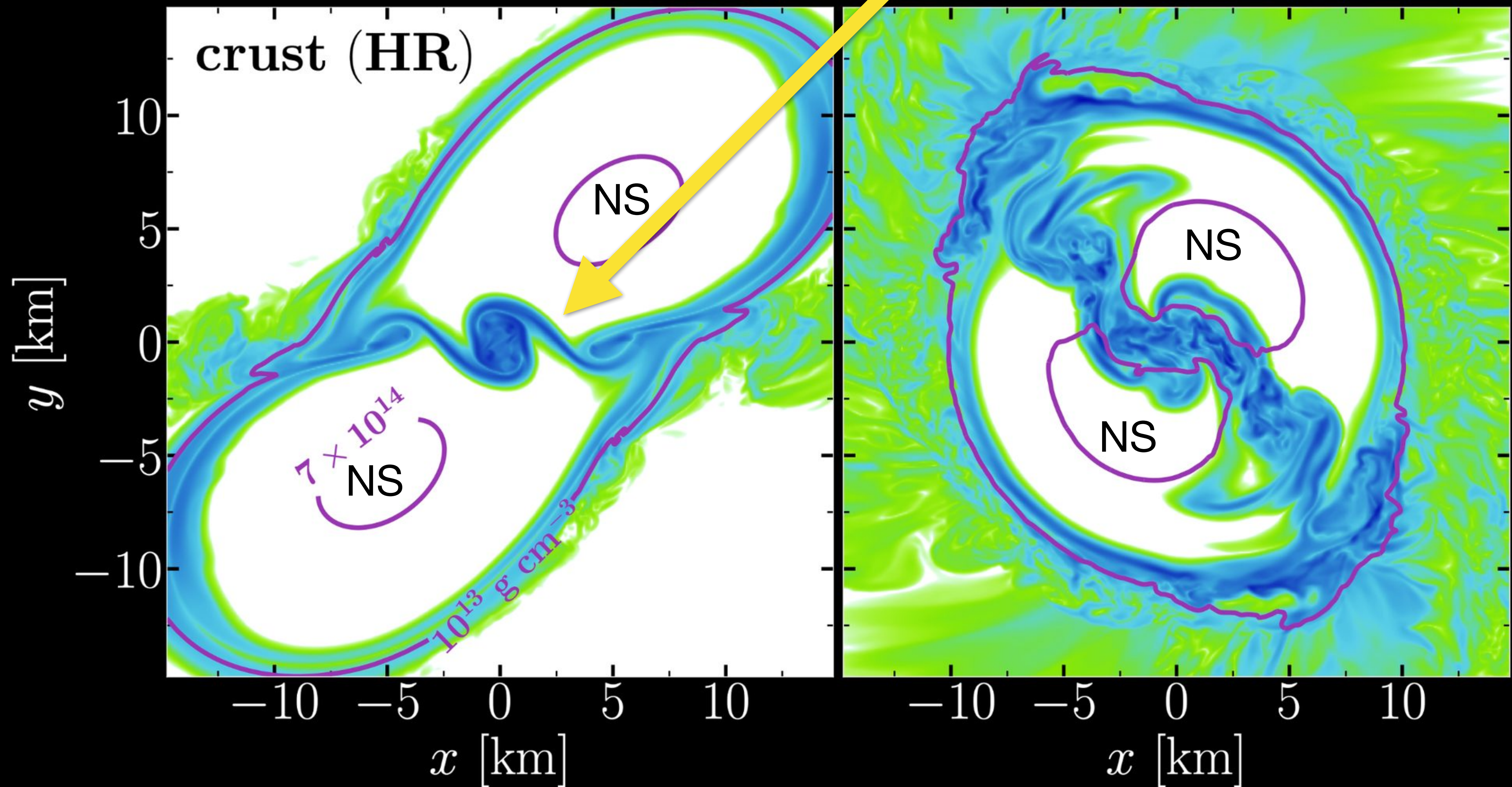
Kelvin Helmholtz instability (KHI) billows
come from nonlinear evolution



One of the most extreme magnetic dynamos

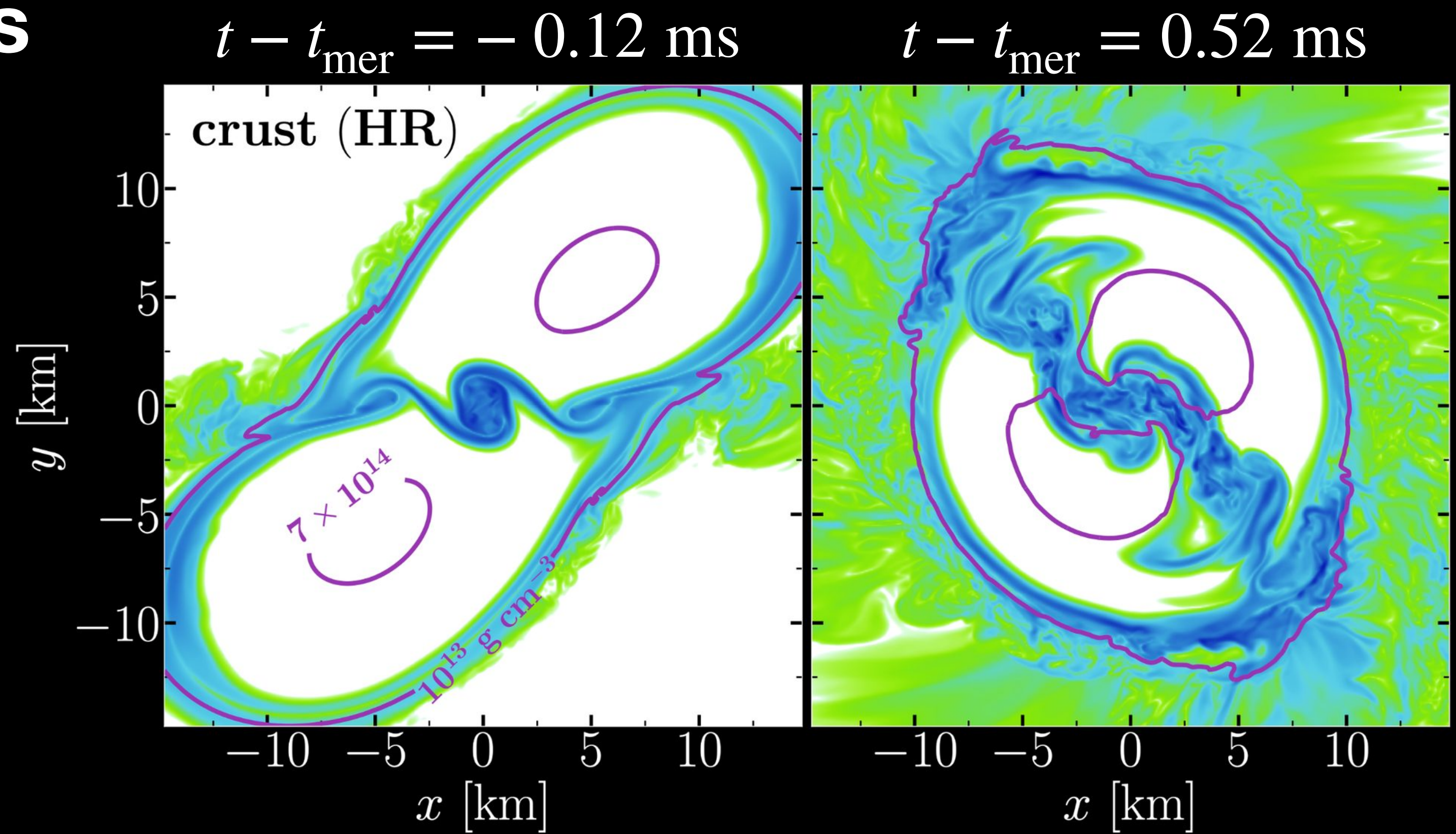
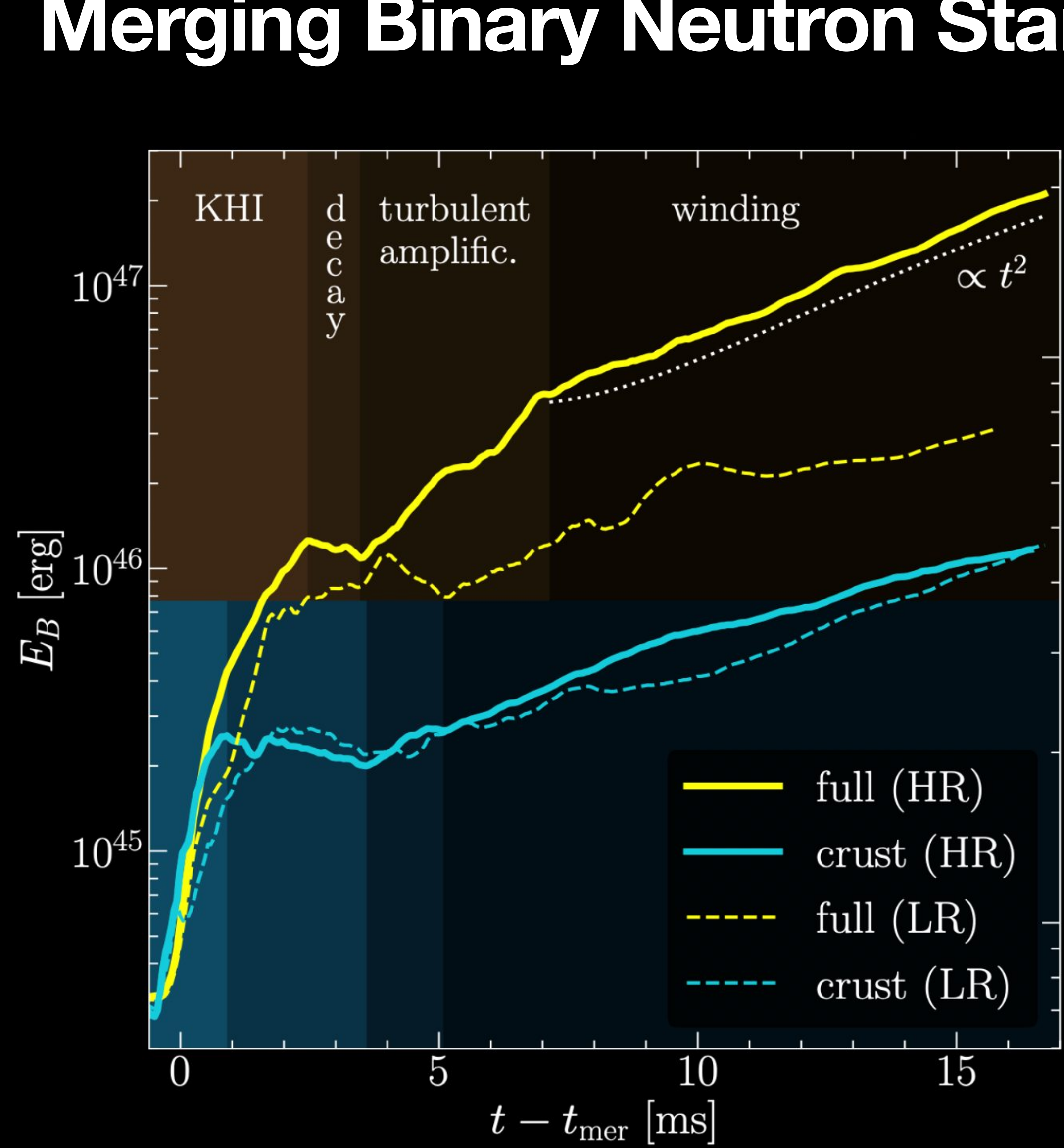
Merging Binary Neutron Stars

Chabanov+ (2023)



One of the most extreme magnetic dynamos

Merging Binary Neutron Stars



fast growth → pause
exponential growth ← → winding

$B \sim 10^{15-16} \text{ G}$

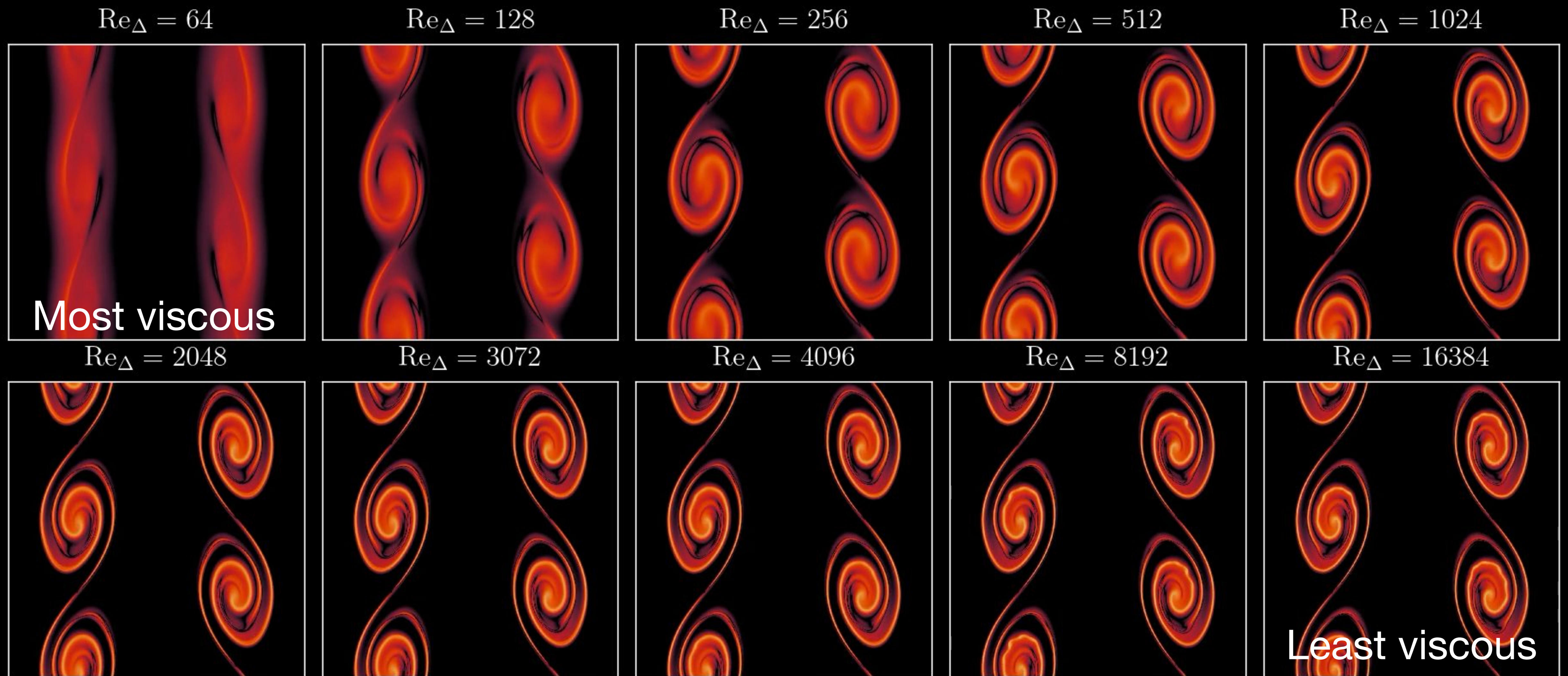
Chabanov+ (2023) *ILES FIL*

One of the most extreme magnetic dynamos

Merging Binary Neutron Stars

1. How does the multi-stage KHI-driven dynamo work?
2. Can we form the right ingredients for jet launching during the KHI
3. Do global calculations resolve the important aspects of the dynamo?
(e.g., remember kinematic growth rates are typically a function of viscosity!!)

Ensemble of 3D visco-resistive KHI dynamo simulations in AthenaK (up to $4,096^3$ cells)

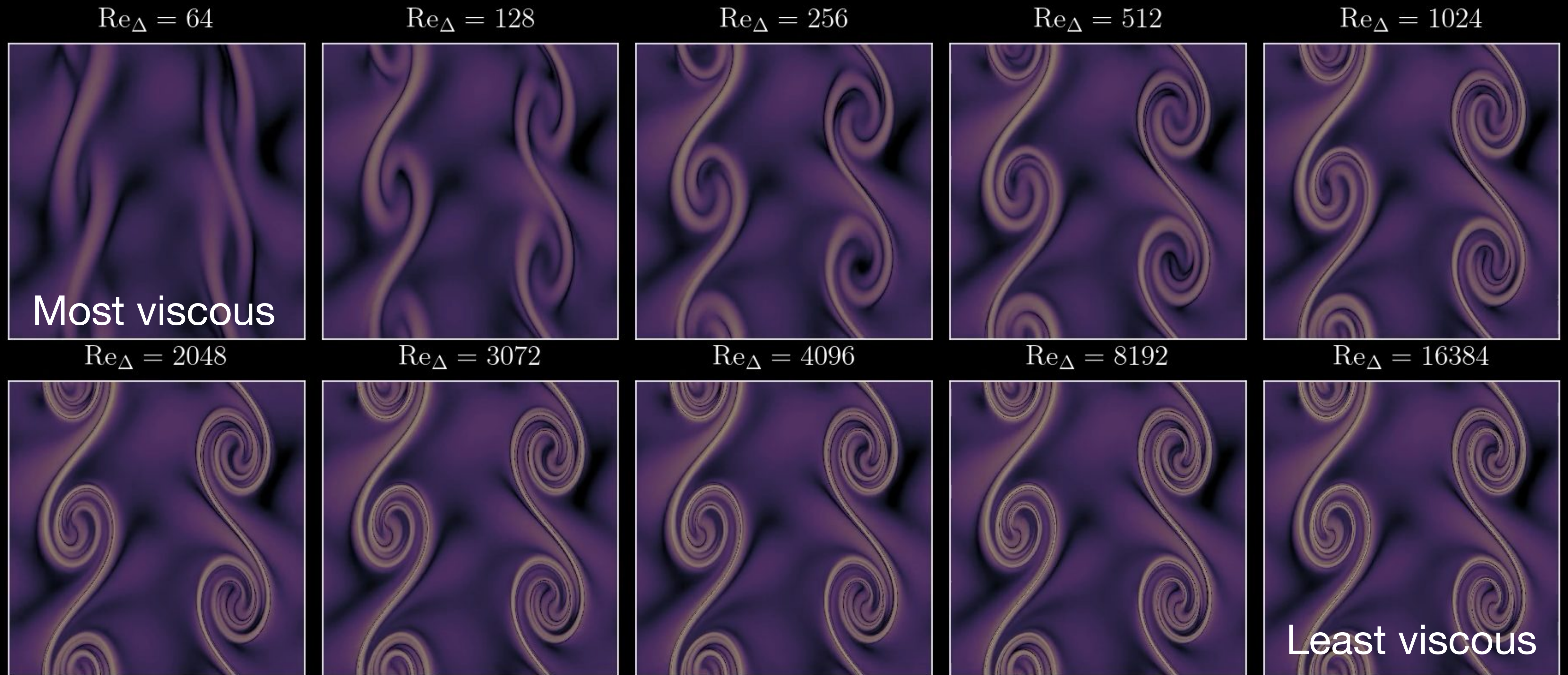


$$\text{Re}_{\Delta} = \frac{\delta_0 \Delta U}{\nu}$$



80M-core-h project
Beattie, in prep

Ensemble of 3D visco-resistive KHI dynamo simulations in AthenaK (up to $4,096^3$ cells)



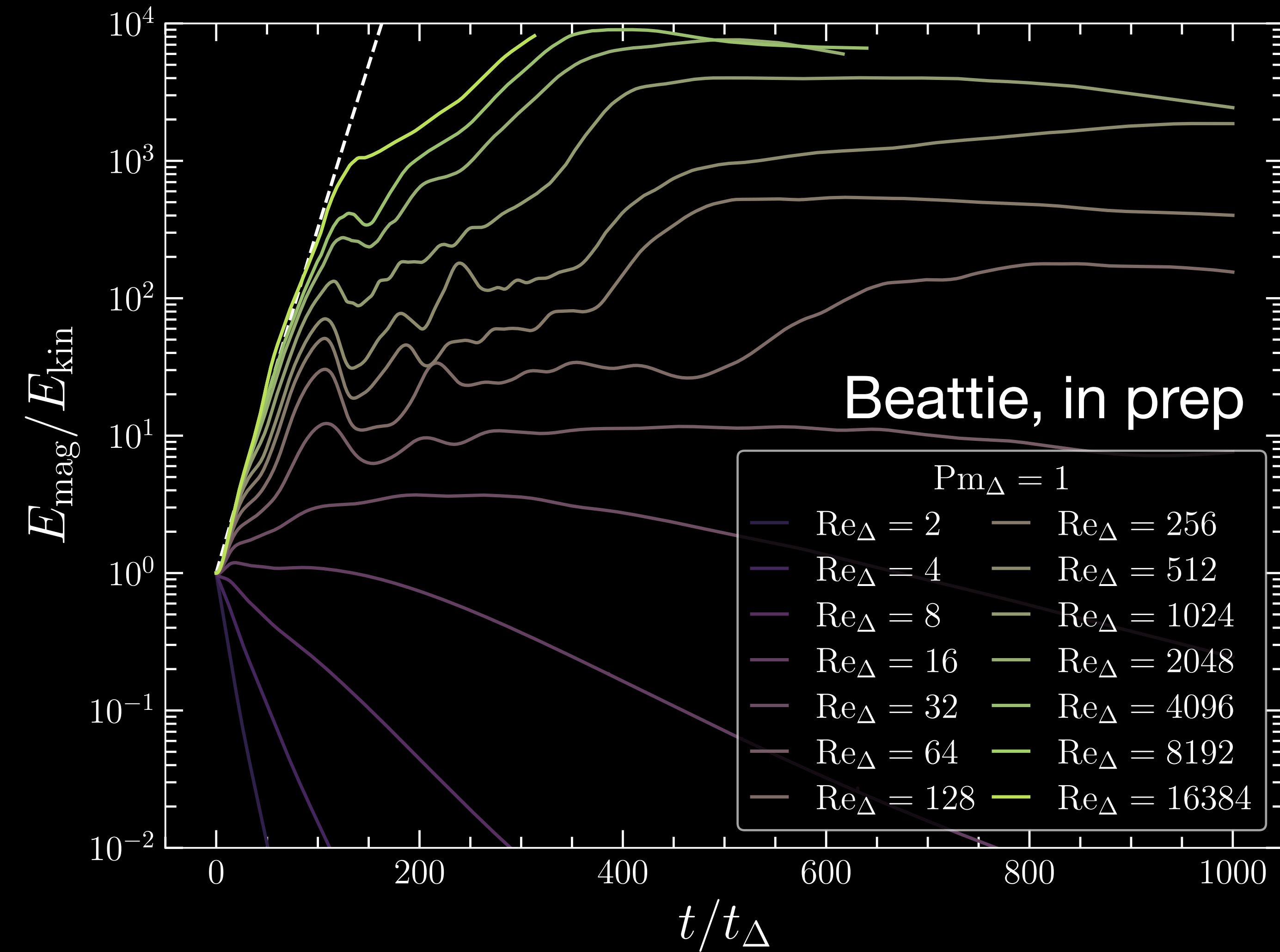
$$\text{Re}_{\Delta} = \frac{\delta_0 \Delta U}{\nu}$$



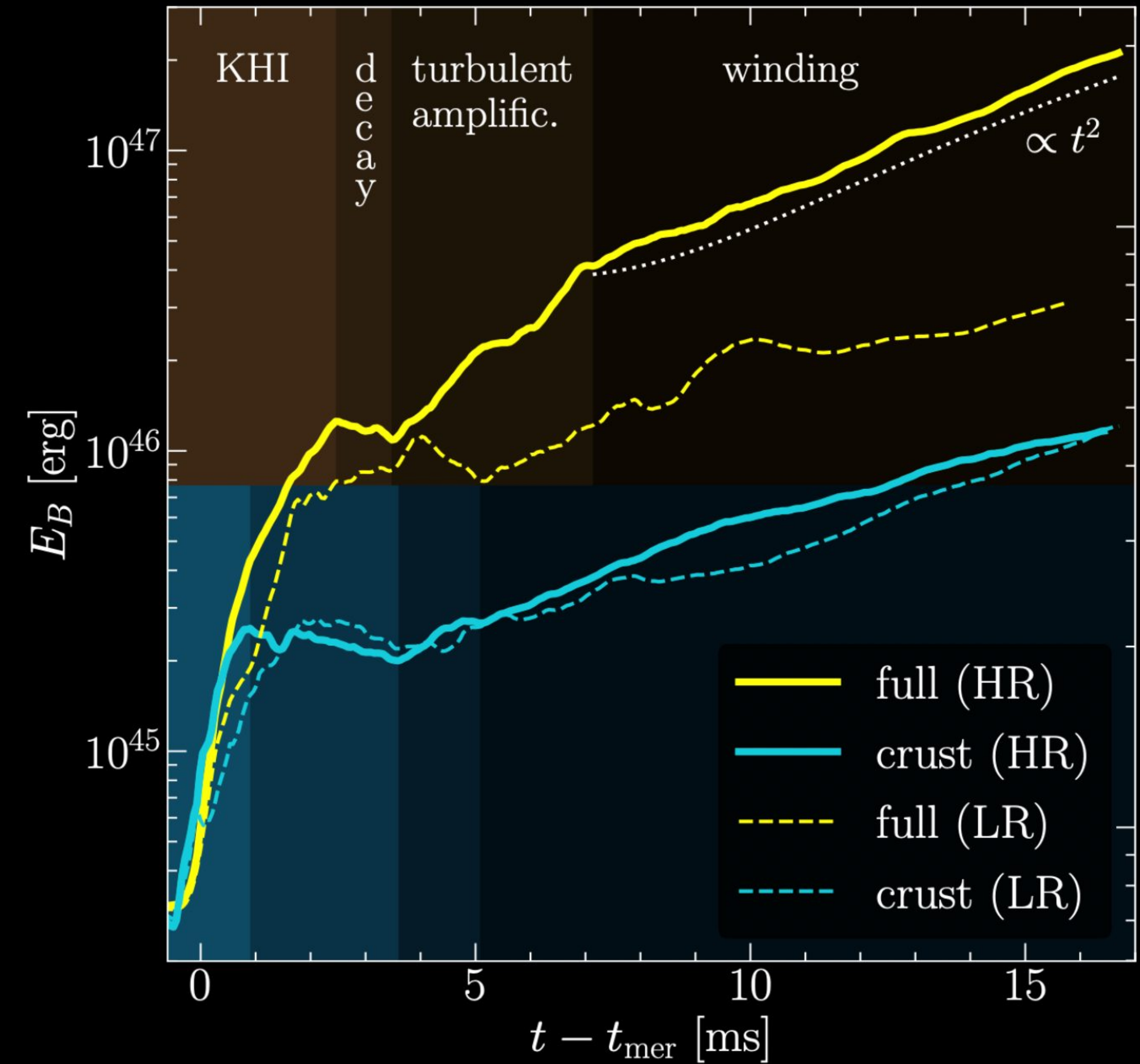
80M-core-h project
Beattie, in prep

Volume integral energies — a great match, qualitatively!

Local calculation

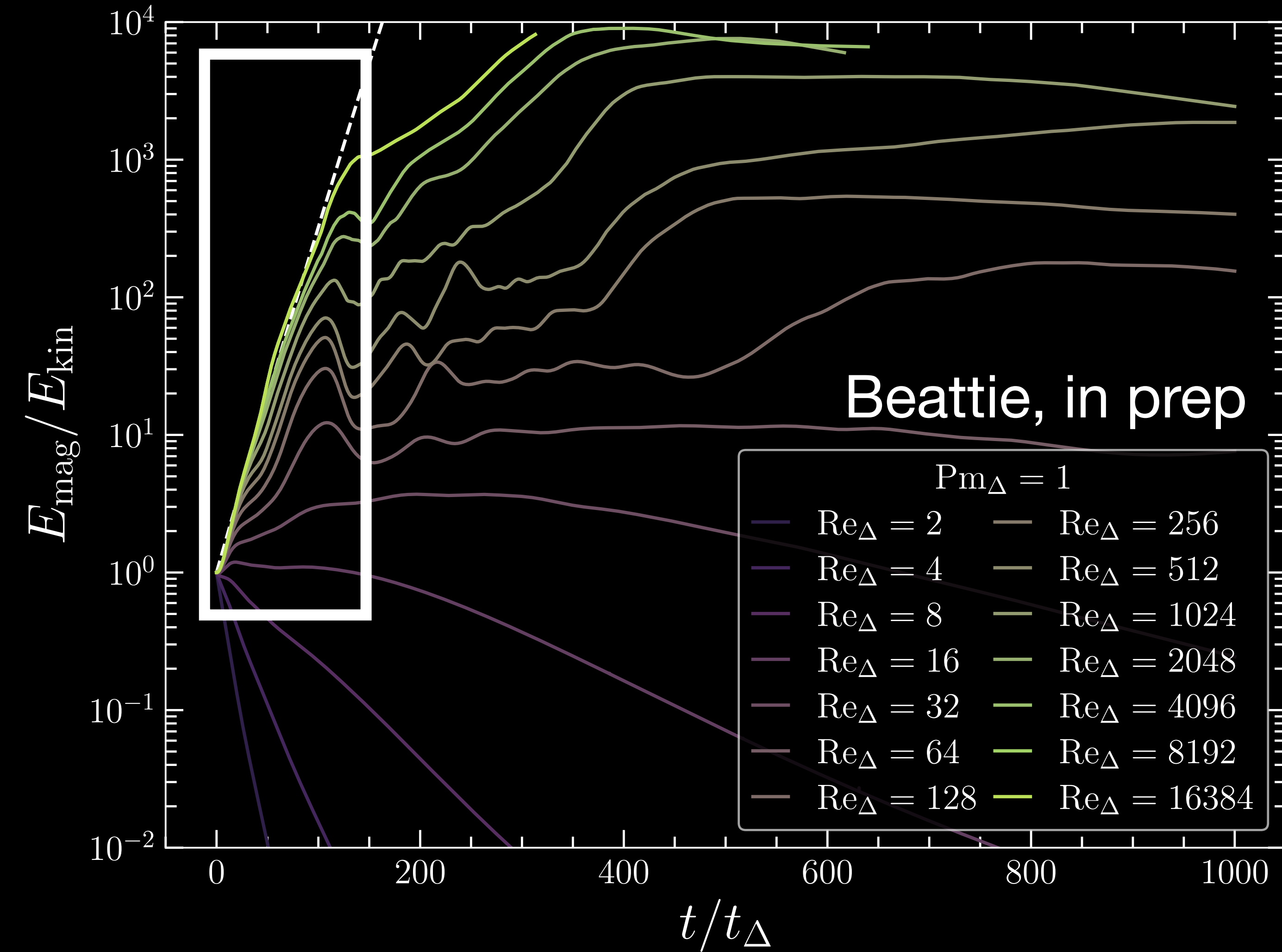


Global calculation

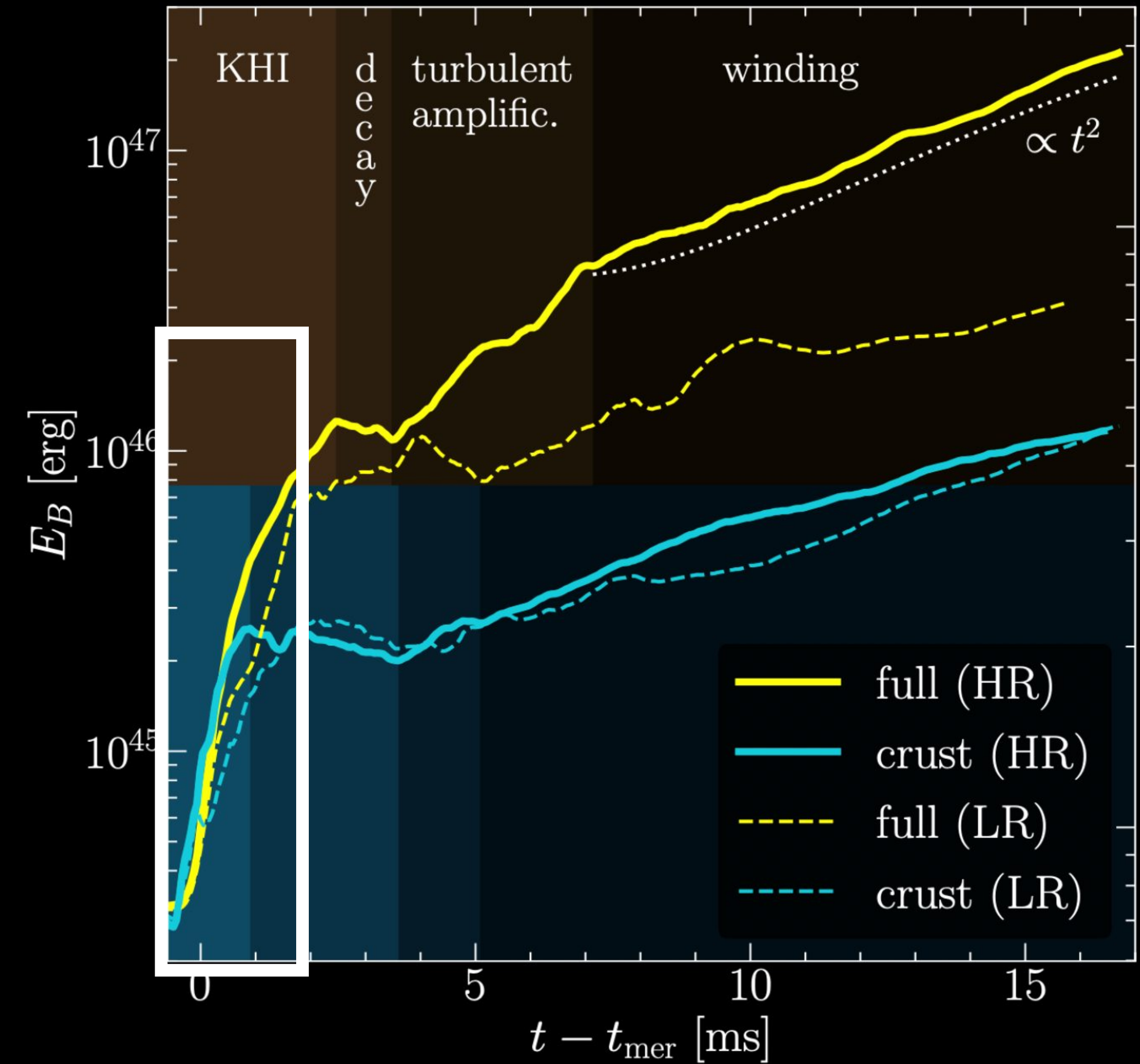


Volume integral energies — stage 1

Local calculation

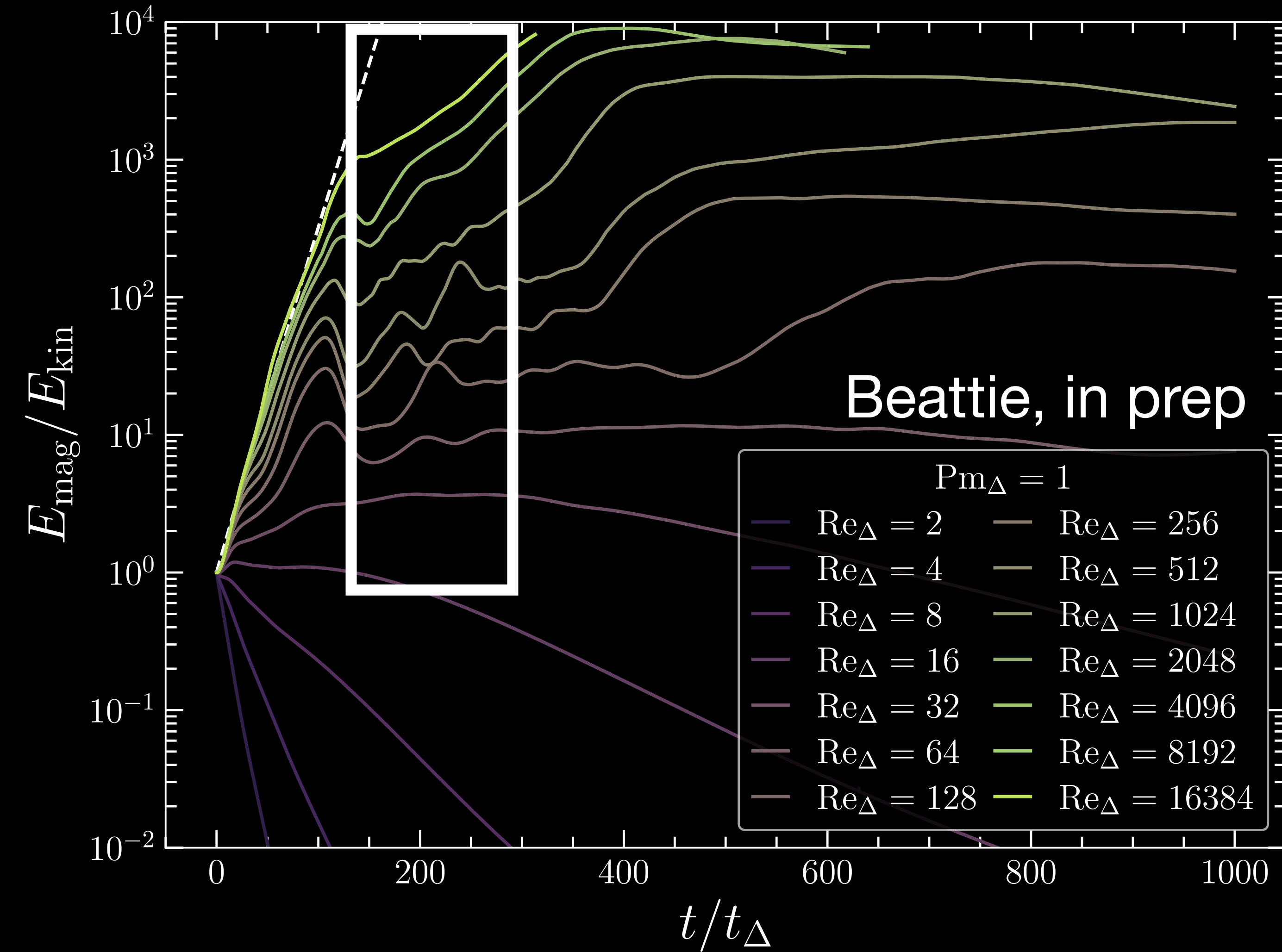


Global calculation

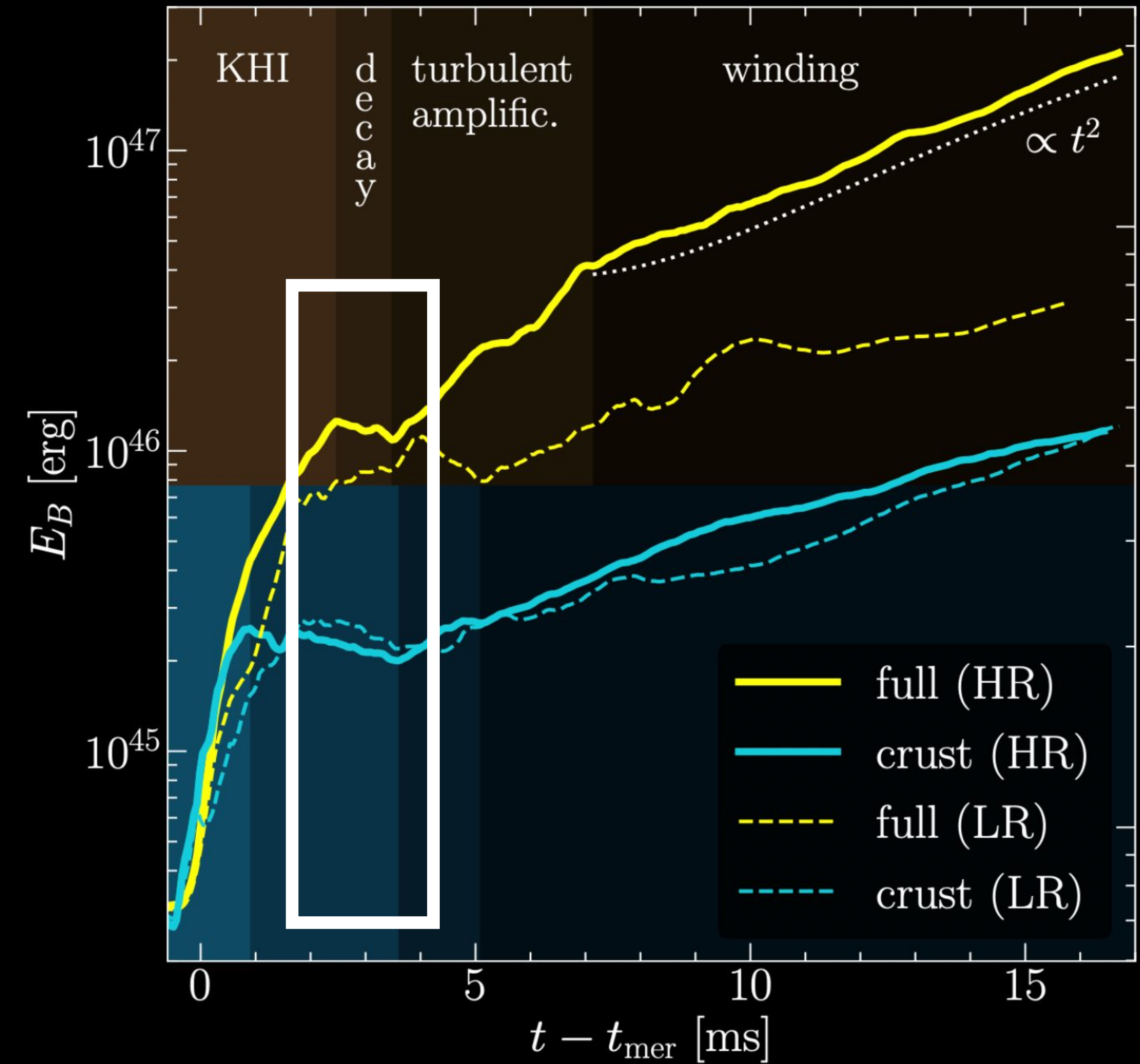


Volume integral energies — stage 2

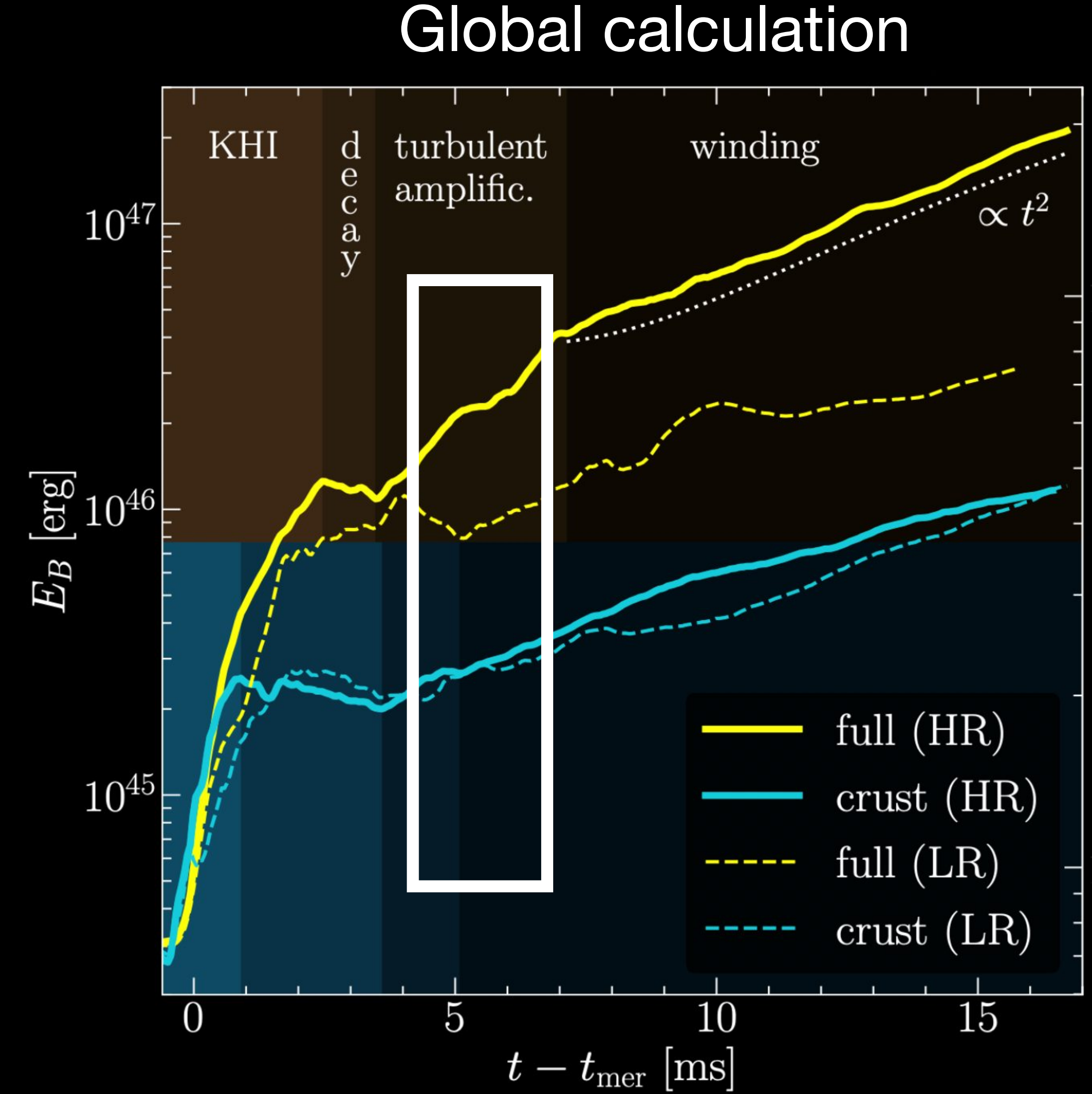
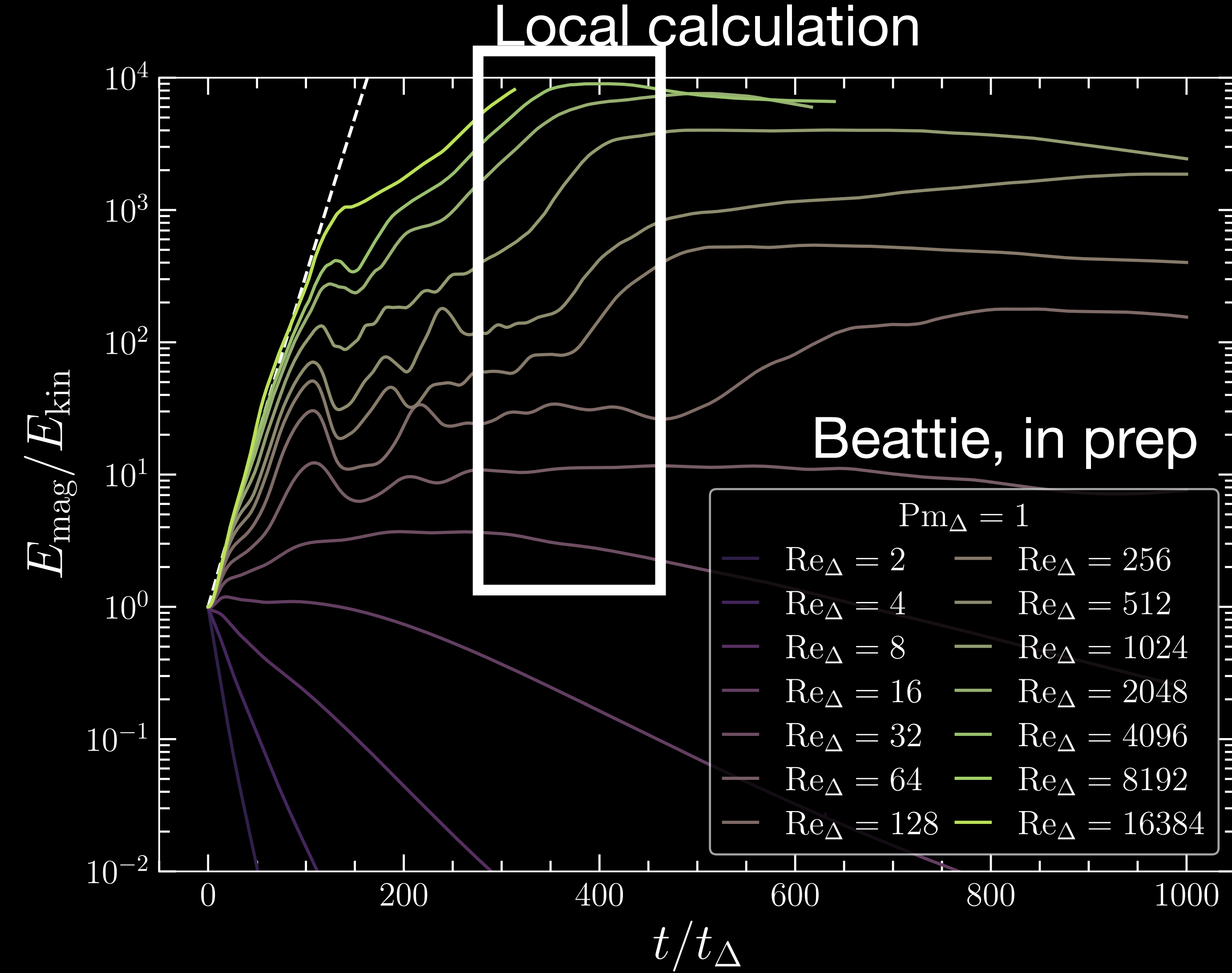
Local calculation



Global calculation

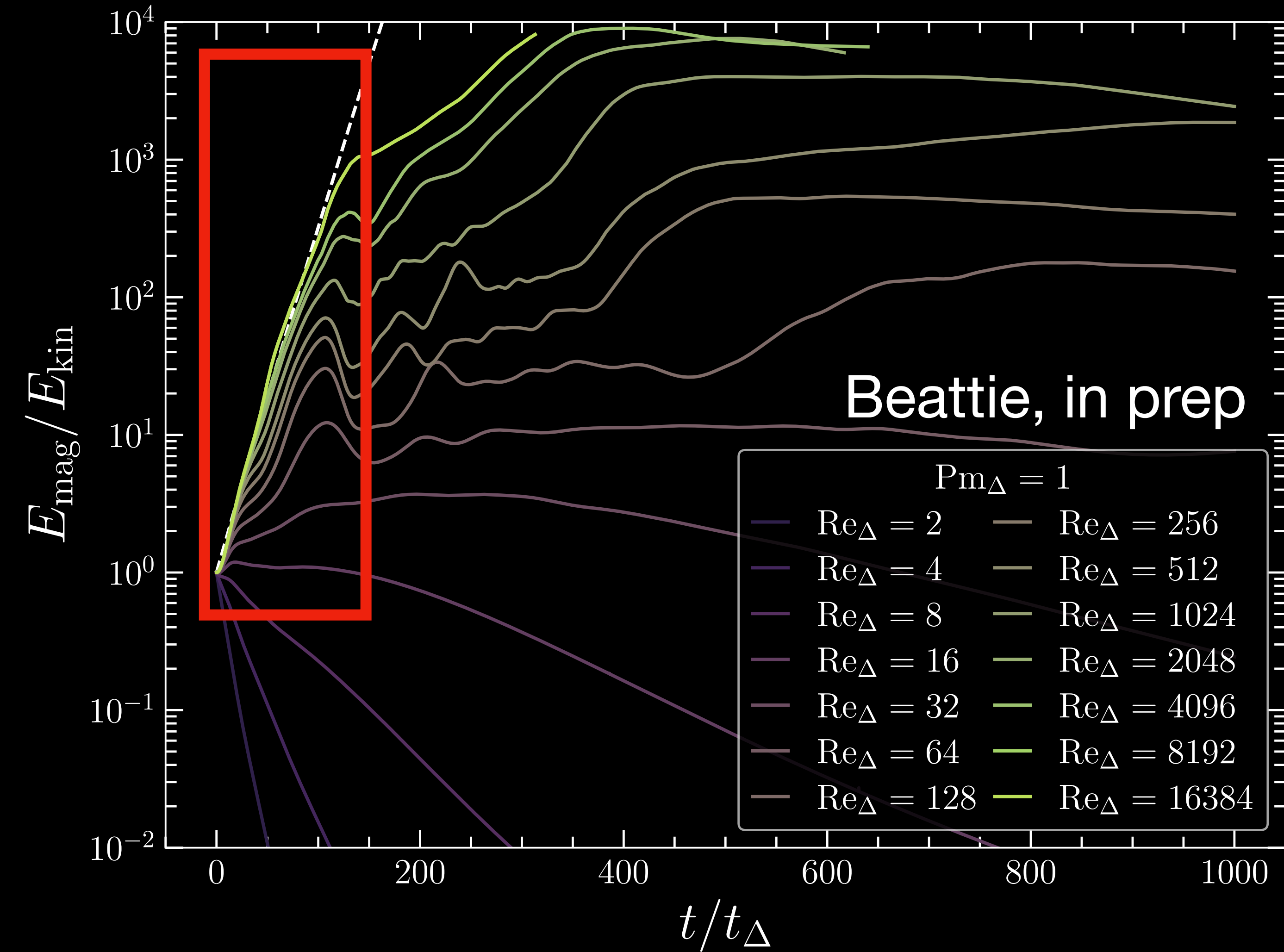


Volume integral energies — stage 3

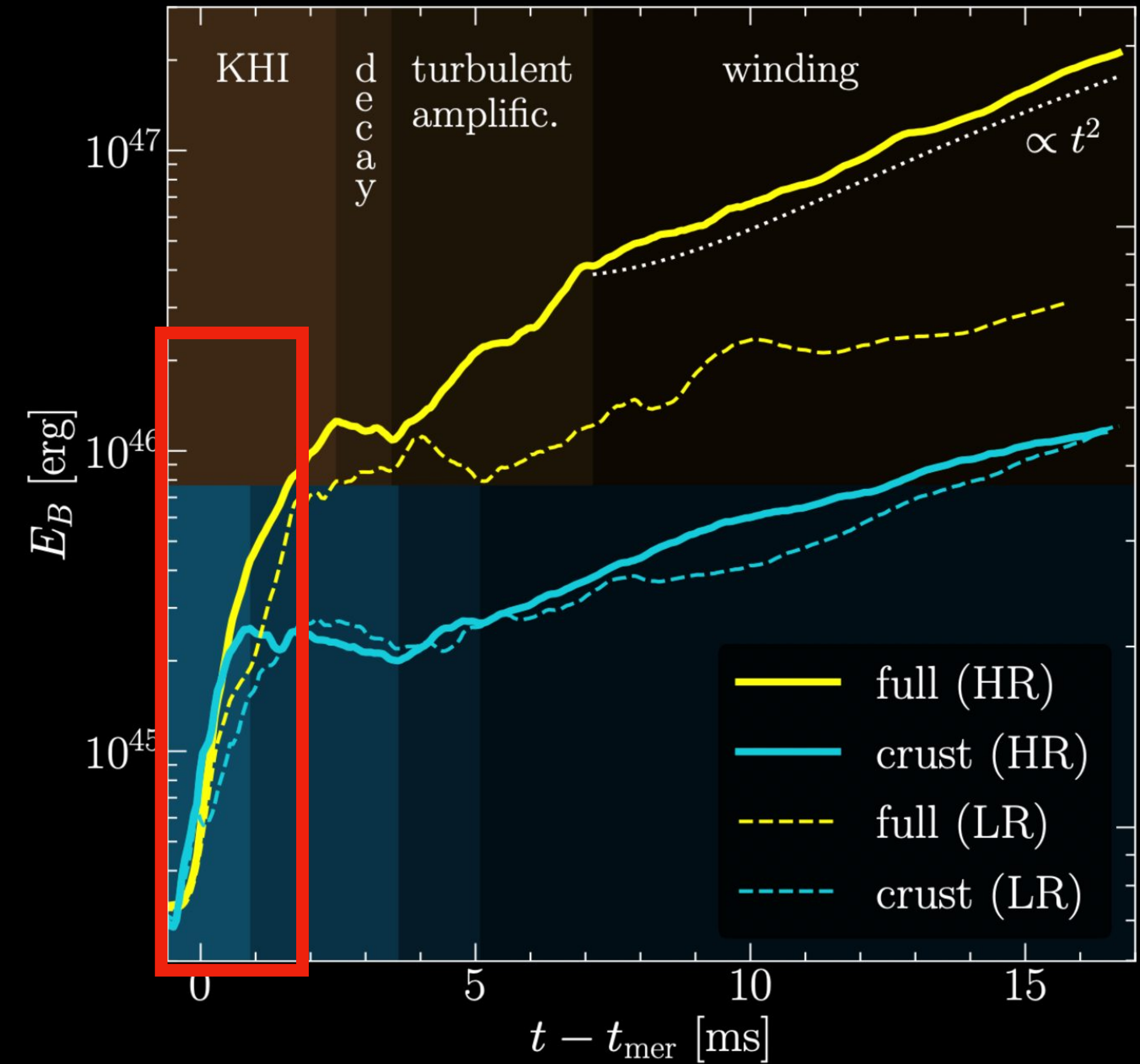


Volume integral energies — stage 1 — begin here

Local calculation



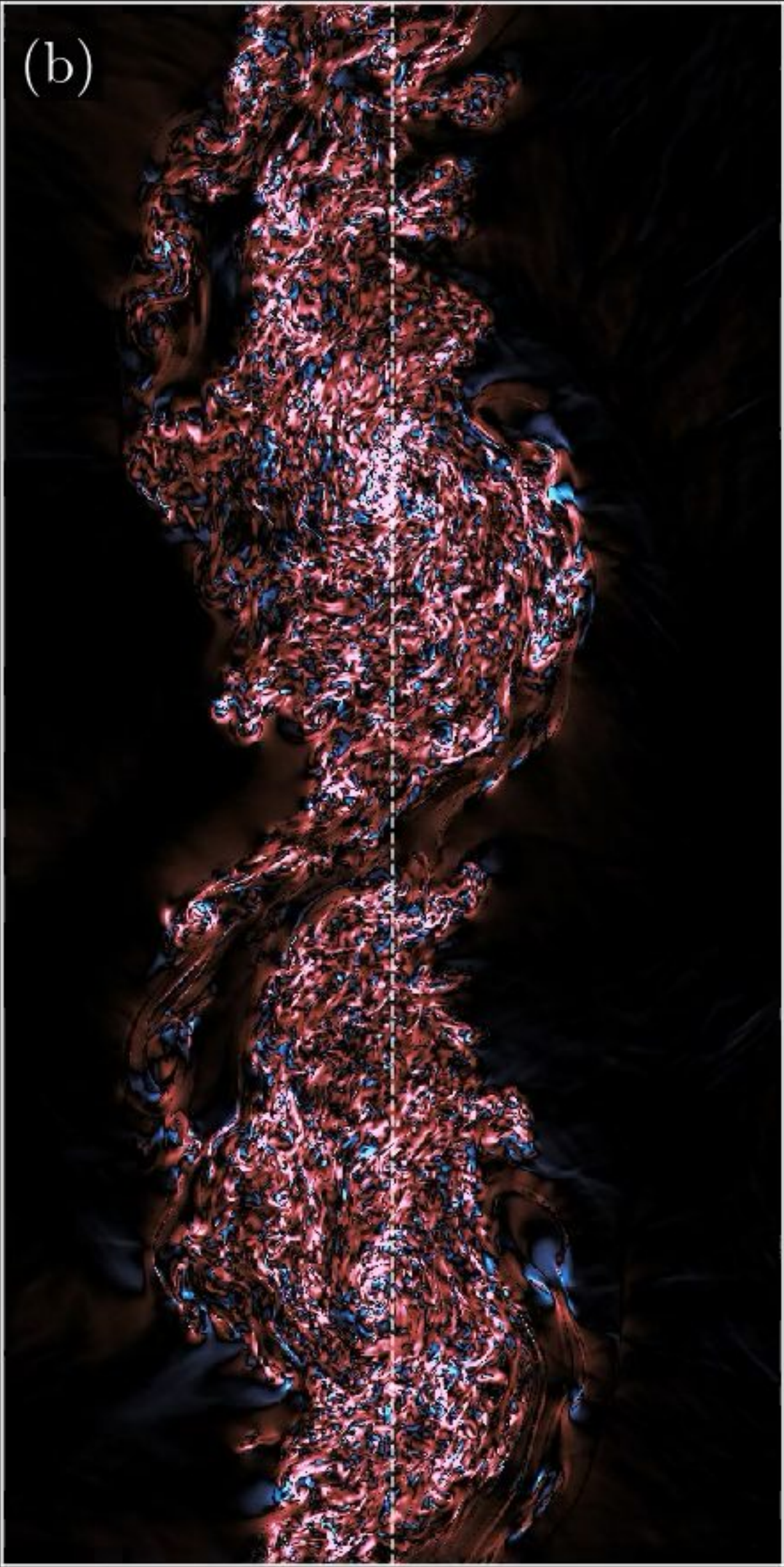
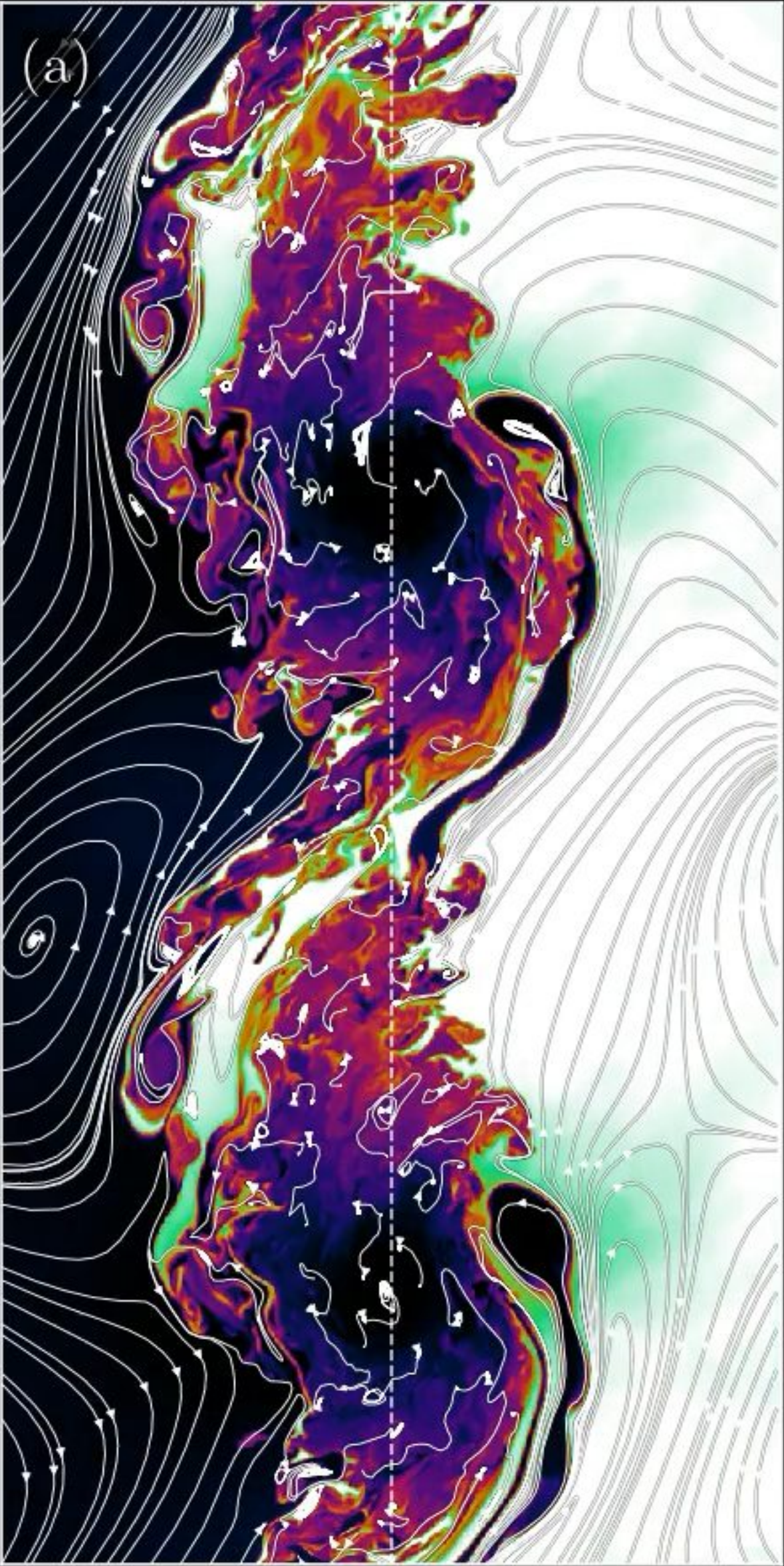
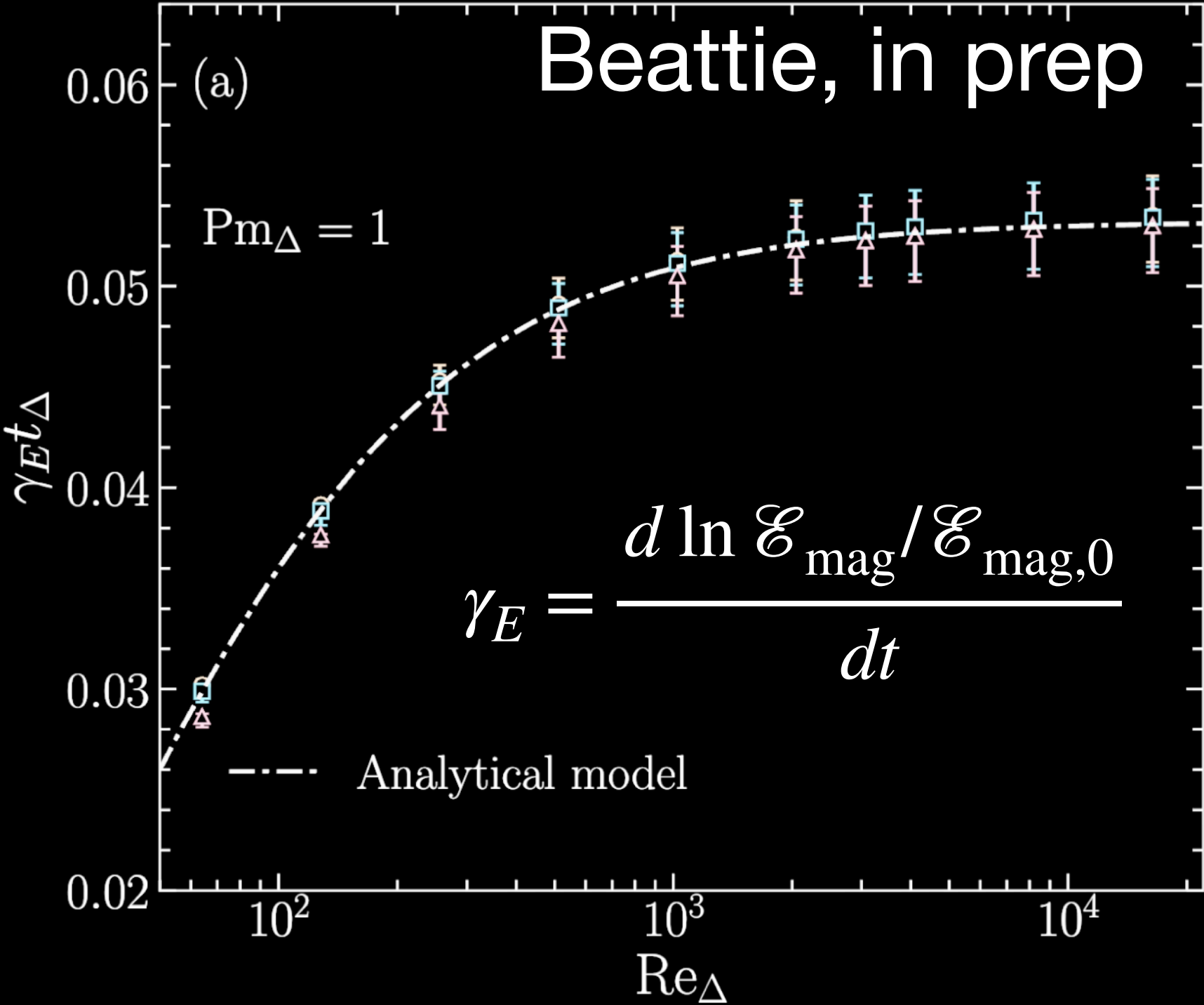
Global calculation



Stage 1: visco-resistive Ω (winding) dynamo

Mass density

Dynamo flux



resistive damping ideal shear

$$\frac{\mathcal{E}_{\text{mag}}}{\mathcal{E}_{\text{mag},0}} = e^{-\alpha\tau/\text{Rm}_\Delta} \left(1 + a\tau + \frac{b\tau^2}{\sqrt{D}} e^{-\beta\tau^3/\text{Rm}_\Delta D} \right)$$

mean shear layer broadening

fixed by seed field

$$\tau = t/t_\Delta$$

Stage 1: visco-resistive Ω (winding) dynamo

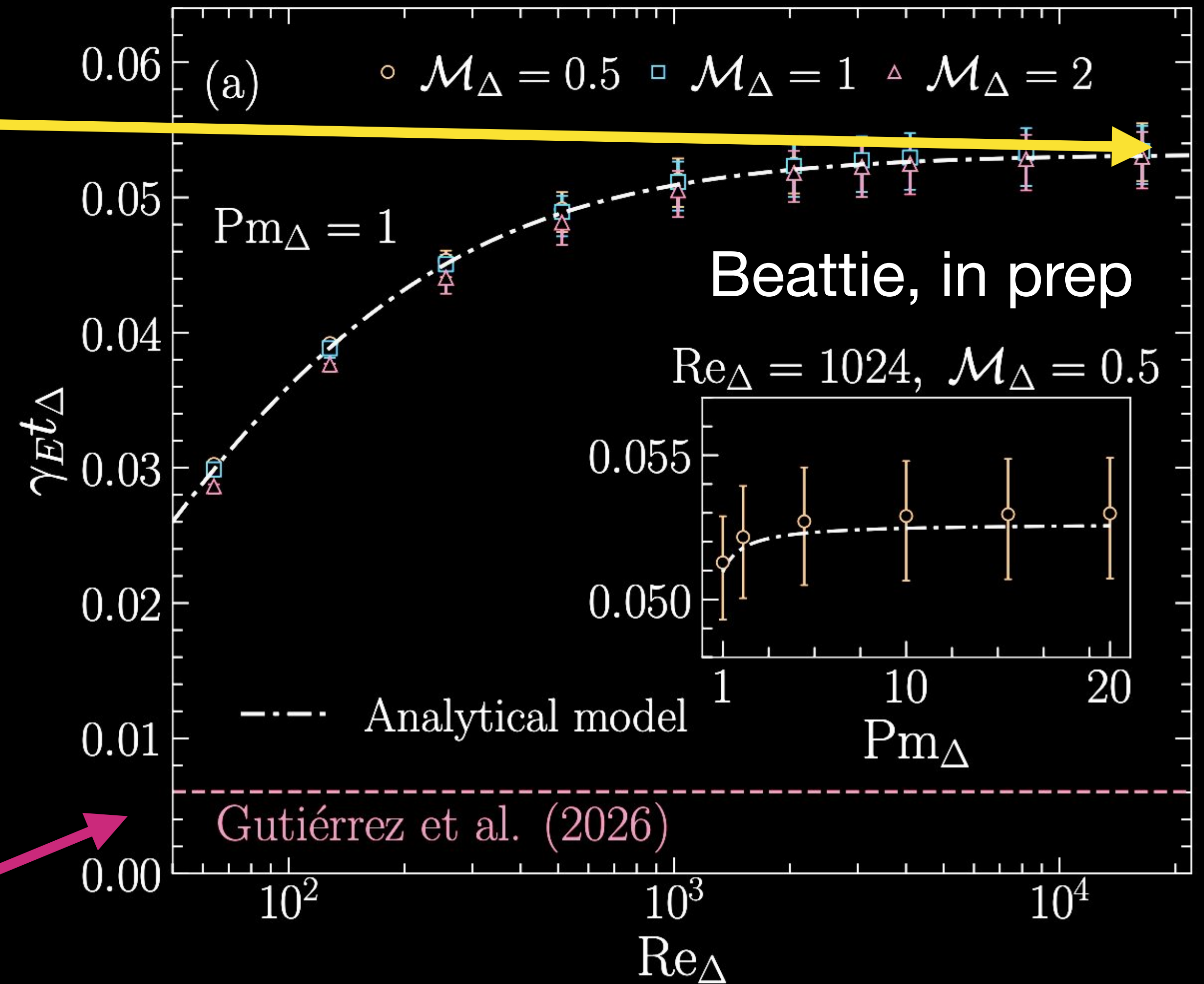
Our $\text{Re}_\Delta \rightarrow \infty$ result:

$$\gamma \sim 10^2 (\text{ms})^{-1}$$

Highest resolution global simulation:

$$\gamma \sim 10 (\text{ms})^{-1}$$

Gutiérrez + (2026)



Extreme challenge for global simulations!
Cannot get the right growth rate!

One of the most extreme magnetic dynamos

Merging Binary Neutron Stars

1. How does the multi-stage KHI-driven dynamo work?

Stage 1: visco-resistive winding, ...

2. Can we form the right ingredients for jet launching during the KHI

3. Do global calculations resolve the important aspects of the dynamo (e.g., remember kinematic growth rate are a function of viscosity)?

Order of magnitude difference in growth rates between resolved local and unresolved global :(

One of the most extreme magnetic dynamos

Merging Binary Neutron Stars

1. How does the multi-stage KHI-driven dynamo work?

Stage 1: visco-resistive winding, ... Stage 2, Stage 3... come and chat!

2. Can we form the right ingredients for jet launching during the KHI

YES! But ask me how.... involves a secondary instability in 3D!

3. Do global calculations resolve the important aspects of the dynamo (e.g., remember kinematic growth rate are a function of viscosity)?

Order of magnitude difference in growth rates between resolved local and unresolved global :(

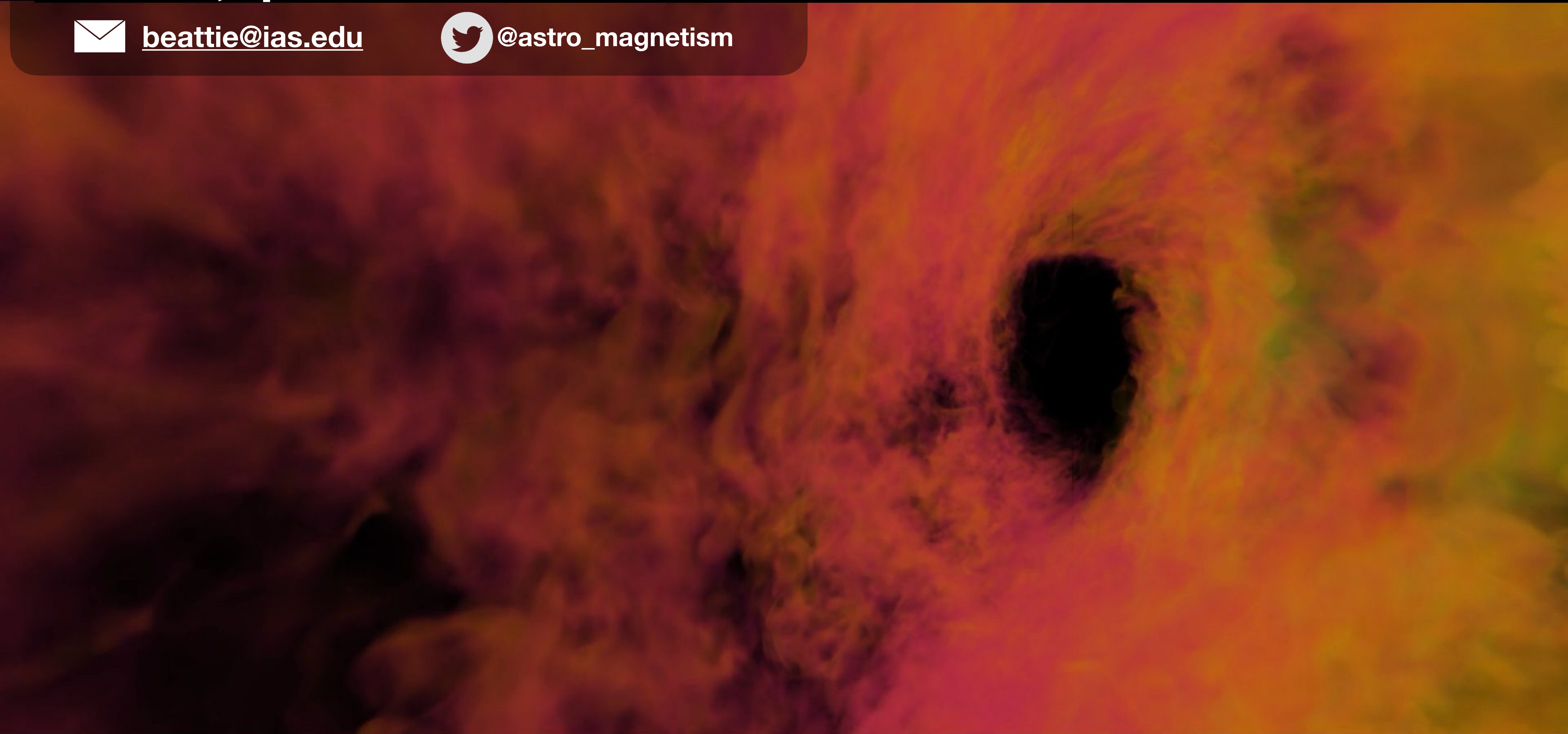
Thanks, questions?



beattie@ias.edu



[@astro_magnetism](https://twitter.com/astro_magnetism)



$\ln(\rho/\rho_0)$

-0.6

-0.3

0

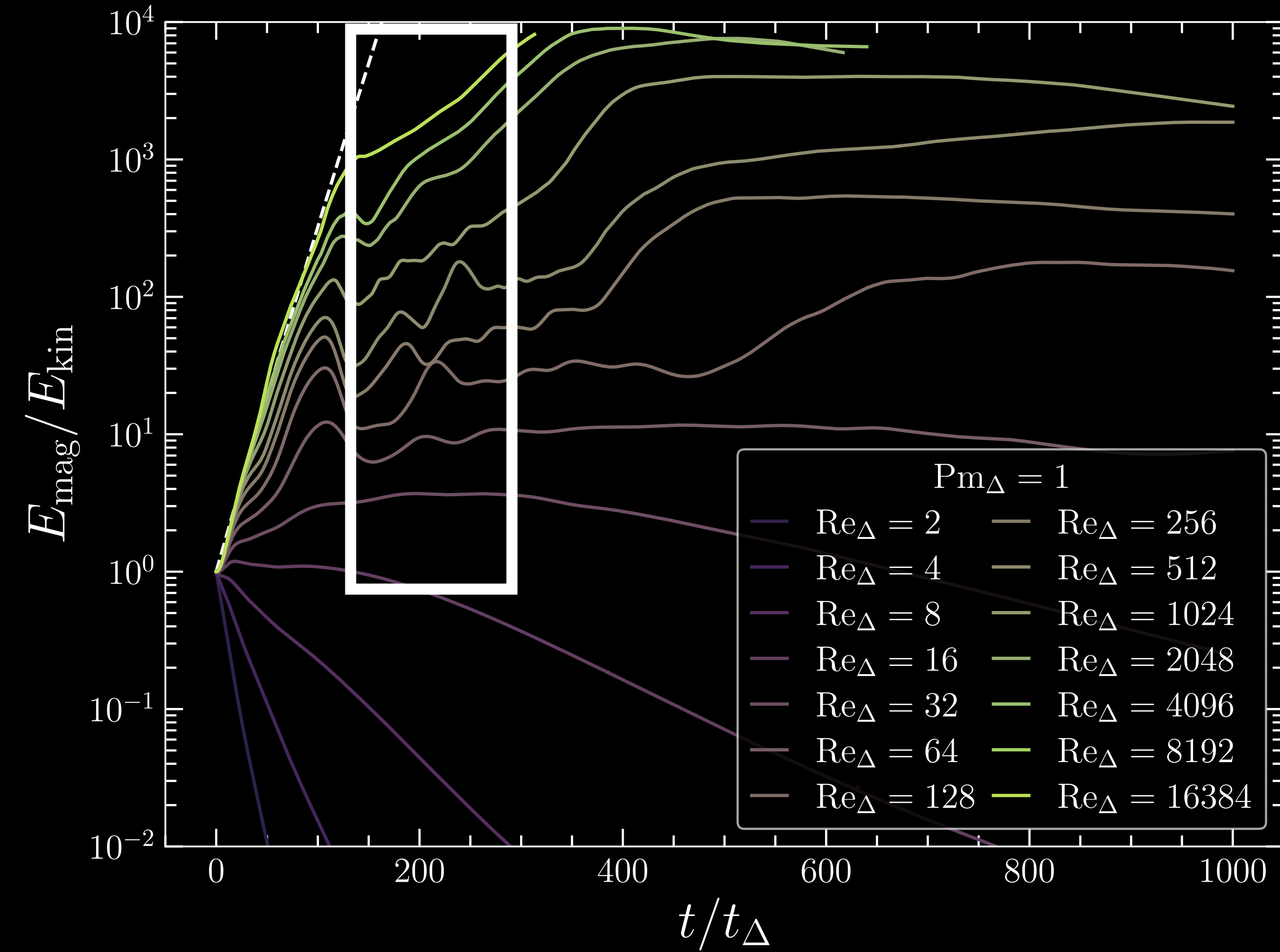
0.3

0.6

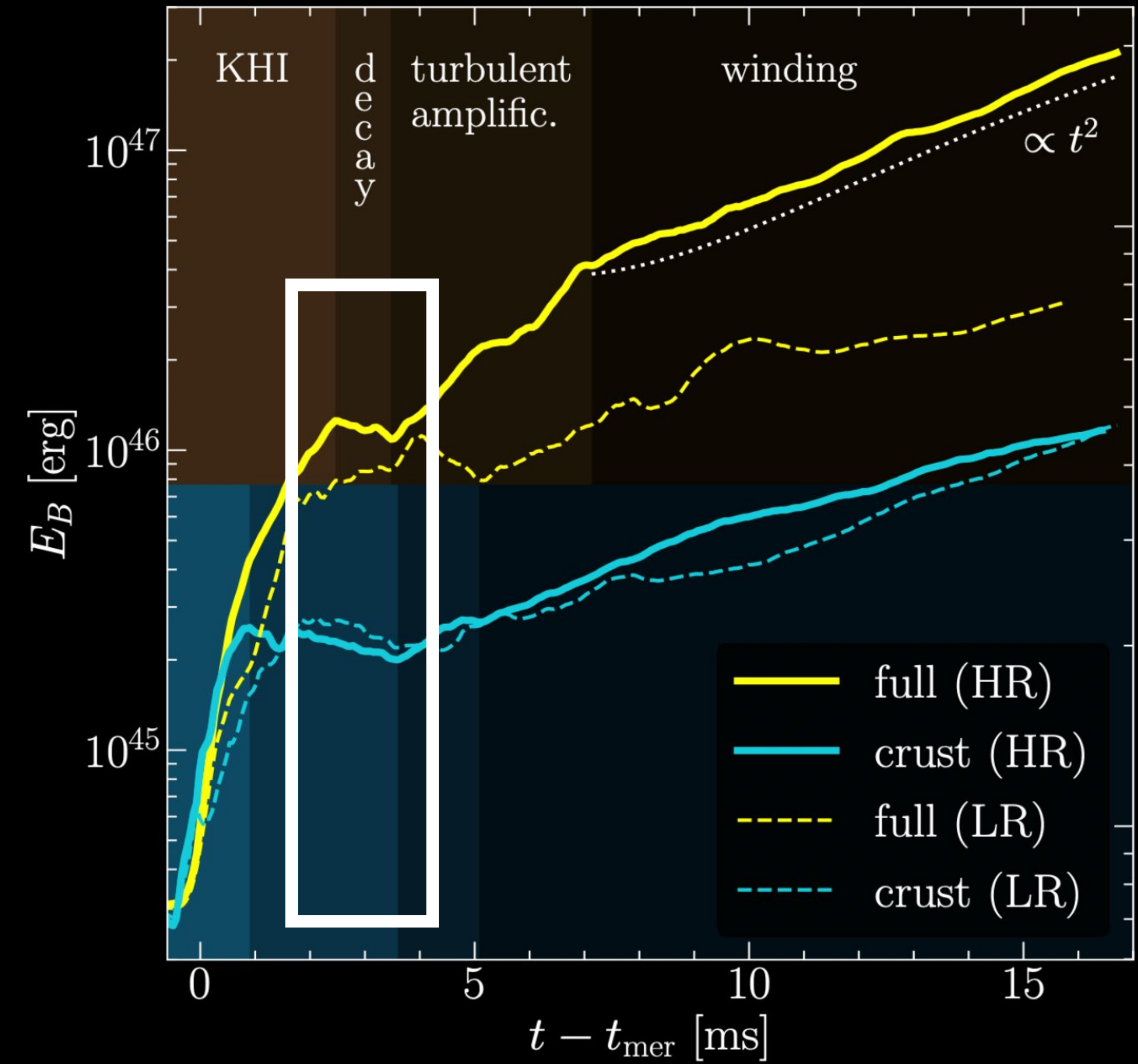
Extra slides

Volume integral energies — stage 2

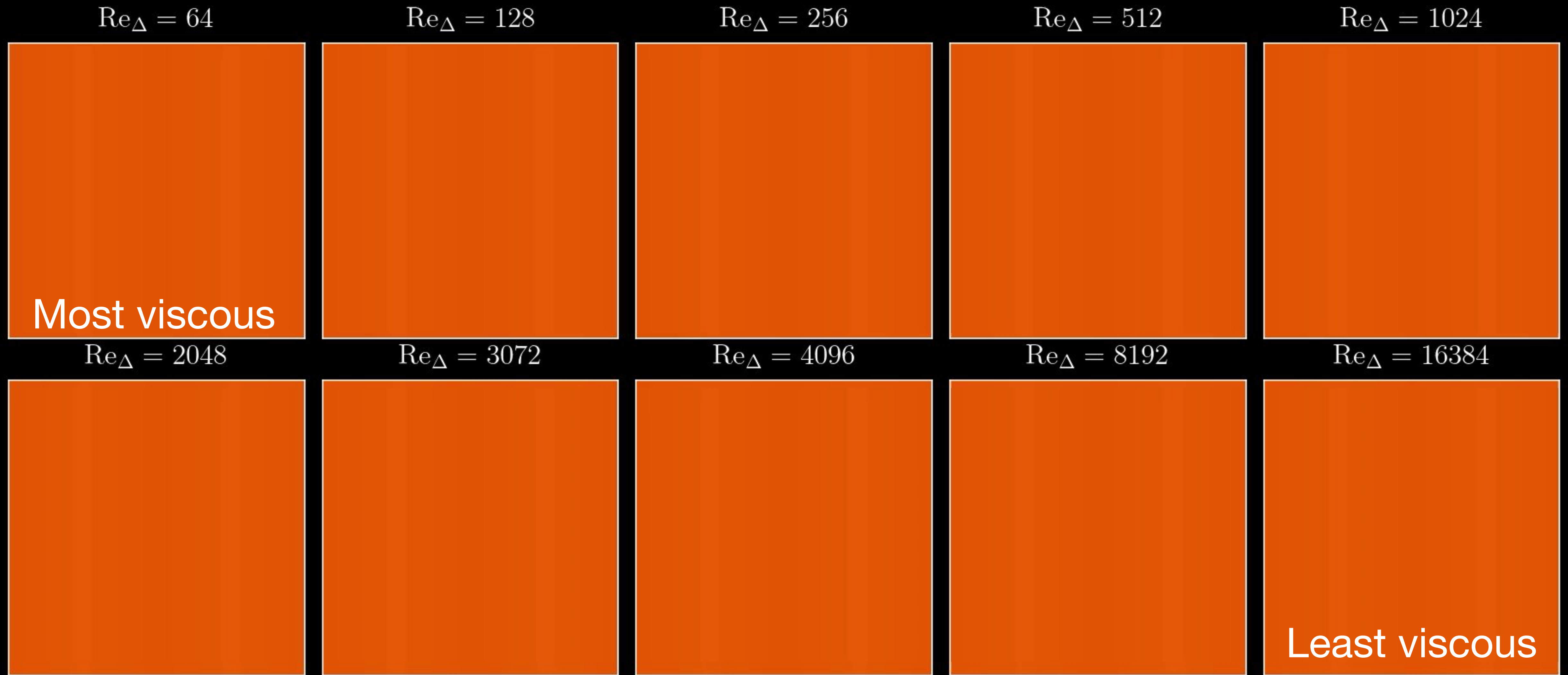
Local calculation



Global calculation



Stage 2: the wait...



$$\text{Re}_\Delta = \frac{\delta_0 \Delta U}{\nu}$$

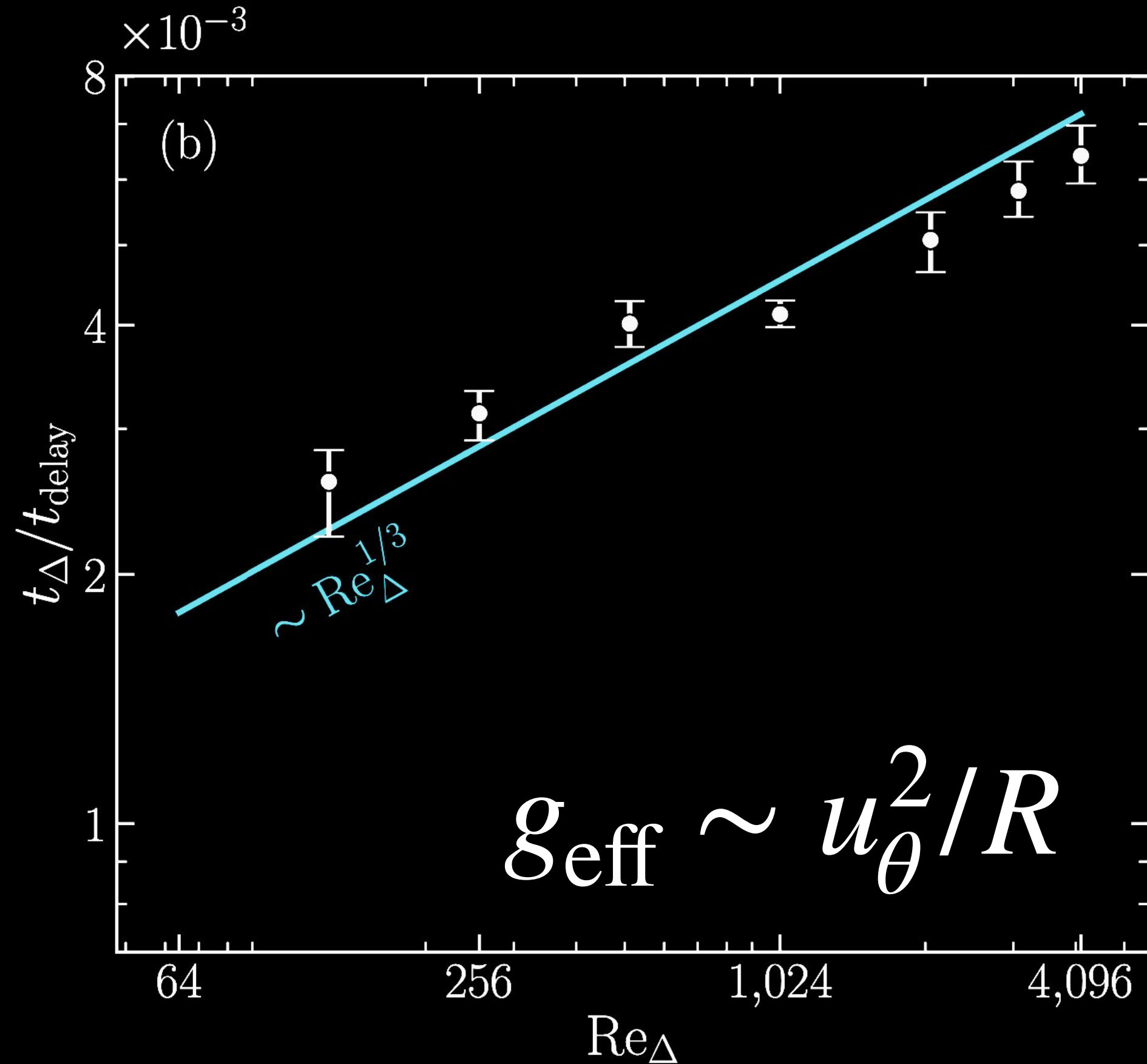


Beattie, in prep

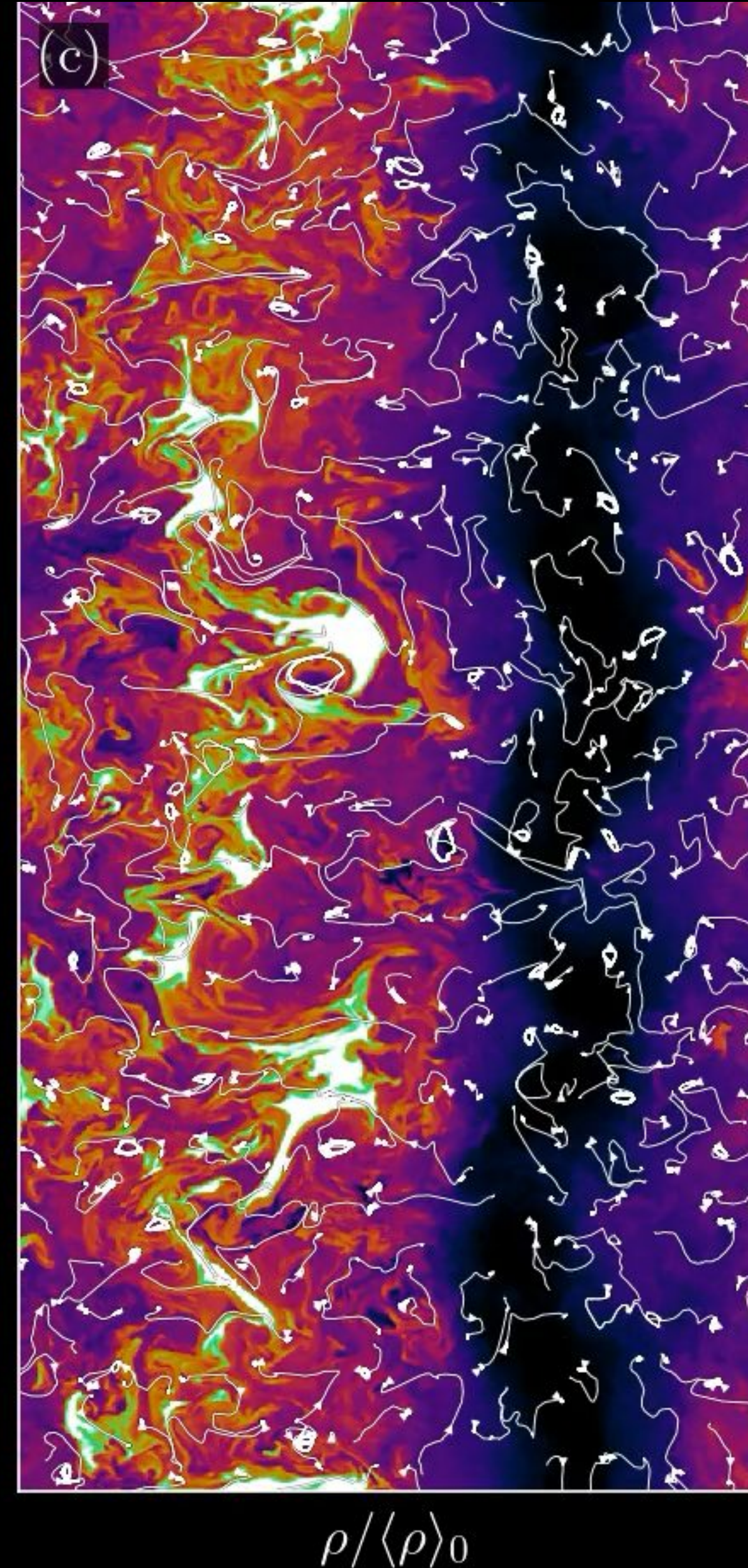
Stage 2: the wait

The fastest growing mode for viscous RTI is:

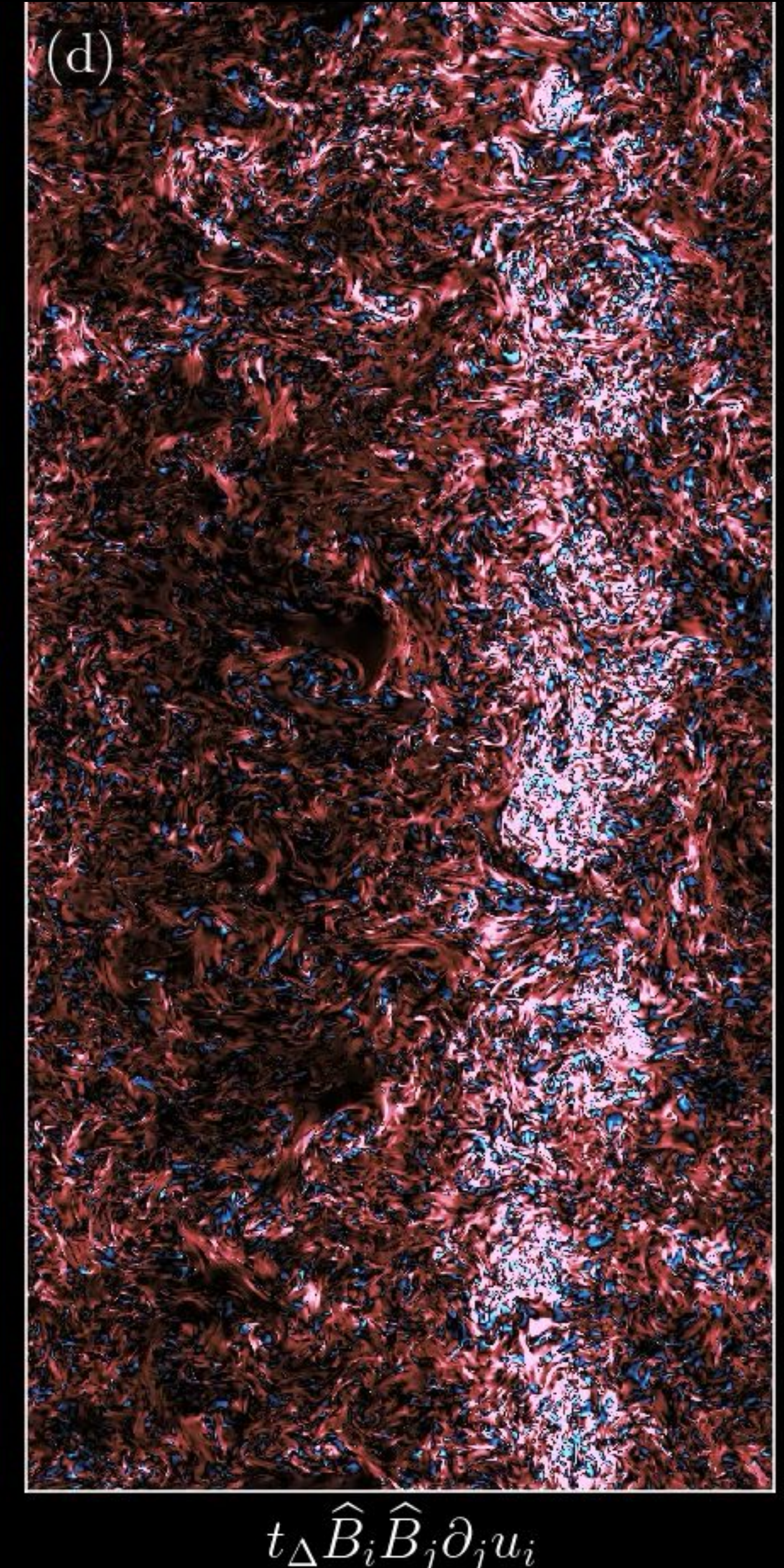
$$\gamma_{\text{RTI,max}} t_{\Delta} = \frac{3}{4^{4/3}} \left[\frac{A g_{\text{eff}} \delta_0}{(\Delta U)^2} \right]^{2/3} \text{Re}_{\Delta}^{1/3}$$



Mass density



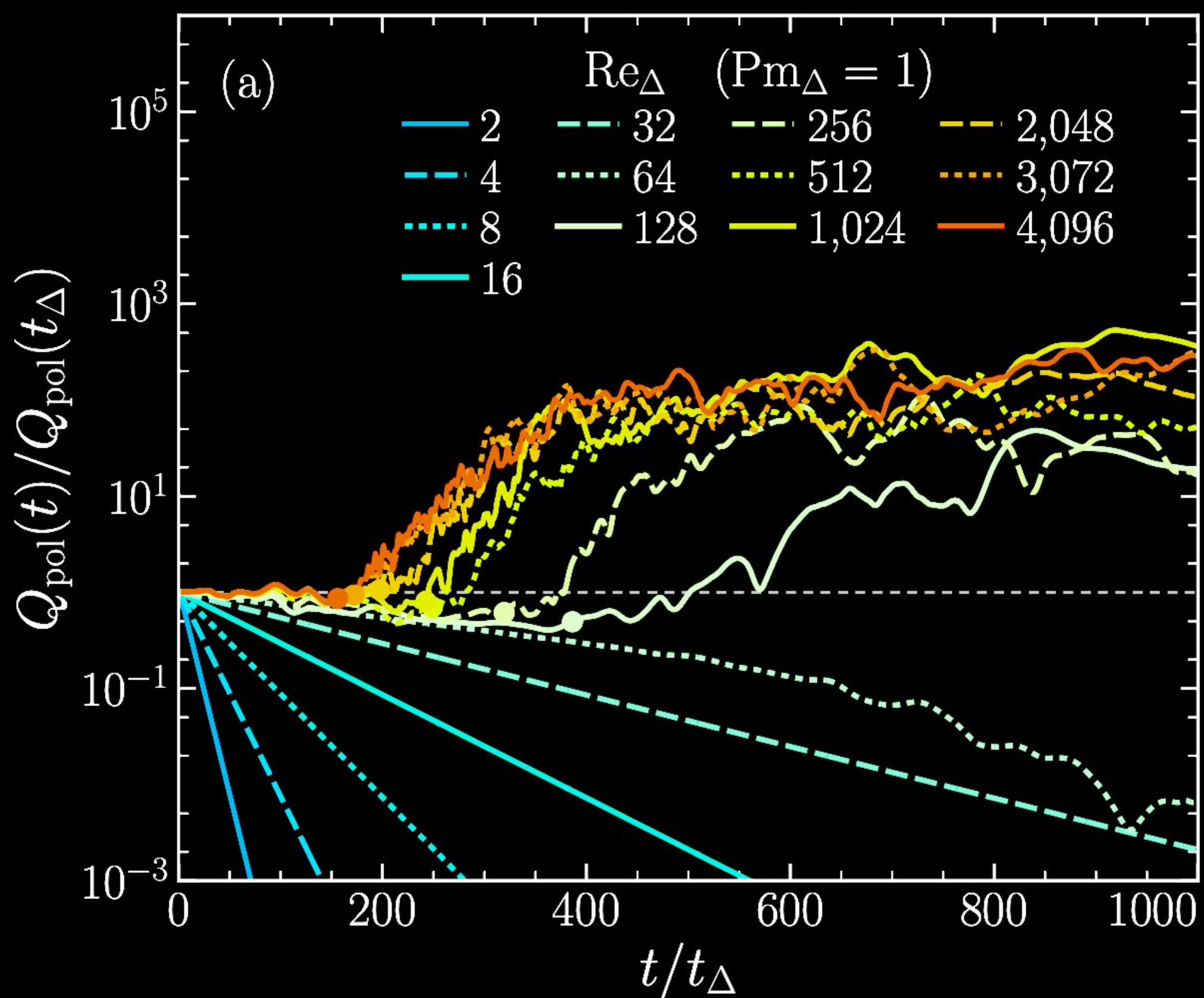
Dynamo flux



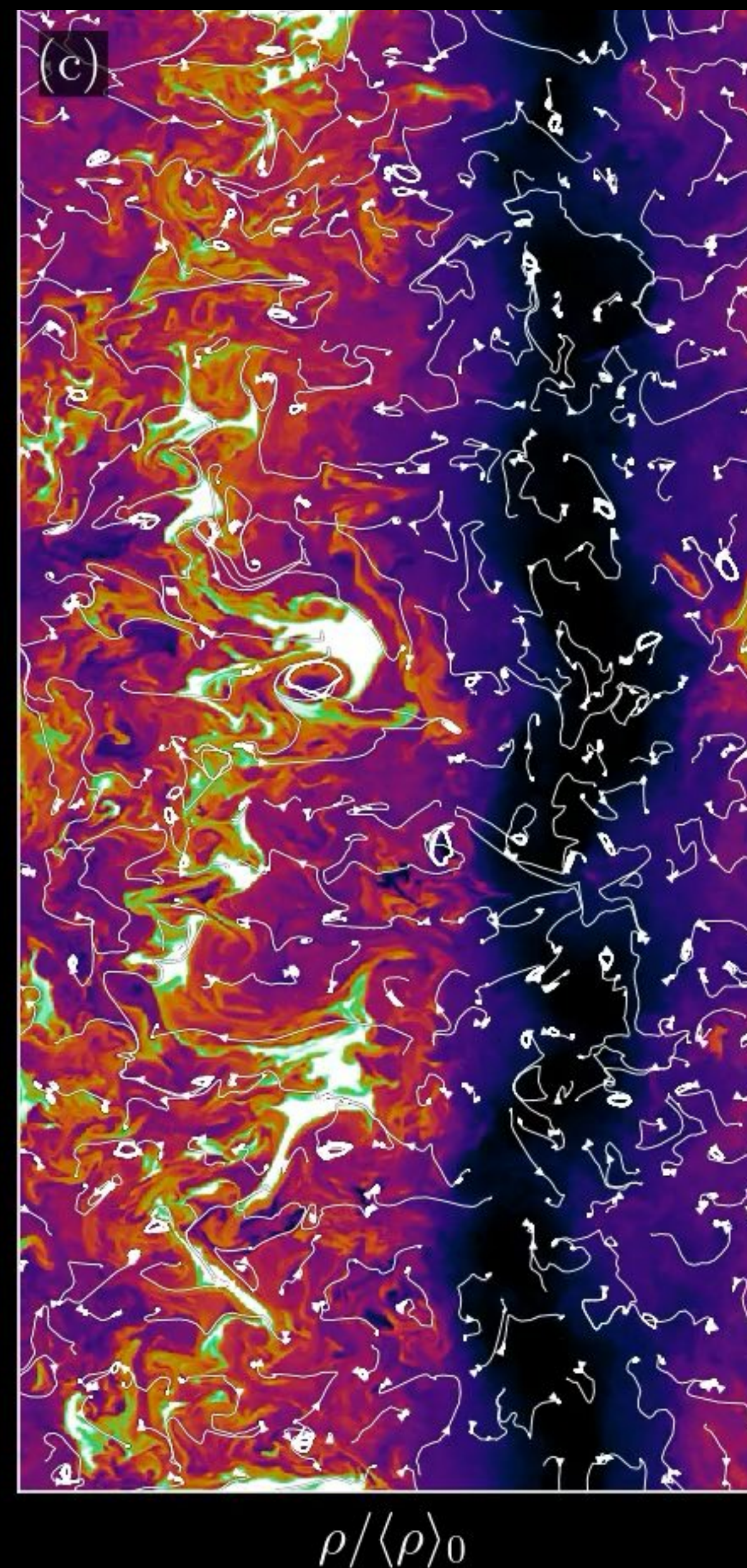
Stage 3: the wait

The fastest growing mode for viscous RTI is:

$$\gamma_{\text{RTI,max}} t_{\Delta} = \frac{3}{4^{4/3}} \left[\frac{A g_{\text{eff}} \delta_0}{(\Delta U)^2} \right]^{2/3} \text{Re}_{\Delta}^{1/3}$$



Mass density



Dynamo flux

